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UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-K

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2022

or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from_____ to _____

Commission File Number 001-37443

Univar Solutions Inc.

(Exact name of registrant as specified in its charter)

Delaware	**26-1251958**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**

3075 Highland Parkway, Suite 200	**Downers Grove,**	**Illinois**	**60515**
(Address of principal executive offices)			**(Zip Code)**

Registrant's telephone number, including area code: (331) 777-6000

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock ($0.01 par value)	**UNVR**	**New York Stock Exchange**

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐ Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

Aggregate market value of common stock held by non-affiliates of the registrant on June 30, 2022: $4.1 billion (see Item 12, under Part III hereof), based on a closing price of registrant's common stock of $24.87 per share.

At February 17, 2023, 158,114,006 shares of the registrant's common stock were outstanding.

Documents Incorporated by Reference

Certain portions of the registrant's Proxy Statement for the Annual Meeting of Stockholders to be held May 4, 2023 and to be filed within 120 days after the registrant's fiscal year ended December 31, 2022 (hereinafter referred to as "Proxy Statement") are incorporated by reference into Part III.

Univar Solutions Inc.

Form 10-K

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SUPPLEMENTAL INFORMATION

In this Annual Report on Form 10-K, "Univar Solutions," "Company," "we," "our" and "us" refer to Univar Solutions Inc., a Delaware corporation, and its subsidiaries included in the consolidated financial statements, except as otherwise indicated or as the context otherwise requires. Our fiscal year ends on December 31, and references to "fiscal" when used in reference to any twelve-month period ended December 31 refer to our fiscal years ended December 31. The term "GAAP" refers to accounting principles generally accepted in the United States of America.

Forward-looking statements and information

Certain parts of this Annual Report on Form 10-K contain forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. Forward-looking statements are generally accompanied by words such as "believes," "expects," "may," "will," "should," "could," "seeks," "intends," "plans," "estimates," "anticipates" or other comparable terms. All forward-looking statements made in this Annual Report on Form 10-K are qualified by these cautionary statements.

Any forward-looking statements represent our views only as of the date of this report and should not be relied upon as representing our views as of any subsequent date, and we undertake no obligation, other than as may be required by law, to update any forward-looking statement. We caution you that forward-looking statements are not guarantees of future performance and that our actual performance may differ materially from those made in or suggested by the forward-looking statements contained in this Annual Report on Form 10-K. Forward-looking statements include, but are not limited to, statements about:

- demand for products, systems and services that meet growing customer sustainability standards, expectations and preferences and our ability to provide such products, systems and services to maintain our competitive position;
- our ability to solve customer technical challenges and accelerate product development cycles;
- our ability to sell specialty products at higher profit;
- our liquidity outlook and the funding thereof, and cash requirements and adequacy of resources to fund them;
- the impact of ongoing tax guidance and interpretations;
- the impact of public-health emergencies, weather events and economic conditions on our end markets, operations, financial condition and operating results;
- our expense control and cost reduction plans and other strategic plans and initiatives;
- our human capital management strategies;
- significant factors that may adversely affect us and our industry;
- the outcome and effect of ongoing and future legal proceedings;
- market conditions and outlook;
- return of capital to stockholders;
- future contributions to, and withdrawal liability in connection with, our pension plans and cash payments for postretirement benefits; and
- future capital expenditures and investments.

Potential factors that could affect such forward-looking statements include, among others:

- general economic conditions, particularly fluctuations in industrial production and consumption and the timing and extent of economic downturns, increased competitive pressures, including as a result of competitor consolidation;
- significant changes in the business strategies of producers or in the operations of our customers;
- delivery failures or hazards and risks related to our operations and the hazardous materials we handle, potential inability to obtain adequate insurance coverage;
- increased competitive pressures, including as a result of competitor consolidation;
- potential supply chain disruptions;
- significant changes in the pricing, demand and availability of chemicals;
- potential cybersecurity incidents, including security breaches;
- our indebtedness, the restrictions imposed by, and costs associated with, our debt instruments, and our ability to obtain additional financing;
- the broad spectrum of laws and regulations that we are subject to, including extensive environmental, health and safety laws and regulations and changes in tax laws;
- an inability to generate sufficient working capital;
- transportation related challenges, including increases in transportation and fuel costs, changes in our relationship with third party transportation providers, and ability to attract and retain qualified drivers;
- accidents, safety failures, environmental damage, product quality issues;
- ongoing litigation, potential product liability claims and recalls, and other environmental, legal and regulatory risks;

- challenges associated with international operations;
- exposure to interest rate and currency fluctuations;
- an inability to integrate the business and systems of companies we acquire, including failure to realize the anticipated benefits of such acquisitions;
- possible impairment of goodwill and intangible assets;
- our ability to attract or retain a qualified and diverse workforce;
- negative developments affecting our pension plans and multi-employer pensions;
- labor disruptions associated with the unionized portion of our workforce;
- our ability to execute on our ability to execute on our initiatives and goals related to environmental, social, and governance (ESG) matters and the increasing legal and regulatory focus on ESG; and
- the other factors described in “Risk Factors” in Item 1A of this Annual Report on Form 10-K.

PART I

ITEM 1. BUSINESS

General

In 1924, George Van Waters and Nat Rogers opened a brokerage business in Seattle, Washington, buying and selling naval supplies, paint, raw materials and cotton linters. Nearly 100 years later, and after many different incarnations of the business, we closed our initial public offering in 2015. In 2019, we acquired Nexeo Solutions, Inc. (“Nexeo”), a leading global chemicals distributor. The acquisition expanded and strengthened our presence in North America and provided the opportunity to create the largest North American sales force in chemical and ingredients distribution.

Today we are a leading global distributor of commodity and specialty chemicals and ingredients, as well as a provider of value-added services, to customers across a wide range of diverse industries. We purchase chemicals and ingredients from producers worldwide to warehouse, repackage, blend, dilute, transport and sell those materials safely to nearly 100,000 customer locations across approximately 120 countries. We provide application development and technical advice to customers formulating new products and have a network of Solution Centers to support customer development activities. We operate an extensive worldwide chemical and ingredient distribution network, including approximately 9,700 employees and approximately 600 facilities, serviced by the following: approximately 4,900 tractors, tankers and trailers; and approximately 2,500 railcars, 120 rail and barge terminals and 20 deep sea terminals, focused on timely and safe delivery to a spectrum of customers, large and small. Our purpose is to help keep our communities healthy, fed, clean and safe.

Chemical and ingredient producers rely on us to safely warehouse, repackage, transport, market and sell their products as a way to expand their market access, enhance their geographic reach and lower their cost to serve and grow their business. Customers who purchase products and services from us benefit from a lower total cost of ownership, as they are able to simplify their sourcing process by outsourcing functions to us, such as “just-in-time delivery,” product availability and selection, packaging, mixing and blending. They also rely on us for safe and secure delivery and the off-loading of chemicals, compliant with increasing local and federal regulations. Additionally, customers and suppliers depend on our deep end market knowledge and technical expertise to provide formulation and recipe development services to help meet the latest market trends and adhere to regulatory requirements across our global network of Solution Centers.

The effects of market conditions on our operations are discussed in Part II, Item 7, “Management’s Discussion and Analysis of Financial Condition and Results of Operations.”

Recent Developments

In July 2022, we acquired Vicom Distribución Productos Quimicos, S.L. ("Vicom"), a leading regional specialty chemical distributor in Spain and Portugal.

In December 2021, we acquired Sweetmix Distribuidora de Materias Primas Industriais Ltda (“Sweetmix”), a food ingredients and coatings, adhesives, sealants and elastomers (“CASE”) specialty chemical distribution company in Brazil.

In April 2021, we sold the Distrupol business within our EMEA segment.

In December 2020, we acquired the business of Zhuhai Techi Chem Silicone Industry Corporation (“Techi Chem”), a leading distributor of specialty silicone solutions used primarily for the CASE market within the China marketplace.

At the beginning of the fourth quarter of 2020, we decided to wind down our Canadian Agriculture wholesale distribution business, which was operationally completed by December 31, 2020.

In November 2020, we sold our Canadian Agriculture services business and in September 2020, we sold our industrial spill and emergency response businesses.

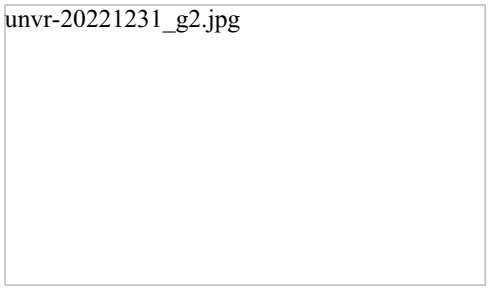
See “Note 3: Business combinations” and “Note 4: Dispositions” in Item 8 of this Annual Report on Form 10-K for additional information regarding the acquisitions and dispositions noted above.

Segments

Our business is organized and managed in four geographical segments: Univar Solutions USA (“USA”), Univar Solutions Europe and the Middle East and Africa (“EMEA”), Univar Solutions Canada (“Canada”) and Univar Solutions Latin America (“LATAM”), which includes developing businesses in Latin America and the Asia-Pacific region. For additional information on our geographical segments, see “Note 20: Segments” in Item 8 of this Annual Report on Form 10-K.

The following charts reflect the breakdown by segment of 2022 net sales of $11.5 billion and gross profit (exclusive of depreciation) of $2.8 billion.





* See definition of gross profit (exclusive of depreciation) under "Non-GAAP Financial Measures" in Item 7 of this Annual Report on Form 10-K for more information.

USA

We are the largest commodity and specialty chemical and ingredient distributor in the United States ("US"). Our locations span the US, with personnel strategically located where customers and suppliers need them and ready to provide agile, reliable support for their businesses. We serve as a marketing and sales arm for our suppliers, helping to bring their chemicals and ingredients to market in a manner that creates additional value for our customers' end products. We blend, mix and repackage bulk chemicals for shipment by our transportation fleet, as well as common carriers, and utilize our network of terminal and supply locations to optimize bulk shipment deliveries. In the US, we use centralized shared information technology systems, raw materials procurement, logistics, route operations and producer relationship management in an effort to benefit from economies of scale and improve cost efficiency.

In addition to our broad commodity and specialties and services offerings, we distribute ingredients and provide specialized expertise. Leveraging our Solution Centers, we seek to provide value-added technical services such as formulation and recipe development, benchmark prototyping, product performance testing, chemical analysis, custom blending and other technical services.

Our offerings service a majority of the Life Sciences (formerly Consumer Solutions) and Industrial sectors in the US and specific end markets, which are discussed further below. Our sales force is organized through a combination of geographic and end market factors. Our bulk and local chemical distribution customers are serviced primarily by regional teams and our ingredients and specialties customers are serviced primarily by our end market and product-technically focused teams.

We believe our close proximity to customers, installed asset base, transportation and digital assets, and our extensive product knowledge and end market expertise, all serve as competitive advantages and provide sustainable value to our suppliers and customers.

EMEA

We maintain a strong presence in the United Kingdom and continental Europe with sales offices in 18 countries. This segment also includes two sales offices in the Middle East and Africa. Within this segment and where possible globally, we leverage centralized or shared information technology systems, raw materials procurement, logistics, route operations and producer relationships management to benefit from economies of scale and improve cost efficiency.

We support commodity and specialty chemical and ingredient distribution to customers primarily in our Industrial Solutions, Life Sciences and General Industrial end markets, with the heaviest focus in the CASE, food ingredients, beauty and personal care, pharmaceutical, and homecare and industrial cleaning submarkets. These strategies and customers are supported by Solution Centers throughout the EMEA region, with key centers of excellence locations in Paris, France and a state-of-the-art formulation laboratory opened in Essen, Germany in 2022.

Our technical sales force supports customers at a country level through strong end market expertise and key account management capabilities. We believe our scale and focused regional, industry and product expertise provide competitive advantages and sustainable value to our suppliers and customers.

Canada

We have distribution sites and Solution Centers located in all key geographies throughout Canada supporting commodity and specialty chemical and ingredient distribution across customers in our General Industrial, Industrial Solutions, Refining & Chemical Processing and Life Sciences end markets.

More specifically, in Eastern Canada, we have deep product knowledge in submarkets such as food ingredients, beauty and personal care, pharmaceutical ingredients & finished products, CASE, chemical manufacturing, homecare and industrial cleaning and mining. In Western Canada, our submarket expertise in forestry and energy (e.g., midstream gas pipeline, oil sands processing and oil refining) further complements our broad offerings within the country.

Our sales force is deployed through specialized account management across Canada to serve our focused customer end markets within ingredients and specialties, and through a geographic sales district model to support the bulk and local chemical distribution end markets. We leverage shared information systems and technology within Canada.

LATAM

Our LATAM operations are regionally focused with distribution sites and Solution Centers located principally in Brazil, Mexico and Colombia. We work to meet the needs of our customers through a team of chemical sales, product management and supply chain professionals. As previously noted, our presence in Brazil grew in December 2021 with the acquisition of Sweetmix.

Our offerings support commodity and specialty chemical and ingredient distribution customers primarily in our General Industrial, Industrial Solutions and Life Sciences end markets, with submarkets largely focused on CASE, beauty and personal care, chemical manufacturing and food ingredients. We also provide formulation services for crop protection manufacturers in Brazil.

Products and End Markets

We source and inventory commodity and specialty chemicals and ingredients in large quantities such as barge loads, railcars or full truck loads directly from producers and break down the bulk quantities to repackage, market, sell and distribute smaller quantities to our customers.

In addition to selling and distributing commodity and specialty chemicals and ingredients, we use our transportation and warehousing infrastructure, along with our broad knowledge of chemicals and hazardous materials handling to provide important distribution and specialized services for our producers and our customers. We also have state-of-the-art Solution Centers at locations across the globe, consisting of formulation labs, development and research centers, and taste kitchens, with specialized industry expertise and innovative technical capabilities to help our customers address their technical challenges and accelerate product development cycles. At the heart of our business model are our technically trained professionals with deep industry experience.

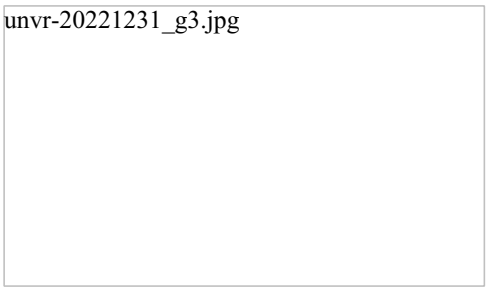
Chemicals and Services

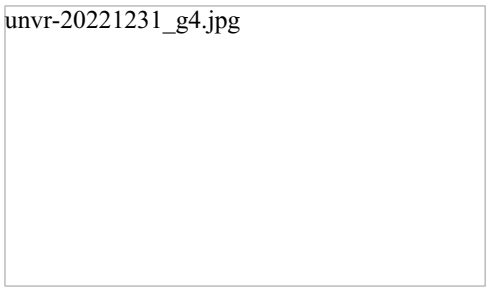
Chemicals and Services represent the largest portion of our business by sales, volume and gross profit (exclusive of depreciation). Our product portfolio principally includes a wide range of organic and inorganic chemistries, surfactants, inorganic compounds and general chemicals which are used extensively throughout most end markets. Investments in salesforce effectiveness systems are focused on a customer centric selling model, which coupled with our facility, transportation and digital asset bases, and our extensive supplier partnerships, all support growth within the chemicals distribution landscape.

Ingredients and Specialties

Ingredients and Specialties represent a more value added, higher-growth portion of the market tailored to specific end market requirements. For each Ingredients and Specialties market, we have dedicated sales, marketing, product management and technical professionals with deep industry knowledge of individual submarkets like food ingredients or pharmaceuticals, serving approximately 500 strategic specialty chemical suppliers and approximately 42,000 customer locations. These specialty products are typically sold in lower volumes, but at a higher gross profit (exclusive of depreciation) than commodity products. We believe growth can be achieved by leveraging salesforce effectiveness, delivering solutions through industry expertise, technical capabilities, formulation support and our comprehensive supply chain infrastructure.

The following charts reflect the breakdown of 2022 net sales and gross profit (exclusive of depreciation) for Chemicals and Services as compared to Ingredients and Specialties.





* See definition of gross profit (exclusive of depreciation) under "Non-GAAP Financial Measures" in Item 7 of this Annual Report on Form 10-K for more information.

Our collective product portfolio serves the following end markets: Industrial Solutions, Life Sciences, General Industrial, Services & Other Markets and Refining & Chemical Processing. A further description of these end markets, and the respective submarkets, is as follows:

Industrial Solutions

- *Coating, adhesives, sealants and elastomers ("CASE" as previously defined)*. We sell resins, pigments, solvents, thickeners, dispersants and other additives used to make paints, inks and coatings. Our product line includes epoxy resins, polyurethanes, titanium dioxide, fumed silica, esters, plasticizers, silicones and specialty amines.
- *Homecare & industrial cleaning.* We offer an extensive range of quality ingredients for cleaners, detergents and disinfectant products. We distribute chemicals manufactured by many of the industry's leading producers of enzymes, surfactants, solvents, dispersants, thickeners, bleaching aides, builders, sealants, acids, alkalis and other chemicals that are used as ingredients and processing aids in the manufacturing of cleaning and sanitation products.
- *Lubricants & metalworking.* Our broad and diverse range of products includes base stocks and performance-enhancing additives for both lubricants and metalworking fluids.

Life Sciences

- *Beauty & personal care.* We are a full-line distributor in the beauty and personal care industry, providing a wide variety of specialty and basic chemicals and ingredients used in skin and hair care products.
- *Food ingredients.* We distribute a diverse portfolio of commodity and specialty products that are sold into the food industry. The major food and beverage markets we serve are meat processing, baked goods, dairy, grain mill products, processed foods, carbonated soft drinks, fruit drinks and alcoholic beverages.
- *Pharmaceutical ingredients & finished products.* Our portfolio includes products along the medicinal production chain, where we offer a broad portfolio of excipients, solvents, reactants, active pharmaceutical ingredients and intermediates to pharmaceutical ingredient producers.

General Industrial

- *Chemical manufacturing.* We distribute a full suite of chemical products in support of the chemical manufacturing industry (organic, inorganic and polymer chemistries).
- *Industrial & municipal water treatment.* We provide the chemistries and products used to sanitize, balance and supplement municipal and industrial water.
- *Forestry, lumber & paper.* We serve the forest industry in the US and Canada, supplying a complete range of chemical products for use at all stages of production, from sap stain prevention to pulp and paper manufacturing.
- *Mining.* Within the mining industry in the US and Canada, we provide the chemistries necessary to leach ore body, mitigate dust, as well as balance and manage tailing ponds and mining water sources.

Services & Other Markets

- *Chemical waste removal.* Our ChemCare waste management business collects both hazardous and non-hazardous waste products at customer locations in the US and Canada, and then works with select vendors in the waste disposal business to safely transport these materials to licensed third party treatment, storage and disposal facilities.
- *Specialized Formulation and Blending*. Leveraging our technical expertise, we work to utilize our blending and mixing capabilities to create specialty chemical formulations to meet specific customer performance demands for products.

Refining & Chemical Processing

- *Midstream & downstream*. We provide chemicals to midstream pipeline and downstream refinery operators primarily in the US and Canada.
- *Upstream oil & gas.* We provide chemicals and services to offshore production, the Canadian Oil Sands and the shale hydraulic-fracturing sector by delivering various chemistries which aid in oil extraction and waste management.



Suppliers

We source materials from approximately 1,800 producers around the globe and we typically maintain relationships with multiple producers in order to protect against disruption in supply and distribution logistics, as well as to ensure competitive pricing of our supply. For the year ended December 31, 2022, our ten largest producers accounted for approximately 41% of our total chemical purchases.

Distribution Channels

We have multiple channels to market, including both warehouse delivery and direct-to-consumer delivery. The principal determinants of the way a customer is serviced include the size, scale and level of customization of a particular order, the nature of the product and the customer, and the location of the product inventories. Our logistics and supply chain expertise allows us to service a wide range of customers by offering multiple package sizes and assistance with unique special handling requirements, safety and quality specifications.

Warehouse distribution

Our warehouse distribution infrastructure is the core of our operations and connects large producers with smaller volume customers whose consumption patterns tend to make them uneconomical to be served directly by producers. As a result, the core customer serviced via our warehouses is a small or medium-volume consumer of chemicals and ingredients. We purchase chemicals and ingredients in truck load or larger quantities from producers based on contracted demands of our customers and our estimates of anticipated customer purchases. Once received, products are stored in one or more of our distribution facilities for sale and distribution in smaller, less-than-truckload quantities to our customers. Our warehouses have various facilities for services such as repackaging, blending and mixing to create specialized solutions needed by our customers in ready-to-use formulations.

Direct distribution

In direct distribution, we sell and service large quantity purchases that are shipped directly from producers through our logistics infrastructure, which provides our customers with sourcing and logistics support services for inventory management and delivery.

Competition

The chemical and ingredient distribution and service markets are highly competitive. Most of the products that we distribute are made to standard specifications and are either produced by or available from multiple sources. Furthermore, chemical and ingredient distribution itself is a fragmented market in which only a small number of competitors have substantial international operations. Our principal international competitors are Brenntag SE, IMCD NV, Barentz International NV and Azelis Group NV, each of which have differentiating strengths across regions and not all of whom have the ability to serve all of the end markets in which we operate. Many other chemical and ingredient distributors focus on a more limited geographic region or smaller subset of products, building strong relationships with local producers and customers that may give them a competitive advantage in their local market.

Chemical and ingredient producers may also sell their products through a direct sales force, digital marketplace or multiple chemical distributors, limiting their use of third party distributors, particularly with respect to higher margin products, or partnering with other chemical and ingredient producers for distribution, each of which could impact our competitive position.

We compete on the basis of service, on-time delivery, packaging type, product breadth and availability, product and market knowledge and insights, safety and environmental compliance, global reach, product price, as well as our ability to provide certain additional value-added services through our comprehensive suite of solution-based offerings.

Human Capital Resources

We recognize that our people represent our most important asset. As part of our effort to drive continued growth and increased profitability, we are intently focused on building a place that attracts and retains top talent and develops the skills necessary to drive differentiation for our business. The below data provides a snapshot of our employee and management demographics as of December 31, 2022:

Employees*	
USA	5,498
EMEA	2,193
Canada	757
LATAM	1,298
Total	9,746
Male	63 %
Female	36 %
Elected to not self-identify gender	<1%

People Managers*	1,753
Male	66 %
Female	34 %
Elected to not self-identify gender	<1%
Ethnically Diverse (US only)	21 %
Executive Officers*	
Male	75 %
Female	25 %
Labor force covered by a collective bargaining agreement	
USA	12 %
EMEA	46 %
Canada	24 %

(*) Information based on self-identification data.

We have developed key recruitment and retention strategies that guide our human capital management approach as part of the overall management of our business. These strategies are advanced through a number of programs and initiatives as outlined below:

Employee Total Rewards and Wellbeing

Our total rewards programs are designed to align the compensation of our employees with their performance and to provide appropriate incentives to attract, retain and motivate employees to achieve superior results. We provide employee wages that are competitive and consistent with employee positions, skill levels, experience, knowledge and geographic location. Full-time US employees are eligible for health, dental and vision insurance, paid and unpaid leaves, a 401(k) plan and life and disability/accident insurance coverage. Employees outside of the US are eligible for a variety of supplemental benefits based on local market practices and all employees worldwide are eligible for an employee assistance program and other well-being programs. We periodically benchmark our total rewards offerings externally to ensure we remain competitive.

Training and Development

All employees have access to on-demand learning resources for their own professional development on a range of diverse topics such as digital transformation skills, professional effectiveness, diversity, equity and inclusion, business skills, productivity, collaboration tools and leadership development. Our internal program offerings range from multi-day, instructor-

led trainings to short, on-demand eLearning programs. As appropriate, additional training hours are delivered and monitored locally, often focusing on material handling and safety for specific roles. The Company's guiding principles and cultural values are incorporated into our Code Handbook, to which all employees certify annually.

Employee Engagement

We regularly conduct confidential employee engagement surveys and focus groups. The feedback gathered from these surveys and focus groups helps guide our decision-making and strategic initiatives. In 2022, at least 85% of our global employee population participated in one or more of these feedback channels, sharing their sentiments on key aspects of working at the Company. The engagement survey results are shared in aggregate with managers, who are then tasked with taking action based on employee feedback. In doing so, our goal is to drive year-over-year improvement in employee engagement, which we believe will deliver qualitative and quantitative benefits for our Company.

Health and Safety

We are serious about safety, which is a key area of focus across our global operations. Specific initiatives include, among others, data driven and causality-based accident prevention work, improved process and facility controls, mandatory general education and role specific safety training, joint management-worker health and safety committees, safety audits, incident investigation and improvement measures. 2022 was our safest year on record with a Total Case Incident Rate (TCIR), the rate of recordable injuries per 200,000 hours worked, of 0.33, compared with 0.40 in 2021 and ahead of our 2025 target of 0.38.

Diversity, Equity & Inclusion

We are committed to fostering a safe, collaborative, supportive and respectful environment that values diverse perspectives, mitigates unconscious bias and enables a culture where employees are able to bring their authentic self to work. In the past year we maintained our score of 100 on the Human Rights Campaign Corporate Equality Index, increased the favorability rating of our inclusive culture amongst our salaried employee population, increased the number of female people managers globally and increased the number of ethnically diverse people managers in the USA. We use our global inclusion councils to help hold ourselves accountable for progress and drive decision-making for the unique needs of our diverse employee groups. Additionally, we seek to drive greater employee engagement, create education opportunities, and instill a greater sense of community through the efforts of our eight Employee Resource Networks.

As a July 2020 signatory of CEO Action for Diversity & Inclusion™ (CEO Action) coalition, we have pledged to take action designed to foster a safe, collaborative, supporting and respectful environment. We are committed to fostering equality and equity across our Company and have a zero-tolerance policy for discrimination and harassment. In accordance with this policy, we strive to recruit and hire employees based on qualifications and fit for a job and without regard to race, religion, nationality, gender, age, disability, sexual orientation or any other status protected by law. We also have systems under which employees can report incidents confidentially or anonymously without fear of reprisal.

Regulatory Matters

We operate in a number of jurisdictions and are subject to numerous international, federal, state and local laws and regulations covering a wide variety of subject matters. We are subject to extensive environmental, health and safety laws and regulations in multiple jurisdictions because we blend, manage, handle, store, sell, transport and arrange for the disposal of chemicals, hazardous materials and hazardous waste. These include, without limitation, laws regulating discharges of hazardous substances into the soil, air and water, blending, managing, handling, storing, selling, transporting and disposing of hazardous substances, investigation and remediation of contaminated properties and protecting the safety of our employees and others. Some of these laws and regulations include the Comprehensive Environmental Response, Compensation and Liability Act ("CERCLA" or "Superfund"), the Toxic Substances Control Act ("TSCA"), the Resource Conservation and Recovery Act ("RCRA") and Registration, Evaluation, Authorization and Restriction of Chemicals ("REACH"), among others. Some of our operations are required to hold environmental permits and licenses to be compliant and certain of our services businesses are also impacted by these laws.

In addition to environmental laws and regulations, we are subject to various other laws and regulations around the world. For example, our business is subject to competition laws in the various jurisdictions where we operate, including the Sherman Antitrust Act and related federal and state antitrust laws in the US, as well as similar foreign laws and regulations. These laws and regulations generally prohibit competitors from fixing prices, boycotting competitors or engaging in other conduct that unreasonably restrains competition, and such laws and regulations may impact potential business relationships or transactions with third parties in the future. We are also required to comply with increasingly complex and changing laws and regulations enacted to protect business and personal data in Europe, the US and other jurisdictions regarding privacy, data protection and data security, including those related to the collection, storage, use, transmission and protection of personal information and other consumer, customer, vendor or employee data. Further, an increasing number of laws and regulations focused on hazardous products and substances could also impact our ability to distribute and sell certain products or require significant capital expenditures to meet regulatory requirements. With respect to the laws and regulations noted above, as well as other

applicable laws and regulations, our compliance programs may under certain circumstances involve material investments in the form of additional processes, training, personnel, information technology and capital.

Information related to government regulation applicable to our business is included in this Annual Report on Form 10-K, including: (i) Part I, Item 1A, "Risk Factors"; (ii) Part II, Item 7, “Management’s Discussion and Analysis of Financial Condition and Results of Operations” and (iii) “Note 2: Significant accounting policies” and “Note 18: Commitments and contingencies” in Item 8 of this Annual Report on Form 10-K.

Sustainability

We expect that there will be a continued increase in demand for products, systems and services that meet growing customer and societal sustainability standards, expectations and preferences. We recognize that our ability to continue to provide these products and services requires our business to further advance environmentally and socially responsible means of operating, reflecting the challenges and opportunities presented through increased legal requirements, climate parameters and market developments.

We are focused on demonstrating our commitment to these expectations through our global sustainability goals, which help us focus on reducing our contribution to global climate change while addressing remediation, reducing resource use, progressing responsible sourcing and bringing more sustainable solutions to market. In 2018, we became a signatory to the United Nations Global Compact, reaffirming our ongoing commitments to responsible business, and in 2019 adopted ‘Advancing a Circular Economy’ as a goal to 2021. In 2021, we developed and shared longer term Environmental, Social and Governance (“ESG”) goals to 2025, 2030 and 2050. Throughout 2022 we continued to implement our strategies to drive progress on these goals.

We incorporate ESG initiatives into our business values and priorities of safety, sustainability and value creation. We continually strive to improve our industry leading safety record, reduce our environmental impact and increase transparency. In 2022, we demonstrated our ongoing commitment to ESG practices and our purpose to help keep our communities healthy, fed, clean and safe with net sales of 8% related to food ingredients, 7% related to homecare and industrial cleaning, 5% related to pharmaceutical ingredients and finished products, 4% related to industrial and municipal water treatment and 2% related to chemical waste removal. In 2022, we dedicated resources to supporting our customers on their sustainability journeys, enhanced our sustainable and natural products portfolio and focused on developing sustainable formulations and services.

Intellectual Property

We consider intellectual property, particularly trade secrets and know-how, as important to our success. We hold some patents and own numerous trademarks in multiple jurisdictions. Further, we have various patent and trademark applications pending in jurisdictions worldwide. Although we consider our patents, trademarks, copyrights and trade secrets to constitute valuable assets, we do not regard any of our businesses as being materially dependent upon an individual patent, trademark, copyright or trade secret.

Significant Customers

No single customer accounted for 10% or more of net sales in any of the years presented.

Other

No material part of our business is subject to renegotiation of profits or termination of contracts or subcontracts at the election of any government.

Information about our Executive Officers

See Part III, Item 10, "Directors, Executive Officers and Corporate Governance".

Available Information

We maintain a website at www.univarsolutions.com and make available free of charge at this website our Annual Report on Form 10-K, quarterly reports on Form 10-Q, current reports on Form 8-K and all amendments to those reports as soon as reasonably practicable after they are electronically filed with or furnished to the SEC. Our Corporate Governance Guidelines, our Code Handbook, our Officers Code of Ethics and the charters of the Audit, Compensation and Governance & Corporate Responsibility Committees of our Board of Directors also are available on our website, where they can be printed free of charge. The information on our website is not, and will not be deemed to be, a part of this Annual Report on Form 10-K, or incorporated into any of our other filings with the SEC, except where we expressly incorporated such information. If you wish to receive a paper copy of any exhibit to our reports filed with or furnished to the SEC, the exhibit may be obtained by writing to: Corporate Secretary, Univar Solutions Inc., 3075 Highland Parkway Suite 200, Downers Grove, Illinois 60515.

Dissemination of Company Information

We intend to make future announcements regarding Company developments and financial performance through our website, www.univarsolutions.com, as well as through press releases, filings with the Securities and Exchange Commission, conference calls and webcasts.

ITEM 1A. RISK FACTORS

The following are certain risk factors that could affect our business, financial condition and results of operations. These risk factors should be considered along with the forward-looking statements contained in this Annual Report on Form 10-K because these factors could cause our actual results or financial condition to differ materially from those projected in forward-looking statements. Although the risks are organized by headings, and each risk is discussed separately, many are interrelated. The following discussion is not an all-inclusive listing of risks, although we believe these are the more material risks that we face. If any of the following occur, our business, financial condition and results of operations could be adversely affected.

Business and Economic Risks

We are affected by general economic conditions, particularly fluctuations in industrial production and consumption, and an economic downturn could adversely affect our business, financial condition and results of operations.

We sell chemicals that are used in manufacturing processes and as components of or ingredients in other products. Our sales are correlated with and affected by fluctuations in the levels of industrial production, manufacturing output and general economic activity. Producers of commodity and specialty chemicals are likely to reduce their output in periods of significant contraction in industrial and consumer demand, while demand for the products we distribute depends largely on trends in demand in the end markets our customers serve. A majority of our sales are in North America and Europe and our business is therefore susceptible to downturns in those economies, as well as, to a lesser extent, the economies in the rest of the world.

Our profit margins, as well as overall demand for our products and services, could decline as a result of a large number of factors outside our control, including the impact of pandemics and other public-health emergencies, economic recessions, reduced customer demand (whether due to changes in production processes, consumer preferences, the industries in which the customer operates, laws and regulations affecting the chemicals industry and the manner in which they are enforced, or other factors), inflation, fluctuations in interest and currency exchange rates, and changes in the fiscal or monetary policies of governments in the regions in which we operate. For example, the COVID-19 pandemic caused growth in certain of our end markets, such as in homecare and industrial cleaning and pharmaceuticals, while other end markets, such as the upstream refining market, were materially and adversely affected.

General economic conditions and macroeconomic trends, as well as the creditworthiness of our customers, could affect overall demand for chemicals. Any overall decline in the demand for chemicals could significantly reduce our sales and profitability. If the creditworthiness of our customers declines, we would face increased credit risk. In addition, volatility and disruption in financial markets could adversely affect our sales and results of operations by limiting our customers' ability to obtain financing necessary to maintain or expand their own operations.

A historical feature of past economic weakness has been significant destocking of inventories, including inventories of chemicals used in industrial and manufacturing processes. It is possible that an improvement in our net sales in a particular period may be attributable in part to restocking of inventories by our customers and represent a level of sales or sales growth that will not be sustainable over the longer term. Further economic weakness could lead to insolvencies among our customers or producers, as well as among financial institutions that are counterparties on financial instruments or accounts that we hold. Any of these developments could have a material adverse effect on our business, financial condition and results of operations.

Significant changes in the business strategies of producers or in the operations of our customers could adversely affect our business, financial condition and results of operations.

Significant changes in the business strategies of producers could disrupt our supply. Large chemical manufacturers may elect to sell certain products (or products in certain regions) directly to customers or utilize digital marketplaces, bypassing distributors such as us. While we do not believe that our results depend materially on access to any individual producer's products, a reversal of the trend toward more active use of distributors would likely result in increasing margin pressure or products becoming unavailable to us.

In addition, unpredictable events may have a significant impact on the industries in which many of our customers operate, reducing demand for products that we normally distribute in significant volumes. Significant disruptions of supply and disruptions in customer industries could have a material adverse effect on our business, financial condition and results of operations.

The markets in which we operate are highly competitive and we may not be able to compete successfully.

The chemical and ingredient distribution and services market is highly competitive. Chemicals can be purchased from a variety of sources, including traders, brokers, wholesalers and other distributors, as well as directly from producers. Many of the products we distribute or finish are essentially fungible with products offered by our competitors, including emerging competitors. The competitive pressure we face is particularly strong in sectors and markets where local competitors have strong positions or where new competitors can easily enter. Increased competition from distributors of products similar to or competitive with ours could result in price reductions, reduced margins and a loss of market share.

We expect to continue to experience significant and increasing levels of competition in the future. We must also compete with smaller companies that have been able to develop strong local or regional customer bases. In certain countries, some of our competitors are more established, benefit from greater name recognition and have greater resources within those countries than we do.

Consolidation of our competitors in the markets in which we operate could place us at a competitive disadvantage and reduce our profitability.

We operate in an industry that is highly fragmented on a global scale, but in which there has been a trend toward consolidation in recent years. Consolidation of our competitors may also further enhance their financial position, provide them with the ability to offer more competitive prices to customers for whom we compete and allow them to achieve increased efficiencies in their consolidated operations that enable them to more effectively compete for customers. This may jeopardize the strength of our positions in one or more of the markets in which we operate and any advantages we currently enjoy due to the comparative scale of our operations. Losing some of those advantages could adversely affect our business, financial condition and results of operations, as well as our growth potential.

Disruption of our supply chain due to various acute or severe causes could have an adverse effect on our business, financial condition and results of operations.

In coordination with our suppliers, our ability to move and sell products is critical to our success. Damage or disruption to our collective supply or distribution capabilities resulting directly or indirectly from pandemics and other public-health emergencies, labor shortages, border closures, natural disasters, weather and physical climate change related events, lack of transportation capacity, increased fuel expenses, global trade flow disruption, increased airport and shipping port congestion, climate change, plant downtime (whether our own or others'), power outages, explosions, information technology system and/or network disruptions, terrorism, strikes or other labor unrest, or other reasons could seriously harm our operations, as well as the operations of our customers and suppliers. Failure to take adequate steps to reduce the likelihood or mitigate the potential impact of any of these events, or to effectively manage such events if they occur, could have a negative impact on our business, results of operations, financial condition and cash flows.

Some of the foregoing examples are hypothetical, but others have been or are being experienced by the Company. For example, during 2020 and 2021, the COVID-19 pandemic, weather events and supply constraints impacted, and we expect will continue to impact, our operations. The combination of supplier shut-downs, port congestion, acute pandemic recovery demand and the Russia-Ukraine conflict have resulted in sustained supply chain constraints, product shortages and chemical price inflation. In some instances, we have placed certain of our products on allocation due to limited supplies.

The prices and costs of the products we purchase may be subject to large and significant price increases. We might not be able to pass such cost increases through to our customers. We could experience financial losses if our inventories of one or more chemicals exceed our sales and the price of those chemicals decreases significantly while in our inventories or if our inventories fall short of our sales and the purchase price of those chemicals increases significantly.

We purchase and sell a wide variety of chemicals, the price and availability of which may fluctuate, and may be subject to large and significant price increases. For example, ongoing shortages across a range of chemicals and ingredients have generally led to fluctuations in chemical prices globally. Many of our contracts with producers include chemical prices that are not fixed or are tied to an index, which allows our producers to change the prices of the chemicals we purchase as the price of the chemicals fluctuates in the market. Changes in chemical prices affect our net sales and cost of goods sold, as well as our working capital requirements, levels of debt and financing costs. We might not always be able to reflect increases in our chemical costs, transportation costs and other costs in our own pricing. Any inability to pass cost increases onto customers may adversely affect our business, financial condition and results of operations.

In order to meet customer demand, we typically maintain significant inventories, and we are therefore subject to a number of risks associated with our inventory levels, including the following:

- declines in the prices of chemicals that are held by us;
- the need to maintain a significant inventory of chemicals that may be in limited supply and therefore difficult to procure;

- buying chemicals in bulk for the best pricing and thereby holding excess inventory;
- responding to the fluctuating demand for chemicals;
- cancellation of customer orders; and
- responding to customer requests for rapid delivery.

In order to manage our inventories successfully, we must estimate demand from our customers and purchase chemicals that substantially correspond to that demand. If we overestimate demand and purchase too much of a particular chemical, we face a risk that the price of that chemical will fall, leaving us with inventory that we cannot sell profitably or have to write down such inventory from its recorded value. If we underestimate demand and purchase insufficient quantities of a particular chemical and prices of that chemical rise, we could be forced to purchase that chemical at a higher price and forego profitability in order to meet customer demand. Our business, financial condition and results of operations could suffer a material adverse effect if either or both of these situations occur frequently or in large volumes.

We require significant working capital, and we expect our working capital needs to increase in the future, which could result in having lower cash available for, among other things, capital expenditures and acquisition financing.

We require significant working capital to purchase chemicals from chemical producers and distributors and sell those chemicals efficiently and profitably to our customers. Our working capital needs may increase if the price of products we purchase and inventory increase. Our working capital needs also increase at certain times of the year, as our customers' requirements for chemicals increase. We need inventory on hand to have product available to ensure timely delivery to our customers. If our working capital requirements increase and we are unable to finance our working capital on terms and conditions acceptable to us, we may not be able to obtain chemicals to respond to customer demand, which could result in a loss of sales.

In addition, the amount of working capital we require to run our business is expected to increase in the future due to expansions in our business activities. Disruptions in our supply chain could also create the need to invest additional working capital in on-hand inventory than in prior years. If our working capital needs increase, the amount of free cash we have at our disposal to devote to other uses will decrease. A decrease in free cash could, among other things, limit our flexibility, including our ability to make capital expenditures and to acquire suitable acquisition targets that we have identified. If increases in our working capital occur and have the effect of decreasing our free cash, it could have a material adverse effect on our business, financial condition and results of operations.

We depend on transportation assets, some of which we do not own, in order to deliver products to our customers.

Although we maintain a significant portfolio of owned and leased transportation assets, including trucks, trailers and rail cars, we also rely on transportation and warehousing provided by third parties (including common carriers and rail companies) to deliver products to our customers. Our access to third party transportation is not guaranteed, and we may be unable to transport chemicals at economically attractive rates in certain circumstances, particularly in cases of adverse market conditions or disruptions to transportation infrastructure. In addition, we could face a challenge with attracting and retaining qualified drivers primarily due to intense market competition, which may subject us to increased payments for driver compensation. If we are unable to continue to attract and retain a sufficient number of drivers, we could face difficulty meeting customer orders or be forced to forego business that would otherwise be available to us, which could adversely affect our profitability and ability to maintain or grow our business. We are also subject to increased costs that we may not always be able to recover from our customers, including fuel prices, as well as charges imposed by common carriers, leasing companies and other third parties involved in transportation.

Accidents, safety failures, environmental damage, product quality issues, delivery failures or hazards and risks related to our operations and the hazardous materials we blend, manage, handle, store, sell, transport or dispose of could damage our reputation and result in substantial damages or remedial obligations.

Our business depends to a significant extent on our customers' and producers' trust in our reputation for reliability, quality, safety and environmental responsibility. Actual or alleged instances of safety deficiencies, mistaken or incorrect deliveries, inferior product quality, exposure to hazardous materials resulting in illness, injury or other harm to persons, property or natural resources, or of damage caused by us or our products, could damage our reputation and lead to customers and producers curtailing the volume of business they do with us. Also, there may be safety, personal injury or other environmental risks related to our products which are not known today. Any of these events, outcomes or allegations could also subject us to substantial legal claims, and we could incur substantial expenses, including legal fees and other costs, in defending such legal claims, which could materially impact our financial position and results of operations.

Actual or alleged accidents or other incidents at our facilities or that otherwise involve our personnel or operations could also subject us to claims for damages by third parties. Because many of the chemicals that we handle are dangerous, we are subject to the ongoing risk of hazards, including leaks, spills, releases, explosions and fires, which may cause property damage, illness, physical injury or death. We sell products used in hydraulic fracturing, a process that involves injecting water, sand and chemicals into subsurface rock formations to release and capture oil and natural gas. The use of such hydraulic fracturing fluids

by our customers may result in releases that could impact the environment and third parties. Several of our distribution facilities are located near, and our transportation routes could take us through, high-density population centers. If any such events occur, whether through our own fault, through preexisting conditions at our facilities, through the fault of a third party or through a natural disaster, terrorist incident or other event outside our control, our reputation could be damaged significantly. We could also become responsible, as a result of environmental or other laws or by court order, for substantial monetary damages or expensive investigative or remedial obligations related to such events, including but not limited to those resulting from third party lawsuits or environmental investigation and cleanup obligations on and off-site. The amount of any costs, including fines, damages and/or investigative and remedial obligations, that we may become obligated to pay under such circumstances could substantially exceed any insurance we have to cover such losses.

Any of these risks, if they materialize, could have a material adverse effect on our business, financial condition and results of operations.

Our business exposes us to significant risks, not all of which are covered by insurance.

Because we are engaged in the blending, managing, handling, storing, selling, transporting and disposing of chemicals, chemical waste products and other hazardous materials, product liability, health impacts, fire damage, safety, cyber security and environmental risks are significant concerns for us. We are also exposed to present and future chemical exposure claims by employees, contractors on our premises and other persons located nearby, as well as related workers' compensation claims. Although we carry insurance to protect us against many risks involved in the conduct of our business, we do not insure against all such risks and the insurance we carry is subject to limitations, including exclusions, deductibles and coverage limits. Due to the variable condition of the insurance market, we have experienced and may experience in the future, increased deductible retention levels and increased premiums. We also may be unable to obtain at commercially reasonable rates in the future adequate insurance coverage for the risks we currently insure against, and certain risks are or could become completely uninsurable or eligible for coverage only to a reduced extent. Increased insurance premiums or the occurrence of significant uncovered losses could have a material adverse effect on our business, financial condition and results of operations.

Changes in tax laws may materially adversely affect our business, prospects, financial condition and operating results.

New income, sales, use, product or other tax laws, statutes, rules, regulations or ordinances could be enacted at any time, which could adversely affect our business, financial condition and results of operations. Further, existing tax laws, statutes, rules, regulations or ordinances could be interpreted, changed, modified or applied adversely to us. Recently enacted changes to the US federal tax regime could impact our tax liability and cash tax payments. Most notably, in July 2022 excise taxes were imposed on a number of commodity chemical products impacting our overall costs.

Our businesses are subject to income taxation in the US as well as internationally. The Company’s operations are subject to multiple, and sometimes conflicting tax laws and regulations. Recently, many countries have adopted, or are considering revisions to their existing tax laws based on recommendations issued by the Organization for Economic Co-operation and Development/G20 Base Erosion and Profit Shifting Project. These changes could materially impact our tax liability because of our organizational structure and significant international operations.

We have in the past and may in the future make acquisitions, ventures and strategic investments, some of which may be significant in size and scope, which have involved in the past and will likely involve in the future numerous risks. We may not be able to address these risks without substantial expense, delay or other operational or financial problems.

Acquisitions or investments have involved in the past and will likely involve in the future various risks, such as:

- integrating the technologies, operations and personnel of any acquired business;
- the potential disruption of our ongoing business, including the diversion of management attention;
- the possible inability to obtain the desired financial and strategic benefits from the acquisition or investment;
- customer attrition arising from preferences to maintain redundant sources of supply;
- producer attrition arising from overlapping or competitive products;
- assumption of contingent or unanticipated liabilities or regulatory liabilities;
- dependence on the retention and performance of existing management and work force of acquired businesses for the future performance of these businesses;
- regulatory risks associated with acquired businesses (including the risk that we may be required for regulatory reasons to dispose of a portion of our existing or acquired businesses); and
- the risks inherent in entering geographic or product markets in which we have limited prior experience.

Future acquisitions and investments may need to be financed in part through additional financing from banks, through public offerings or private placements of debt or equity securities or through other arrangements, and could result in substantial cash expenditures. The necessary acquisition financing may not be available to us on acceptable terms if and when required, particularly if our debt leverage levels make it difficult or impossible for us to secure additional financing for acquisitions.

Our balance sheet includes significant goodwill and intangible assets, the impairment of which could affect our future financial condition and results of operations

We carry significant goodwill and intangible assets on our balance sheet. As of December 31, 2022, our goodwill and intangible assets totaled $2,288.2 million and $167.0 million, respectively. At least annually, the Company assesses goodwill for impairment. If testing indicates that goodwill is impaired, the carrying value is written down based on fair value with a charge against earnings. Weakened demand for a specific product line or business could result in an impairment. Intangible assets are amortized for book purposes over their respective useful lives and are tested for impairment if any event occurs or circumstances change that indicates that carrying value may not be recoverable. Accordingly, any determination requiring the write-off of a significant portion of goodwill or intangible assets could negatively impact the Company's financial condition and results of operations. See “Note 2: Significant accounting policies” in Item 8 of this Annual Report on Form 10-K for a discussion of our 2022 impairment review.

Risks Related to Technology

Our business could be seriously impacted by cybersecurity incidents, including security breaches.

Cyber-attacks or security breaches could compromise confidential, business critical information, private information including without limitation the personally identifiable information ("PII") of our employees or business partners, cause a disruption in the Company’s operations or harm the Company’s reputation. During the normal course of business, we have experienced and expect to continue to experience attempts to compromise our information technology and control systems, network infrastructure and other assets. To date, no cybersecurity incident or attack has had a material impact on our business or results of operations. Additionally, the increase in remote working as a result of the COVID-19 pandemic has resulted in increased cyber-security and fraud risks. There can be no assurance that the Company's cyber-security programs, procedures, controls and intelligence will be sufficient to prevent security breaches from occurring. Moreover such programs are costly to maintain and it is expected that such costs will increase over time. If any security breaches were to occur, they could lead to losses of sensitive information (including without limitation PII), critical infrastructure or capabilities essential to our operations and could have a material adverse effect on our reputation, financial position, results of operations or cash flows, and could result in claims being brought against us.

Risks Related to Our Indebtedness

Our level of indebtedness may adversely affect our business, financial condition and results of operations.

As of December 31, 2022, we had $2,465.8 million of total debt. Our level of indebtedness, as well as any additional debt or other obligations that we may incur in the future, may have material adverse effects on our business, financial condition and results of operations, and could have important consequences for holders of our common stock. Examples of potential issues associated with our level of indebtedness that could materially and adversely affect our business or financial condition include:

- our ability to satisfy obligations to lenders or noteholders may become impaired by downturns in our business or other factors, resulting in possible defaults on and acceleration of our indebtedness;
- additional financing for refinancing of existing indebtedness, working capital, capital expenditures, product and service development, acquisitions, general corporate purposes and other purposes may not be available when required on acceptable terms or at all;
- our assets that currently serve as collateral for our debt may be insufficient, or may not be available, to support future financings;
- a substantial portion of our cash flow from operations may need to be used to repay the principal and interest on our debt;
- we may be increasingly vulnerable to economic downturns and increases in interest rates;
- our flexibility in planning for and reacting to changes in our business and the markets in which we operate may be limited; and
- we may be placed at a competitive disadvantage relative to other companies in our industry with less debt or comparable debt with more favorable terms.

The agreements governing our indebtedness contain operating covenants and restrictions that limit our operations and could lead to adverse consequences if we fail to comply with them.

The agreements governing our indebtedness contain certain operating covenants and other restrictions relating to, among other things, limitations on indebtedness (including guarantees of additional indebtedness) and liens, mergers, consolidations and dissolutions, sales of assets, investments and acquisitions, dividends and other restricted payments, repurchase of shares of capital stock and options to purchase shares of capital stock and certain transactions with affiliates. In addition, our Senior ABL Facility includes a springing financial covenant as set forth in the governing agreement.

Failure to comply with these financial and operating covenants could result from, among other things, changes in our results of operations, the incurrence of additional indebtedness, the pricing of our products, our success at implementing cost reduction initiatives, our ability to successfully implement our overall business strategy or changes in general economic conditions, many of which may be beyond our control. The breach of any of these financial and operating covenants could result in a default under the agreements that govern these facilities that would permit the lenders to declare all amounts outstanding thereunder to be due and payable, together with accrued and unpaid interest. If we are unable to repay such amounts, lenders having secured obligations could proceed against the collateral securing these obligations. This could have serious consequences on our business, financial condition and results of operations and could require us to engage in restructuring activities. In addition, these covenants may restrict our ability to engage in transactions that we believe would otherwise be in the best interests of our business and stockholders. We may also incur future debt obligations that might subject us to additional restrictive covenants that could affect our financial and operational flexibility.

Increases in interest rates would increase the cost of servicing our debt and could reduce our profitability.

Certain of our outstanding debt bears interest at variable rates. As a result, increases in interest rates would increase the cost of servicing our debt and could materially reduce our profitability and cash flows. As of December 31, 2022, approximately 78% of our debt is indexed to the London Inter-Bank Offered Rate ("LIBOR") or the Secured Overnight Financing Rate ("SOFR") as benchmarks for establishing rates and we hold other operational contracts, including leases, that are also indexed to LIBOR or SOFR.

The potential consequences to our current or future debt obligations and hedging instruments from the reform of LIBOR as it converts and is replaced with the SOFR implementation of alternative reference rates, and any interest rate transition process may adversely affect the cost of servicing our debt. For additional information on our indebtedness, debt service obligations and sensitivity to interest rate fluctuations, see “Quantitative and Qualitative Disclosures About Market Risk” in Item 7A of this Annual Report on Form 10-K.

We may have future capital needs and may not be able to obtain additional financing on acceptable terms, or at all.

We have historically relied on debt financing to fund our operations, capital expenditures and expansion. The macroeconomic conditions that affect the markets in which we operate, and our credit ratings, could have a material adverse effect on our ability to secure financing on acceptable terms, if at all. The terms of additional financing may limit our financial and operating flexibility, and if financing is not available when needed, or is not available on acceptable terms, we may be unable to take advantage of business opportunities or respond to competitive pressures, any of which could have a material adverse effect on our business, financial condition and results of operations.

If we raise additional funds through further issuances of equity, convertible debt securities or other securities convertible into equity, our existing stockholders could suffer significant dilution in their percentage ownership of our company, and any new securities we issue could have rights, preferences and privileges senior to those of holders of our common stock.

Litigation, Environmental and Regulatory Risk

As a result of our current and past operations, we are subject to extensive environmental, health and safety laws and regulations, which expose us to risks that could have a material adverse effect on our business, financial condition and results of operations.

We are subject to extensive environmental, health and safety laws and regulations in multiple jurisdictions because we blend, manage, handle, store, sell, transport and arrange for the disposal of chemicals, hazardous materials and hazardous waste. These include laws and regulations governing our management, storage, transportation and disposal of chemicals; product regulation; air, water and soil contamination; activities related to climate change; and the investigation and cleanup of contaminated sites, including any spills or releases that may result from our management, handling, storage, sale, or transportation of chemicals and other products. Compliance with these laws and regulations, and with the permits and licenses we hold, requires that we expend significant amounts for ongoing compliance, investigation and remediation. If we fail to comply with such laws, regulations, permits or licenses, we may be subject to fines, damages and other civil, administrative or criminal sanctions and investigations, including the revocation of permits and licenses necessary to continue our business activities. In addition, future changes in laws and regulations, or the interpretation of existing laws and regulations, could have an adverse effect on us by adding restrictions, reducing our ability to do business, increasing our costs of doing business, reducing our profitability or reducing the demand for our products.

Previous operations, including those of acquired companies, have resulted in contamination at a number of current and former sites, which must be investigated and remediated. We have ongoing investigations and remediation activities, or are contributing to cleanup costs, at approximately 128 currently or formerly owned, operated or used sites or other sites impacted by our operations. We have spent substantial sums on such investigation and remediation and we expect to continue to incur such expenditures in the future. We may incur losses in connection with investigation and remediation obligations that exceed our environmental liabilities. There is no guarantee that our estimates will be accurate, that new contamination will not be

discovered or that new environmental laws or regulations will not require us to incur additional costs. Any such inaccuracies, discoveries or new laws or regulations, or the interpretation of existing laws and regulations, could have a material adverse effect on our business, financial condition and results of operations.

We could be held liable for the costs to investigate, remediate or otherwise address contamination at any real property we have ever owned, leased, operated or used or other sites impacted by our operations. Some environmental laws could impose on us the entire cost of cleanup of contamination present at a site even though we did not cause all of the contamination. These laws often identify parties who can be strictly and jointly and severally liable for remediation. The discovery of previously unknown contamination at current or former sites or the imposition of other environmental liabilities or obligations in the future, including additional investigation or remediation obligations with respect to contamination that has impacted other properties, could lead to additional costs or the need for additional accruals that have a material adverse effect on our business, financial condition and results of operations. In addition, we may be required to pay damages or civil judgments related to third party claims, including those relating to personal injury (including exposure to hazardous materials or chemicals we blend, handle, store, sell, transport or dispose of), product quality issues, property damage or contribution to remedial obligations. We have been identified as a potentially responsible party at certain third party sites at which we have arranged for the disposal of our hazardous wastes. We may be identified as a potentially responsible party at additional sites beyond those for which we currently have financial obligations. Such developments could have a material adverse effect on our business, financial condition and results of operations.

Societal concerns regarding the safety of chemicals in commerce and their potential impact on the environment have resulted in a growing trend towards increasing levels of product safety and environmental protection regulations. These concerns could influence public perceptions, impact the commercial viability of the products we sell and increase the costs to comply with increasingly complex regulations, which could have a negative impact on our business, financial condition and results of operations. Additional findings by government agencies that chemicals pose significant environmental, health or safety risks may lead to their prohibition in some or all of the jurisdictions in which we operate.

Our business exposes us to potential product liability claims and recalls, which could adversely affect our business, financial condition and results of operations.

The repackaging, blending, mixing, manufacturing, sale and distribution of chemical products by us, including products used for food, pharmaceutical and nutritional supplement applications, involve an inherent risk of exposure to product liability claims, product recalls, product seizures and related adverse publicity and reputational harm. A product liability claim, judgment or recall against our customers could also result in substantial and unexpected expenditures for us, affect confidence in our products or services and divert management's attention from other responsibilities. Although we maintain product liability insurance, there can be no assurance that the type or level of coverage is adequate or that we will be able to continue to maintain our existing insurance or obtain comparable insurance at a reasonable cost, if at all. A product recall or a partially or completely uninsured judgment against us could have a material adverse effect on our business, financial condition and results of operations.

Many of the products we sell have "long-tail" exposures, giving rise to liabilities many years after their sale and use. Insurance purchased at the time of sale may not be available when costs arise in the future and producers may no longer be available to provide indemnification.

International Market Risk

Our results of operations could suffer if we are unable to expand into new geographic markets or manage the various risks related to our international activities.

Our profitability and longer-term success may be adversely affected if we fail to continue to expand our penetration in certain foreign markets and to enter new and emerging foreign markets. The profitability of our international operations will largely depend on our continued success in the following areas:

- securing key producer relationships to help establish our presence in international markets;
- hiring and training personnel capable of supporting producers and our customers and managing operations in foreign countries;
- localizing our business processes to meet the specific needs and preferences of foreign producers and customers;
- building our reputation and awareness of our services among foreign producers and customers; and
- implementing new financial, management information and operational systems, procedures and controls to monitor our operations in new markets effectively, without causing undue disruptions to our operations and customer and producer relationships.

In addition, we are subject to risks associated with operating in foreign countries, including:

- varying and often unclear legal and regulatory requirements that may be subject to inconsistent or disparate enforcement, particularly regarding environmental, health and safety issues and security or other certification

requirements, as well as other laws and business practices that favor local competitors, such as exposure to possible expropriation, nationalization, restrictions on investments by foreign companies or other governmental actions;
- less stable supply sources;
- regional conflicts that may result in supply chain disruptions;
- competition from existing market participants that may have a longer history in and greater familiarity with the foreign markets where we operate;
- tariffs, export duties, quotas and other barriers to trade; as well as possible limitations on the conversion of foreign currencies into US dollars or remittance of dividends and other payments by our foreign subsidiaries;
- divergent labor regulations and cultural expectations regarding employment and agency;
- different cultural expectations regarding industrialization, international business and business relationships;
- foreign taxes and related regulations, including foreign taxes that we may not be able to offset against taxes imposed upon us in the US, and foreign tax and other laws limiting our ability to repatriate earnings to the US;
- extended payment terms and challenges in our ability to collect accounts receivable;
- changes in a specific country's or region's political or economic conditions;
- compliance with anti-bribery laws such as the US Foreign Corrupt Practices Act, the UK Bribery Act and similar anti-bribery laws in other jurisdictions, the violation of which could expose us to severe criminal or civil sanctions; and
- compliance with anti-boycott, privacy, economic sanctions, anti-dumping, antitrust, import and export laws and regulations by our employees or intermediaries acting on our behalf, the violation of which could expose us to significant fines, penalties or other sanctions.

Fluctuations in currency exchange rates may adversely affect our results of operations.

We have sizable sales and operations in Canada, Europe, Middle East, Africa, Asia and Latin America. We report our consolidated results in US dollars and the results of operations and the financial position of our local operations are generally reported in the relevant local currencies and then translated into US dollars at the applicable exchange rates. As a result, our financial performance is impacted by currency fluctuations. For additional details on our currency exposure and risk management practices, see "Quantitative and Qualitative Disclosures About Market Risk" in Item 7A of this Annual Report on Form 10-K.

Employee and Benefit Plan Risk

We may be unable to attract or retain a qualified and diverse workforce.

We depend upon the ability and experience of a number of our executive management and other key personnel who have substantial experience with our operations, the chemicals and chemical distribution industries and the selected markets in which we operate. The loss of the services of one or a combination of our senior executives or key employees could have a material adverse effect on our results of operations. We also might suffer an additional impact on our business if one of our senior executives or key employees is hired by a competitor.

In addition, our ability to achieve our operating goals depends upon our ability to recruit, hire, retain and develop qualified and diverse personnel to operate and expand our businesses. We compete with other companies both within and outside of our industry for talented personnel. If we fail to hire, retain and develop a sufficient number of qualified and diverse employees to operate and expand our businesses, our businesses, financial condition, results of operations and cash flows could be harmed. The COVID-19 pandemic and its aftermath contributed to fewer workers participating in the workforce than in February 2020. We expect we will continue to experience labor shortages, which has negatively impacted, and we expect will continue to negatively impact, our businesses, financial condition, results of operations and cash flows.

Negative developments affecting our pension and multi-employer pension plans in which we participate may occur.

We operate a number of pension plans for our employees and have obligations with respect to several multi-employer pension plans sponsored by labor unions in the US. The terms of these plans vary from country to country. The recognition of costs and liabilities associated with the Company's pension plans are affected by assumptions made by management and used by actuaries engaged by us to calculate the benefit obligations and the expenses recognized for these plans. The inputs used in developing the required estimates are calculated using a number of assumptions, which represent management's best estimate of the future. The assumptions that have the most significant impact on costs and liabilities are the discount rate, the expected long-term return on plan assets for the funded plans, retirement rates and mortality rates. Changes to the funded status of our pension plans as a result of updates to actuarial assumptions and actual experience that differs from our estimates are recognized as gains or losses in the period incurred under our "mark to market" accounting policy, and could result in a requirement for additional funding.

As of December 31, 2022, our pension plans were underfunded by $82.7 million. Rising interest rates have positively impacted the funded status of our pension plans. When interest rates decline, funding requirements for our pension plans may become more significant. If our cash flows and capital resources are insufficient to fund our obligations under these pension plans, we

could be forced to reduce or delay investments and capital expenditures, seek additional capital, or incur indebtedness. The Central States, Southeast and Southwest Areas Pension Plan ("Central States Pension Fund") union sponsored multi-employer pension plan in which we participated until 2021 is also underfunded. While we have recorded an estimated withdrawal liability associated with leaving this plan, changes to that estimate could force us to reduce or delay investments and capital expenditures, seek additional capital or incur indebtedness.

A portion of our workforce is unionized and labor disruptions could decrease our profitability.

As of December 31, 2022, approximately 24% of our labor force is covered by a collective bargaining agreement, including approximately 12%, 46% and 24% of our labor force in the US, Europe and Canada, respectively. Approximately 1% of our labor force is covered by a collective bargaining agreement that will expire within one year. These arrangements grant certain protections to employees. We cannot guarantee that we will be able to negotiate these or other collective bargaining agreements or arrangements with works councils on the same or more favorable terms as the current agreements or arrangements, or at all, and without interruptions, including labor stoppages at the facility or facilities subject to any particular agreement or arrangement. A prolonged labor dispute, which could include a work stoppage, could have a material adverse effect on our business, financial condition and results of operations.

Environmental, Social, and Governance Risks

Our ability to execute on our initiatives and goals related to environmental, social, and governance (ESG) matters and the increasing legal and regulatory focus on ESG could have a material adverse effect on our business, financial condition and results of operations.

We have announced a number of ESG initiatives and goals, which may require ongoing investment to execute, and there is no assurance that we will achieve any of these goals or that our initiatives will achieve their intended outcomes. In addition, our ability to implement some of our strategies and goals may be dependent on external factors out of our control such as third-party collaboration, scientific and technological developments, and the availability of economically feasible solutions at scale. A failure to achieve our goals could expose our business to reputational damage and have a material adverse effect on our business, financial condition or results of operations.

US and international regulators, investors and other stakeholders are increasingly focused on ESG matters and new domestic and international laws and regulations relating to ESG matters are under consideration. Such laws and regulations could affect the output of producers of commodity and specialty chemicals as well as demand in the end markets our customers serve. Responding to any such laws and regulations could require us to incur additional expenditures to either purchase new, or modify existing, equipment or processes, and the implementation of new practices and reporting processes could introduce additional compliance risk. In addition, any such laws and regulations could expose us to fines, litigation and increased tax responsibilities, and negatively impact our ability to compete with companies situated in geographies not subject to such limitations. A failure to meet the increasing ESG expectations of our investors and customers could also lead to market share loss.

Risks Related to Our Common Stock

Anti-takeover provisions in our charter documents and Delaware law could discourage, delay or prevent a change in control of our company and may affect the trading price of our common stock.

Our Certificate of Incorporation and Bylaws include a number of provisions that may discourage, delay or prevent a change in our management or control over us that stockholders may consider favorable. For example, our Certificate of Incorporation and By-laws currently:

- authorize the issuance of “blank check” preferred stock that could be issued by our Board of Directors to thwart a takeover attempt;
- limit the ability of stockholders to act by written consent or call a special meeting; and
- establish advance notice requirements for nominations for election to our Board of Directors or for proposing matters that can be acted upon by stockholders at stockholder meetings.

These provisions may prevent our stockholders from receiving the benefit from any premium to the market price of our common stock offered by a bidder in a takeover context. Even in the absence of a takeover attempt, the existence of these provisions may adversely affect the prevailing market price of our common stock if the provisions are viewed as discouraging takeover attempts in the future. These provisions may also facilitate management entrenchment that may delay, deter, render more difficult or prevent a change in our control, which may not be in the best interests of our stockholders.

General Risk Factors

Our business is subject to additional general regulatory requirements, which increase our cost of doing business, could result in claims and enforcement actions and could restrict our business in the future.

Our general business operations are subject to a broad spectrum of international, federal, state and local laws and regulations, including, without limitation, those relating to antitrust, environmental, food and drug, labor and human resources, tax, trade compliance, unclaimed property, transportation, anti-bribery, banking and treasury, privacy and data protection (including the European Union's General Data Protection Regulation), among others. These laws and regulations add cost to our conduct of business and could, in some instances, result in claims or enforcement actions or could reduce our ability to pursue business opportunities. Any changes in the laws and regulations applicable to us, the enactment of any additional laws or regulations, or the failure to comply with, or increased enforcement activity of, such laws and regulations, could significantly impact our products and services and have a material adverse effect on our business, financial condition and results of operations. Additionally, governmental agencies may refuse to grant or renew our operating licenses and permits.

We are exposed to litigation and other legal and regulatory actions and risks in the ordinary course of our business, and we could incur significant liabilities and substantial legal fees.

Especially because of the nature of our business, we are subject to the risk of litigation, other legal claims and proceedings, and regulatory enforcement actions in the ordinary course of our business. Also, there may be safety or personal injury risks related to our products which are not known today. The results of legal proceedings cannot be predicted with certainty. We cannot guarantee that the results of current or future legal proceedings will not materially harm our business, reputation or brand, nor can we guarantee that we will not incur losses in connection with current or future legal proceedings that exceed any provisions we may have set aside in respect of such proceedings or that exceed any applicable insurance coverage. The occurrence of any of these events could have a material adverse effect on our business, financial condition or results of operations.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 2. PROPERTIES

Our principal executive office is located in Downers Grove, Illinois under a lease expiring in June 2024. As of December 31, 2022, we had 292 locations in the US in 45 states and one territory and 287 locations outside of the US in 31 countries. Our warehouse facilities are comprised of owned, leased and third party warehouses and our office space is generally leased. Our facilities focus on the storing, repackaging and blending of chemicals and ingredients for distribution. Such facilities do not require substantial investments in equipment, can be opened fairly quickly and are replaceable with little disruption. As such, we believe that none of our facilities on an individual basis is material to the operation of our business.

ITEM 3. LEGAL PROCEEDINGS

See “Note 18: Commitments and contingencies” in Item 8 of this Annual Report on Form 10-K for information regarding legal proceedings, the content of which is incorporated by reference to this Item 3.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market Information for Common Stock

Our common stock is listed on the New York Stock Exchange under the symbol UNVR.

Holders of Record

As of December 31, 2022, based on the number of participants reflected in a security position listing provided to us by the Equiniti Trust Company ("EQ"), there were nine holders of record of our common stock. Because such EQ participants include banks, brokers and other institutions holding shares of our common stock on behalf of their customers, we do not know the actual number of unique stockholders represented by these record holders.

Issuer Purchases of Equity Securities

Information on our common stock purchases during the fourth quarter of 2022 is provided below:

Period	Total Number of Shares Purchased	Average Price Paid per Share	Total Number of Shares Purchased as Part of a Publicly Announced Plan or Program [1]	Maximum Approximate Dollar Value of Shares that may yet be Purchased Under the Plan or Program (millions) [1]
October 1-31, 2022	—	$ —	—	$ 1,245.1
November 1-30, 2022 [2]	5,918,353	26.06	5,918,353	1,040.9
December 1-31, 2022	—	—	—	1,040.9
Total	5,918,353	$ 26.06	5,918,353	

(1) On November 1, 2021, the Company announced that its Board of Directors had authorized a share repurchase program of up to $500.0 million of its outstanding common stock, which expires on October 27, 2026. On November 1, 2022, the Company announced that its Board of Directors had approved an increase in the amount of authorized repurchases under the program of $1.0 billion, resulting in a total authorized repurchase amount of $1.5 billion.

(2) On November 1, 2022, the Company entered into an accelerated share repurchase agreement ("ASR") with Goldman Sachs & Co. LLC ("Goldman") to repurchase $200.0 million of its common stock. On November 3, 2022, the Company paid $200.0 million to Goldman and received an initial delivery of approximately 5.8 million shares of its common stock, which represented 75% of the notional value of the ASR divided by the closing price of the Company's common stock on November 1, 2022. The final number of shares repurchased under the ASR was based on the daily volume-weighted average prices for Rule 10b-18 eligible transactions in the Company's common stock during the term of the ASR, less a discount and subject to adjustment pursuant to the terms of the ASR. In February 2023, the final settlement was determined to be an additional delivery of approximately 0.5 million shares of the Company's common stock from Goldman.

Stock Performance

The following graph shows a comparison of cumulative total stockholder return, calculated on a dividend reinvested basis, for the Company, the S&P 500 and the S&P 500 Chemical Index from December 31, 2017 through December 31, 2022. The graph assumes $100 was invested in each of the Company's common stock, the S&P 500 and the S&P 500 Chemical Index as of the market close on December 31, 2017. Note that historic stock price performance is not necessarily indicative of future stock price performance.



	December 31,					
	2017	2018	2019	2020	2021	2022
Univar Solutions Inc.	$ 100	$ 57	$ 78	$ 61	$ 92	$ 103
S&P 500	100	96	126	149	192	157
S&P 500 Chemical Index	100	88	108	127	160	142

Dividend Policy

We did not pay a cash dividend on our common stock in 2022, 2021 or 2020. Our Board of Directors may consider declaring and paying dividends in the future; however, there can be no assurance that our Board of Directors will determine to do so. Additionally, our credit facilities contain certain limitations on our ability to pay dividends on our common stock.

ITEM 6. [RESERVED]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

This Management's Discussion and Analysis of Financial Condition and Results of Operations ("MD&A") is based on financial data derived from the financial statements prepared in accordance with US GAAP and certain other financial data that is prepared using non-GAAP measures. For a reconciliation of each non-GAAP financial measure to its most comparable GAAP measure, see the "Analysis of Segment Results" and "Non-GAAP Financial Measures" sections within this Item. Refer to "Non-GAAP Financial Measures" within this Item for more information about our use of Non-GAAP financial measures.

Our MD&A is provided in addition to the accompanying consolidated financial statements and notes to assist readers in understanding our results of operations, financial condition and cash flows. This section of this Annual Report on Form 10-K discusses year-to-year comparisons between 2022 and 2021. For discussions of year-to-year comparisons between 2021 and 2020 that are not included in this Annual Report on Form 10-K, please refer to "Management's Discussion and Analysis of Financial Condition and Results of Operations" in Part II, Item 7 of the Company's Annual Report on Form 10-K for the year ended December 31, 2021, filed with the SEC on February 25, 2022.

Overview

Univar Solutions Inc. is a leading global solutions provider to users of specialty ingredients and chemicals and provider of value-added services to customers across a wide range of diverse industries. We purchase chemicals and ingredients from thousands of producers worldwide and warehouse, repackage, blend, dilute, transport and sell them to nearly 100,000 customer locations across approximately 120 countries.

Our operations are structured into four reportable segments that represent the geographic areas under which we operate and manage our business. As previously defined within Item 1 of this Annual Report on Form 10-K, our segments are USA, EMEA, Canada and LATAM, which includes developing businesses in Latin America and the Asia-Pacific region.

Factors Affecting Comparability of Results

Acquisitions and Divestitures

In July 2022, we acquired Vicom, a leading regional specialty chemicals distributor in Spain and Portugal.

In December 2021, we acquired Sweetmix, a food ingredients and CASE specialty chemical distribution company in Brazil.

In April 2021, we sold the Distrupol business within our EMEA segment. The sale of this business did not meet the criteria to be classified as discontinued operations in our consolidated financial statements.

See "Note 3: Business combinations" and "Note 4: Dispositions" in Item 8 of this Annual Report on Form 10-K for additional information regarding the acquisitions and divestiture noted above.

Constant Currency

We believe providing information on a non-GAAP constant currency basis offers valuable supplemental information regarding our results of operations, consistent with how we evaluate our performance. Currency impacts on consolidated and segment results have been derived by translating current period financial results in local currency using the average exchange rate for the prior period to which the financial information is being compared.

Inflation Reduction Act

On August 16, 2022, the Inflation Reduction Act ("IRA") was enacted into US law. Effective for tax years beginning after December 31, 2022, the IRA imposes a 15% corporate minimum tax, a 1% excise tax on share repurchases, and creates and extends certain tax-related energy incentives. Management does not expect the tax-related provisions of the IRA to have a material impact on the Company's consolidated financial statements.

Market Conditions

We sell commodity and specialty chemicals and ingredients that are used in manufacturing processes and as components in other products. Our sales are correlated with and affected by seasonal fluctuations and cycles in the levels of industrial production, manufacturing output and general economic activity. The current business environment in the markets in which we operate consists of complex dynamics. A combination of factors has impacted and disrupted global trade flows, which has resulted in differing pricing conditions, in different end markets, in different geographies. The duration of these dislocated and imbalanced conditions is unknown.

These market factors have also impacted the transportation market and, coupled with continued energy price pressure, driver shortages and inflation, have resulted in higher operating costs. Additionally, shortages across a range of chemicals and ingredients have generally led to fluctuations in chemical prices globally, with corresponding impacts to sales and profits.

Investments in working capital to bridge supply disruption and account for higher chemical prices may impact our ability to achieve forecasted cash flows in the short-term.

In such a dynamic environment, we believe remaining connected with our customers to understand demand and supply impacts on their operations is critical to our success. We believe our value as a distributor is heightened in the current environment as we work to meet demand requirements through our extensive network, installed asset base, transportation and digital assets, and supplier partnerships, supported by our network of Solution Centers and technically-trained professionals with deep industry knowledge.

A summary of our sales channel and underlying end market performance as of December 31, 2022, with corresponding impacts from the current environment are as follows (percentages represent 2022 Consolidated Net sales):

Chemicals and Services (66%) - Our Chemicals and Services sales channel saw strong growth over the course of the year with many of the core industries seeing accelerated demand and resilient pricing. Our continued focus on putting the customer at the center of all we do has led to above-market growth in chemical manufacturing, mining and energy. Our complete offering of water treatment chemistries allows us to provide both municipalities and industrial customers with solutions for managing and purifying water. We are beginning to accelerate our offerings into the North American electronics market, with high-purity chemistries necessary for manufacturing microchips. Additionally, we saw growth in agricultural chemistries, pulp and paper and in general manufacturing. Our services business had top-line growth as we improved our service capabilities and we saw a slight rebound in automotive manufacturing. The recent rise in interest rates causes us some concern that global industrial production decline may dampen the demand for our products; however, we anticipate a higher value on increased carrying costs of working capital.

Ingredients and Specialties (34%) - Our Ingredients and Specialties sales channel saw strong growth throughout the year in both our end markets as well as in a number of our differentiated chemistries. We saw particularly strong growth in our Pharmaceuticals business, which has continued to grow since the pandemic on increasing demand for high-purity actives, excipients and other necessary ingredients. Our food ingredients business saw strength throughout the year on security of supply and a partnership with our suppliers and customers to provide solutions that cater to consumer trends. Our beauty and personal care business saw another year of growth on product management and new supplier authorizations. Within our CASE business, we saw product tightness and inflation for much of the year. The slowdown in housing construction resulting from higher interest rates during the back half of the year dampened demand; however, our service performance and product offering allowed for top-line growth. The lubricants and metalworking business finished the year with market share gain as well as margin expansion on good pricing discipline. Finally, the homecare and industrial cleaning business finished the year with sales growth on pricing discipline and further supplier development.

See “Risk Factors” in Part I, Item 1A of this Annual Report on Form 10-K for further information on our Business and Economic Risks.

Executive Summary

Driven by agile commercial execution, chemical price inflation and growing customer demand, Univar Solutions delivered strong year-over-year growth despite constrained inventory and supply and transportation challenges. With the Nexeo acquisition integration and system migration completed, we look to continue to leverage our digital investments to drive organic growth through market share capture and margin gains through leverage of our cost structure.

Management is focused on the following financial objectives:

- Growth above consensus GDP;
- Consolidated Adjusted EBITDA growth and corresponding Adjusted EBITDA margin expansion;
- Increased return on invested capital;
- Maintaining targeted leverage levels;
- Growing inorganically through bolt-on acquisitions; and
- Return of capital to stockholders.

Achievement of these objectives is expected through the following operational initiatives:

- Application of a consistent global strategy, while driving local execution;
- Leverage of our global transportation and digital asset footprint;
- Solutions to support suppliers' and customers' ESG objectives;
- Delivery of an effortless experience for customers that drives preference and retention;
- Focus on technical and application development offerings;
- Unlocking our global Solution Centers' value for suppliers and customers;
- Leverage and expansion of our key supplier relationships, with the goal of enhancing service to key global customers;

- Capitalize on US re-industrialization and near-shoring trends; and
- Execute accretive bolt-on acquisition opportunities in regions and markets where we believe there are synergistic opportunities for growth, cross-selling and cost rationalization.

Results of Operations

	Year ended December 31,					
	2022		2021			
(in millions)	Dollars	% of Net sales	Dollars	% of Net sales	Favorable (unfavorable)	% Change
Net sales	$ 11,475.3	100.0 %	$ 9,535.5	100.0 %	$ 1,939.8	20.3 %
Cost of goods sold (exclusive of depreciation)	8,704.1	75.9 %	7,142.3	74.9 %	(1,561.8)	21.9 %
Operating expenses:						
Outbound freight and handling	476.8	4.2 %	403.7	4.2 %	(73.1)	18.1 %
Warehousing, selling and administrative	1,248.5	10.9 %	1,191.8	12.5 %	(56.7)	4.8 %
Other operating expenses, net	32.8	0.3 %	107.5	1.1 %	74.7	(69.5)%
Depreciation	131.7	1.1 %	150.9	1.6 %	19.2	(12.7)%
Amortization	48.2	0.4 %	52.5	0.6 %	4.3	(8.2)%
Impairment charges	0.6	— %	3.0	— %	2.4	(80.0)%
Total operating expenses	1,938.6	16.9 %	1,909.4	20.0 %	(29.2)	1.5 %
Operating income	832.6	7.3 %	483.8	5.1 %	348.8	72.1 %
Other (expense) income:						
Interest income	4.3	— %	4.3	— %	—	— %
Interest expense	(107.2)	(0.9)%	(101.5)	(1.1)%	(5.7)	5.6 %
Gain on sale of business	—	— %	88.2	0.9 %	(88.2)	(100.0)%
Other income, net	26.5	0.2 %	110.4	1.2 %	(83.9)	(76.0)%
Total other (expense) income	(76.4)	(0.7)%	101.4	1.1 %	(177.8)	(175.3)%
Income before income taxes	756.2	6.6 %	585.2	6.1 %	171.0	29.2 %
Income tax expense	210.9	1.8 %	124.6	1.3 %	(86.3)	69.3 %
Net income	$ 545.3	4.8 %	$ 460.6	4.8 %	$ 84.7	18.4 %

	Year ended December 31,					
	2022		2021			
(in millions)	Dollars	% of Net sales	Dollars	% of Net sales	Favorable (unfavorable)	% Change
Gross profit (exclusive of depreciation):						
Net sales	$ 11,475.3	100.0 %	$ 9,535.5	100.0 %	$ 1,939.8	20.3 %
Cost of goods sold (exclusive of depreciation)	8,704.1	75.9 %	7,142.3	74.9 %	(1,561.8)	21.9 %
Gross profit (exclusive of depreciation)	$ 2,771.2	24.1 %	$ 2,393.2	25.1 %	$ 378.0	15.8 %

Net sales

Net sales increased $1,939.8 million, or 20.3%, for the year ended December 31, 2022. On a constant currency basis, net sales increased by $2,302.2 million, or 24.1%. The increase was primarily due to our pricing discipline in inflationary markets and market share gains. Refer to the “Analysis of Segment Results” for additional information.

Gross profit (exclusive of depreciation)

Gross profit (exclusive of depreciation) increased $378.0 million, or 15.8%, to $2,771.2 million for the year ended December 31, 2022. On a constant currency basis, gross profit (exclusive of depreciation) increased $459.2 million or 19.2%. The increase in gross profit (exclusive of depreciation) was primarily attributable to our pricing discipline in inflationary markets, operational execution and market share gains, partially offset by higher input cost inflation. Gross margin decreased from 25.1% for the year ended December 31, 2021 to 24.1% for the year ended December 31, 2022. Refer to the “Analysis of Segment Results” and “Non-GAAP Financial Measures” for additional information.

Operating expenses

Outbound freight and handling

Outbound freight and handling expenses increased $73.1 million, or 18.1%, for the year ended December 31, 2022. On a constant currency basis, outbound freight and handling expenses increased $83.1 million, or 20.6%. Refer to the “Analysis of Segment Results” for additional information.

Warehousing, selling and administrative

Warehousing, selling and administrative expenses ("WS&A") increased $56.7 million, or 4.8%, for the year ended December 31, 2022. On a constant currency basis, WS&A increased $94.6 million, or 7.9%. The increase is primarily attributable to higher operating costs and variable compensation, partially offset by an environmental recovery, net synergies and higher environmental remediation costs in the prior year. Refer to the “Analysis of Segment Results” for additional information.

Other operating expenses, net

Other operating expenses, net decreased $74.7 million, or 69.5%, for the year ended December 31, 2022. Refer to “Note 8: Supplemental financial information” in Item 8 of this Annual Report on Form 10-K for additional information.

Depreciation and Amortization

Depreciation expense decreased $19.2 million, or 12.7%, for the year ended December 31, 2022, primarily due to certain assets reaching the end of their depreciable lives. On a constant currency basis, depreciation expense decreased $16.0 million, or 10.6%, for the year ended December 31, 2022, primarily due to certain assets reaching the end of their depreciable lives.

Amortization expense decreased $4.3 million, or 8.2%, for the year ended December 31, 2022, primarily due to certain intangible assets reaching the end of their amortizable lives.

Other (expense) income

Interest expense

Interest expense increased $5.7 million, or 5.6%, for the year ended December 31, 2022, primarily due to higher average interest rates on floating rate debt. Refer to “Note 11: Debt” in Item 8 of this Annual Report on Form 10-K for additional information.

Gain on sale of business

Gain on sale of business was $88.2 million for the year ended December 31, 2021, primarily due to the sale of our Distrupol business and also included adjustments to the sale price of the Canadian Agriculture services business. Refer to “Note 4: Dispositions” in Item 8 of this Annual Report on Form 10-K for additional information.

Other income, net

Other income, net decreased $83.9 million, or 76.0%, for the year ended December 31, 2022, primarily due to a lower pension mark to market gain in the current year and a fair value adjustment on warrants in the prior year. Refer to “Note 8: Supplemental financial information” in Item 8 of this Annual Report on Form 10-K for additional information.

Income tax expense

Income tax expense was $210.9 million for the year ended December 31, 2022, resulting in an effective income tax rate of 27.9%. Our effective income tax rate was higher than the US federal statutory rate of 21.0%, primarily due to higher rates on foreign earnings, US tax on foreign earnings, US state income taxes and non-deductible employee costs.

Income tax expense was $124.6 million for the year ended December 31, 2021, resulting in an effective income tax rate of 21.3%. Our effective income tax rate was higher than the US federal statutory rate of 21.0% primarily due to higher rates on foreign earnings, US tax on foreign earnings, US state income taxes and non-deductible employee costs offset by the favorable impact of the Distrupol divestiture.

Results of Reportable Business Segments

Our operations are structured into four reportable segments that represent the geographic areas under which we operate and manage our business. These segments are USA, EMEA, Canada and LATAM, which includes developing businesses in Latin America and the Asia-Pacific region. Management believes Adjusted EBITDA is an important measure of operating performance, which is used as the primary basis for the chief operating decision maker to evaluate the performance of each of our reportable segments.

We believe certain other financial measures that are not calculated in accordance with US GAAP provide relevant and meaningful information concerning our ongoing operating results. These financial measures include gross profit (exclusive of depreciation), gross margin and Adjusted EBITDA margin. Such non-GAAP financial measures are referred to from time to time in this report but should not be viewed as a substitute for GAAP measures of performance and should be considered along with the comparable US GAAP measures. See “Note 20: Segments” in Item 8 of this Annual Report on Form 10-K, “Analysis of Segment Results” within this Item and "Non-GAAP Financial Measures" within this Item for additional information.

Analysis of Segment Results

USA

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Net sales:				
External customers	$ 7,533.5	$ 6,024.0	$ 1,509.5	25.1 %
Inter-segment	129.5	92.1	37.4	40.6 %
Total net sales	7,663.0	6,116.1	1,546.9	25.3 %
Cost of goods sold (exclusive of depreciation)	5,799.9	4,578.6	(1,221.3)	26.7 %
Outbound freight and handling	359.4	291.0	(68.4)	23.5 %
Warehousing, selling and administrative	794.4	748.3	(46.1)	6.2 %
Adjusted EBITDA	$ 709.3	$ 498.2	$ 211.1	42.4 %

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Gross profit (exclusive of depreciation):				
Net sales	$ 7,663.0	$ 6,116.1	$ 1,546.9	25.3 %
Cost of goods sold (exclusive of depreciation)	5,799.9	4,578.6	(1,221.3)	26.7 %
Gross profit (exclusive of depreciation)	$ 1,863.1	$ 1,537.5	$ 325.6	21.2 %

External sales increased $1,509.5 million, or 25.1%, for the year ended December 31, 2022, primarily due to our pricing discipline in inflationary markets and market share gains.

Gross profit (exclusive of depreciation) increased $325.6 million, or 21.2%, for the year ended December 31, 2022, primarily due to pricing discipline in inflationary markets, operational execution and market share gains, partially offset by input cost inflation. Gross margin decreased from 25.5% for the year ended December 31, 2021 to 24.7% for the year ended December 31, 2022, primarily impacted by input cost inflation, partially offset by our pricing discipline in inflationary markets.

Outbound freight and handling expenses increased $68.4 million, or 23.5%, for the year ended December 31, 2022, primarily due to higher delivery costs caused by supply chain constraints.

WS&A increased $46.1 million, or 6.2%, for the year ended December 31, 2022, primarily due to higher operating costs and variable compensation, partially offset by higher environmental remediation costs in the prior year. Net synergies and an environmental recovery also favorably impacted the current year. As a percentage of external sales, WS&A decreased from 12.4% for the year ended December 31, 2021 to 10.5% for the year ended December 31, 2022 primarily due to the increase in sales.

Adjusted EBITDA increased $211.1 million, or 42.4%, for the year ended December 31, 2022, driven by higher gross profit (exclusive of depreciation), partially offset by higher outbound freight and handling expenses as well as higher WS&A. Adjusted EBITDA margin increased from 8.3% for the year ended December 31, 2021 to 9.4% for the year ended December 31, 2022, reflecting the business operating leverage.

EMEA

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Net sales:				
External customers	$ 2,064.7	$ 1,971.1	$ 93.6	4.7 %
Inter-segment	10.6	4.3	6.3	146.5 %
Total net sales	2,075.3	1,975.4	99.9	5.1 %
Cost of goods sold (exclusive of depreciation)	1,594.8	1,488.6	(106.2)	7.1 %
Outbound freight and handling	65.9	64.0	(1.9)	3.0 %
Warehousing, selling and administrative	238.0	252.2	14.2	(5.6)%
Adjusted EBITDA	$ 176.6	$ 170.6	$ 6.0	3.5 %

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Gross profit (exclusive of depreciation):				
Net sales	$ 2,075.3	$ 1,975.4	$ 99.9	5.1 %
Cost of goods sold (exclusive of depreciation)	1,594.8	1,488.6	(106.2)	7.1 %
Gross profit (exclusive of depreciation)	$ 480.5	$ 486.8	$ (6.3)	(1.3)%

External sales increased $93.6 million, or 4.7%, for the year ended December 31, 2022. On a constant currency basis, external sales increased $422.2 million, or 21.4%, primarily due to our pricing discipline in inflationary markets and market share gains, partially offset by the effects of the Distrupol divestiture in the prior year.

Gross profit (exclusive of depreciation) decreased $6.3 million, or 1.3%, for the year ended December 31, 2022. On a constant currency basis, gross profit (exclusive of depreciation) increased $67.6 million, or 13.9%, primarily due to pricing discipline in inflationary markets, operational execution and market share gains, partially offset by the effects of the Distrupol divestiture in the prior year. Gross margin decreased from 24.7% for the year ended December 31, 2021 to 23.3% for the year ended December 31, 2022 primarily due to input cost inflation, partially offset by our pricing discipline in inflationary markets.

Outbound freight and handling expenses increased $1.9 million, or 3.0%, for the year ended December 31, 2022. On a constant currency basis, outbound freight and handling expenses increased $10.7 million, or 16.7%, primarily due to higher delivery costs caused by supply chain constraints.

WS&A decreased $14.2 million, or 5.6%, for the year ended December 31, 2022. On a constant currency basis, WS&A increased $20.9 million, or 8.3%, primarily due to higher variable compensation and operating costs. As a percentage of external sales, WS&A decreased from 12.8% for the year ended December 31, 2021 to 11.5% for the year ended December 31, 2022 due to the impact of fluctuations in foreign currency exchange rates, partially offset by higher variable compensation and operating costs.

Adjusted EBITDA increased $6.0 million, or 3.5%, for the year ended December 31, 2022. On a constant currency basis, Adjusted EBITDA increased $36.0 million, or 21.1%, primarily due to higher gross profit (exclusive of depreciation), partially offset by higher WS&A as well as higher outbound freight and handling expenses. Adjusted EBITDA margin decreased from 8.7% for the year ended December 31, 2021 to 8.6% for the year ended December 31, 2022 primarily due to lower gross margin, partially offset by business operating leverage.

Canada

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Net sales:				
External customers	$ 1,120.5	$ 930.0	$ 190.5	20.5 %
Inter-segment	10.5	4.4	6.1	138.6 %
Total net sales	1,131.0	934.4	196.6	21.0 %
Cost of goods sold (exclusive of depreciation)	864.7	701.3	(163.4)	23.3 %
Outbound freight and handling	38.4	36.2	(2.2)	6.1 %
Warehousing, selling and administrative	108.2	92.7	(15.5)	16.7 %
Adjusted EBITDA	$ 119.7	$ 104.2	$ 15.5	14.9 %

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Gross profit (exclusive of depreciation):				
Net sales	$ 1,131.0	$ 934.4	$ 196.6	21.0 %
Cost of goods sold (exclusive of depreciation)	864.7	701.3	(163.4)	23.3 %
Gross profit (exclusive of depreciation)	$ 266.3	$ 233.1	$ 33.2	14.2 %

External sales increased $190.5 million, or 20.5%, for the year ended December 31, 2022. On a constant currency basis, external sales increased $233.6 million, or 25.1%, primarily due to our pricing discipline in inflationary markets and market share gain.

Gross profit (exclusive of depreciation) increased $33.2 million, or 14.2%, for the year ended December 31, 2022. On a constant currency basis, gross profit (exclusive of depreciation) increased $43.4 million, or 18.6%, primarily due to our pricing discipline in inflationary markets, operational execution and market share gains, partially offset by input cost inflation. Gross margin decreased from 25.1% for the year ended December 31, 2021 to 23.8% for the year ended December 31, 2022, driven by input cost inflation, partially offset by our pricing discipline in inflationary markets.

Outbound freight and handling expenses increased $2.2 million, or 6.1%, for the year ended December 31, 2022. On a constant currency basis, outbound freight and handling expenses increased $3.7 million, or 10.2%, primarily due to higher delivery costs caused by supply chain constraints.

WS&A increased $15.5 million, or 16.7%, for the year ended December 31, 2022. On a constant currency basis, WS&A increased $19.6 million, or 21.1%. The increase was primarily due to higher operating costs and variable compensation. WS&A as a percentage of external sales decreased from 10.0% for the year ended December 31, 2021 to 9.7% for the year ended December 31, 2022 primarily due to the increase in sales.

Adjusted EBITDA increased $15.5 million, or 14.9%, for the year ended December 31, 2022. On a constant currency basis, Adjusted EBITDA increased $20.1 million, or 19.3%, primarily due to higher gross profit (exclusive of depreciation), partially offset by increased WS&A. Adjusted EBITDA margin decreased from 11.2% for the year ended December 31, 2021 to 10.7% for the year ended December 31, 2022 due to lower gross margin, partially offset by business operating leverage.

LATAM

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Net sales:				
External customers	$ 756.6	$ 610.4	$ 146.2	24.0 %
Inter-segment	0.5	0.7	(0.2)	(28.6)%
Total net sales	757.1	611.1	146.0	23.9 %
Cost of goods sold (exclusive of depreciation)	595.8	475.3	(120.5)	25.4 %
Outbound freight and handling	13.0	12.5	(0.5)	4.0 %
Warehousing, selling and administrative	85.3	66.4	(18.9)	28.5 %
Adjusted EBITDA	$ 63.0	$ 56.9	$ 6.1	10.7 %

(in millions)	Year ended December 31, 2022	2021	Favorable (unfavorable)	% Change
Gross profit (exclusive of depreciation):				
Net sales	$ 757.1	$ 611.1	$ 146.0	23.9 %
Cost of goods sold (exclusive of depreciation)	595.8	475.3	(120.5)	25.4 %
Gross profit (exclusive of depreciation)	$ 161.3	$ 135.8	$ 25.5	18.8 %

External sales increased $146.2 million, or 24.0%, for the year ended December 31, 2022.On a constant currency basis, external sales increased $137.0 million, or 22.4%, primarily due to our pricing discipline in inflationary markets and the Sweetmix acquisition, which contributed 9.1% of the increase.

Gross profit (exclusive of depreciation) increased $25.5 million, or 18.8%, for the year ended December 31, 2022.On a constant currency basis, gross profit (exclusive of depreciation) increased $22.6 million, or 16.6%, primarily due to our pricing discipline in inflationary markets and the Sweetmix acquisition, which contributed 7.7% of the increase, partially offset by input cost inflation. Gross margin decreased from 22.2% for the year ended December 31, 2021 to 21.3% for the year ended December 31, 2022, primarily due to input cost inflation.

Outbound freight and handling expenses increased $0.5 million, or 4.0%, for the year ended December 31, 2022. On a constant currency basis, outbound freight and handling expenses increased $0.3 million, or 2.4%.

WS&A increased $18.9 million, or 28.5%, for the year ended December 31, 2022. On constant currency basis, WS&A increased $17.4 million, or 26.2%, primarily due to increased corporate cost allocation as a result of a SAP implementation and higher operating costs. As a percentage of external sales, WS&A increased from 10.9% for the year ended December 31, 2021 to 11.3% for the year ended December 31, 2022, primarily due to increased corporate cost allocation as a result of a SAP implementation and higher operating costs.

Adjusted EBITDA increased $6.1 million, or 10.7%, for the year ended December 31, 2022. On a constant currency basis, Adjusted EBITDA increased $4.8 million, or 8.4%, primarily due to higher gross profit (exclusive of depreciation), partially offset by increased WS&A. Adjusted EBITDA margin decreased from 9.3% for the year ended December 31, 2021 to 8.3% for the year ended December 31, 2022, primarily due to lower gross margin and increased WS&A.

Liquidity and Capital Resources

The Company's primary sources of liquidity are cash generated from operations and borrowings under its committed Senior ABL Credit Facility. As of December 31, 2022, liquidity for the Company was $1,060.5 million, comprised of $385.3 million of cash and cash equivalents and $675.2 million of available borrowings under our credit facility. The credit facility is guaranteed by certain significant subsidiaries and secured by such parties' eligible accounts receivable, inventory and cash with a maximum borrowing capacity of $1.6 billion. Significant reductions in our accounts receivable, inventory and cash would reduce our availability to access liquidity under the credit facility. We have no active financial maintenance covenants in our credit agreements; however, there is a springing fixed charge coverage ratio ("FCCR") under the revolving credit facility of 1.0x, applicable only if availability is less than or equal to 10% of the borrowing capacity. If the FCCR was applicable, the calculation would have been 6.5x as of December 31, 2022.

Our primary short-term liquidity and capital resource needs are to finance operating expenses, working capital, capital expenditures, other liabilities including environmental remediation and interest, possible business acquisitions, share repurchases and general corporate purposes. The majority of our debt obligations mature in 2026 and beyond. To the extent that our cash balances from time to time exceed amounts that are needed to fund our immediate liquidity requirements, we will consider alternative uses of some or all of such excess cash. Such alternatives may include, among others, the redemption or repurchase of debt securities or other borrowings through open market purchases, privately negotiated transactions or otherwise. Refer to "Note 11: Debt" in Item 8 of this Annual Report on Form 10-K for additional information related to our debt obligations. Management continues to balance its focus on sales and earnings growth with continuing efforts in cost control and working capital management.

Access to debt capital markets has historically provided the Company with sources of liquidity beyond normal operating cash flows. We do not anticipate having difficulty in obtaining financing from those markets in the future with our history of favorable results in the debt capital markets and strong relationships with global financial institutions. However, our ability to continue to access the debt capital markets with favorable interest rates and other terms will depend, to a significant degree, on maintaining our current ratings assigned by the credit rating agencies.

We may from time to time refinance or take steps to reduce debt or interest costs. The amount of debt, if any, that may be reduced or refinanced will depend on market conditions, trading levels of our debt, our cash position, compliance with debt covenants and other considerations. On October 27, 2022, we entered into the Second Amended and Restated ABL Credit Agreement ("Senior ABL Facility"), which provided a five-year senior secured ABL credit facility in an aggregate principal amount of $1.6 billion and a five-year senior secured term loan facility in an aggregate principal amount of $200 million. The Senior ABL Facility amends and restates in full the amended and restated ABL facility entered into by the Company on February 28, 2019. In connection with the entry into the Senior ABL Facility, the Company terminated its existing European ABL Credit Agreement and the Euro ABL Facility thereunder.

Our defined benefit pension plans had an underfunded status of $82.7 million and $122.3 million as of December 31, 2022 and 2021, respectively. Based on current projections and minimum funding requirements, we expect to make cash contributions of $16.0 million to our defined benefit pension plans in 2023. The amount and timing of any such requirement in future years is uncertain given the implicit uncertainty regarding the future developments of factors described in "Risk Factors" in Item 1A of this Annual Report on Form 10-K.

We expect our 2023 capital expenditures to be approximately $150 million to $160 million for maintenance and growth, including safety, cost improvements and ESG investments. Interest payments for 2023 are expected to be $120 million to $130 million. We expect to fund our capital expenditures and interest payments with cash from operations or cash on hand.

We believe funds provided by our primary sources of liquidity will be adequate to meet our liquidity, debt repayment obligation and capital resource needs for at least the next 12 months under current operating conditions.

Cash Flows

The following table presents a summary of our cash flows:

(in millions)	Year ended December 31, 2022	2021
Net cash provided by operating activities	$ 546.4	$ 290.3
Net cash (used) provided by investing activities	(161.7)	23.6
Net cash used by financing activities	(232.7)	(424.6)

Operating Activities

Cash provided by operating activities increased $256.1 million for the year ended December 31, 2022. The increase was primarily due to higher net income, exclusive of non-cash items, and the timing of changes in trade working capital, partially offset by the other, net cash flow item.

The change in net income, exclusive of non-cash items, increased $274.7 million from $500.4 million for the year ended December 31, 2021 to $775.1 million for the year ended December 31, 2022. Cash used by other, net increased $180.2 million as compared to the prior year, primarily attributable to tax payments, accrued compensation and timing differences related to other assets and liabilities.

Cash used by trade working capital, which includes trade accounts receivable, net, inventories and trade accounts payable, decreased $139.3 million as compared to the prior year. The year-over-year decrease in cash used by trade working capital was due to a favorable change in trade accounts receivable, net related to the timing of sales and cash collections, partially offset by an unfavorable change in trade accounts payable primarily attributable to the timing of inventory purchases and payments.

Investing Activities

Investing cash flows for the year ended December 31, 2022 included capital expenditures of $153.8 million, cash paid for the Vicom acquisition of $12.9 million, cash paid for the Sweetmix acquisition of $3.8 million and proceeds of $7.8 million from the sale of property, plant and equipment. Investing cash flows for the year ended December 31, 2021 included proceeds of $136.7 million from the sale of the Distrupol business, proceeds of $29.0 million from the sale of property, plant and equipment, capital expenditures of $110.9 million and cash paid for the Sweetmix acquisition of $28.7 million (net of cash acquired of $1.2 million).

Financing Activities

Financing cash flows for the year ended December 31, 2022 included share repurchases of $409.1 million, long-term debt repayments of $87.7 million, long-term debt issuances of $199.6 million and net proceeds under revolving credit facilities of $51.8 million. Financing cash flows for the year ended December 31, 2021 included long-term debt repayments of $1,440.5 million, share repurchases of $50.0 million, long-term debt issuances of $995.0 million, net proceeds under revolving credit facilities of $32.4 million and proceeds from the exercise of warrants of $27.1 million.

Contractual Obligations and Commitments

Our significant contractual obligations and commitments as of December 31, 2022 were as follows:

	Payments Due by Period				
(in millions)	Total	2023	2024 - 2025	2026 - 2027	Thereafter
Finance leases [1]	$ 114.9	$ 28.7	$ 48.4	$ 31.5	$ 6.3
Long-term debt, including current maturities [2]	2,380.0	14.0	28.0	1,403.0	935.0
Interest [3]	545.8	124.0	219.3	183.3	19.2
Operating leases [1]	265.9	53.2	70.1	49.5	93.1
Estimated environmental liability payments [4]	90.9	36.5	19.3	9.8	25.3
Other [5]	63.1	3.7	16.4	12.5	30.5
Total [6]	$ 3,460.6	$ 260.1	$ 401.5	$ 1,689.6	$ 1,109.4

(1) See “Note 19: Leasing” in Item 8 of this Annual Report on Form 10-K for additional information.
(2) See “Note 11: Debt” in Item 8 of this Annual Report on Form 10-K for additional information.
(3) Interest payments on debt are calculated for future periods using projected interest rates and contractual maturities as of December 31, 2022. Projected interest payments include the related effects of interest rate and cross currency swap contracts. Certain of these projected interest payments may differ in the future based on changes in floating interest rates, foreign currency fluctuations or other factors or events.
(4) See "Note 18: Commitments and contingencies" in Item 8 of this Annual Report on Form 10-K for additional information.
(5) Commitments related to multi-employer pension plan withdrawal obligations and acquisitions.
(6) This table excludes our defined benefit pension contributions. Based on current projections and minimum funding requirements, we expect to make cash contributions of $16.0 million to our defined benefit pension plans in the year ended December 31, 2023. The amount and timing of any such requirement in future years is uncertain given the implicit uncertainty regarding the future developments of factors described in “Risk Factors” in Item 1A of this Annual Report on Form 10-K and “Note 12: Employee benefit plans” in Item 8 of this Annual Report on Form 10-K.

We enter into certain unconditional purchase commitments in the normal course of business. These commitments do not exceed our projected operational requirements and generally do not extend past one year.

We expect that we will be able to fund our obligations and commitments with cash flows from operations. To the extent we are unable to fund these obligations and commitments with cash flows from operations, we intend to fund them with proceeds from available borrowing capacity under our credit facility or future financings. With the exception of outstanding letters of credit of $127.7 million, we had no material off-balance sheet arrangements as of December 31, 2022.

Critical Accounting Estimates

Preparation of our financial statements requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenues and expenses. Our significant accounting policies are described in “Note 2: Significant accounting policies” in Item 8 of this Annual Report on Form 10-K. We consider an accounting estimate to be critical if that estimate requires that we make assumptions about matters that are highly uncertain at the time we make that estimate and if different estimates that we could reasonably have used or changes in accounting estimates that are reasonably likely to occur could materially affect our consolidated financial statements. The accounting assumptions and estimates discussed below are those that we consider most critical to an understanding of our financial statements because they inherently involve significant judgments and estimates. By their nature, these judgments and estimates are subject to an inherent degree of uncertainty. Actual results could differ from our estimates.

Goodwill

Total goodwill as of December 31, 2022 and 2021 was $2,288.2 million and $2,310.4 million, respectively. We perform an annual goodwill impairment assessment at the reporting unit level each year as of October 1, or more frequently if potential impairment indicators exist. The analysis may include both qualitative and quantitative factors to assess the likelihood of an impairment. A reporting unit’s carrying value used in an impairment test represents the assignment of various assets and liabilities, including corporate allocations, based on the enterprise approach.

Qualitative factors include industry and market considerations, overall financial performance and other relevant events and factors affecting the reporting units. Additionally, as part of this assessment, we may perform a quantitative analysis to support the qualitative factors above by applying sensitivities to assumptions and inputs used in measuring a reporting unit’s fair value.

Our quantitative impairment test considers both the income approach and the market approach to estimate a reporting unit’s fair value. Significant assumptions include forecasted EBITDA, market segment growth rates and discount rates based on a reporting unit's weighted average cost of capital. The use of different assumptions, estimates or judgments could significantly impact the estimated fair value of a reporting unit, and therefore impact a reporting unit's fair value in excess of carrying value assessment.

Through qualitative assessments performed on the USA, EMEA, Canada, Latin America and Asia-Pacific reporting units, we concluded that it was more likely than not that each reporting unit's fair value exceeded its carrying value. As such, a quantitative assessment was not performed for any of our reporting units. No goodwill impairment losses were recorded during the years ended December 31, 2022, 2021, or 2020.

Environmental Liabilities

We recognize environmental liabilities for probable and reasonably estimable losses associated with environmental remediation. The estimated environmental liability includes incremental direct costs of investigations, remediation efforts and post-remediation monitoring. Total environmental liabilities at December 31, 2022 and 2021 were $90.9 million and $88.1 million, respectively. See "Note 18: Commitments and contingencies" in Item 8 of this Annual Report on Form 10-K for additional information.

Our environmental liabilities are subject to numerous uncertainties that affect our ability to estimate our costs, or our share of costs if multiple parties are responsible. These uncertainties involve the legal, regulatory and enforcement parameters governing environmental assessment and remediation, the nature and extent of contamination at the sites, the extent and cost of assessment and remediation efforts required, our insurance coverage for the sites and, in the case of sites with multiple responsible parties, the number and financial strength of those parties. In addition, our determination as to whether a loss is probable may change, particularly as new facts emerge as to the causes of contamination. We evaluate each environmental site as new information and facts become available and make adjustments to accruals based upon our assessment of these factors, using technical experts, legal counsel and other specialists.

Defined Benefit Pension Plans

We sponsor defined benefit pension plans in the US and other countries. The valuation for these plans depends on assumptions made by management, which are used by actuaries we engage to calculate the projected and accumulated benefit obligations and the annual expense recognized for the plans. Significant assumptions include discount rates and expected rate of return on plan assets. Changes in assumptions and plan experience result in the recognition of gains and losses in earnings, as our accounting policy is to recognize changes in the fair value of plan assets and projected benefit obligation in the fourth quarter of each year (the "mark to market" adjustment), unless an earlier remeasurement is required. For the years ended December 31, 2022 and 2021, we recorded mark to market gains of $17.2 million and $75.9 million, respectively. See "Note 12: Employee benefit plans" in Item 8 of this Annual Report on Form 10-K for additional information.

Changes in the assumed discount rate and expected rate of return on plan assets would have the following effects:

		Increase (decrease) in	
(in millions)	Change	2023 Net benefit cost	2022 Projected benefit obligation
Discount rate	100 bps decrease	$ (2.5)	$ 99.8
Discount rate	100 bps increase	1.7	(87.9)
Expected rate of return on plan assets	100 bps decrease	5.0	N/A
Expected rate of return on plan assets	100 bps increase	(4.4)	N/A

Income Taxes

We are subject to income taxes in the jurisdictions in which we sell products and earn revenues. We record income taxes under the asset and liability method. Under this method, deferred tax assets and liabilities are recognized based on the future tax consequences of temporary differences between the financial statement carrying values of existing assets and liabilities and their respective tax bases. Deferred tax assets and liabilities are measured using the enacted tax rates expected to apply in the years in which the temporary differences are expected to reverse. A reduction of the carrying values of deferred tax assets by a valuation allowance is required if, based on the available evidence, it is more likely than not that such assets will not be realized. In evaluating our ability to realize deferred tax assets, in full or in part, we consider all available positive and negative evidence, including past operating results, forecasted and appropriate character of future taxable income, the duration of statutory carryforward periods, our experience with operating loss and tax credit carryforwards not expiring unused and feasible tax strategies. We have a valuation allowance on certain deferred tax assets, primarily related to foreign tax credits and net operating loss carry forwards. We account for unrecognized tax benefits through an assessment of whether a position is more likely than not to be sustained upon examination by taxing authorities based on its technical merits. We record liabilities for uncertain tax positions taken or expected to be taken in a tax return.

Recently Issued Accounting Pronouncements

See "Note 2: Significant accounting policies" in Item 8 of this Annual Report on Form 10-K.

Non-GAAP Financial Measures

We monitor the results of our reportable segments separately for the purposes of making decisions about resource allocation and performance assessment, and evaluate performance using Adjusted EBITDA. Additionally, the Company uses Adjusted EBITDA in setting performance incentive targets to align management compensation measurement with operational performance.

We define Adjusted EBITDA as the sum of consolidated net income; depreciation; amortization; net interest expense; income tax expense; impairment charges; (gain) loss on sale of business; other operating expenses, net and other income (expense), net (for both, see "Note 8: Supplemental financial information" in Item 8 of this Annual Report on Form 10-K for additional information). For 2020, Adjusted EBITDA also included an adjustment to remove a Brazil VAT charge. See below for a reconciliation of net income, the most comparable measure calculated in accordance with GAAP, to Adjusted EBITDA.

We believe other financial measures, as defined below, that do not comply with US GAAP provide relevant and meaningful information concerning the ongoing and future operating results of the Company.

- Gross profit (exclusive of depreciation): net sales less cost of goods sold (exclusive of depreciation);
- Gross margin: gross profit (exclusive of depreciation) divided by external sales on a segment level and by net sales on a consolidated level; and
- Adjusted EBITDA margin: Adjusted EBITDA divided by external sales on a segment level and by net sales on a consolidated level.

We evaluate our results of operations on both an as reported and a constant currency basis. The constant currency presentation is a non-GAAP financial measure, which excludes the impact of fluctuations in foreign currency exchange rates. We believe providing information on a constant currency basis provides valuable supplemental information regarding our results of operations, consistent with how we evaluate our performance. We calculate constant currency percentages and other information by converting our financial results in local currency for a period using the average exchange rate for the prior period to which we are comparing.

The non-GAAP financial measures noted above are not calculated in accordance with GAAP and should not be considered a substitute for net income or any other measure of financial performance presented in accordance with GAAP. They are included as a complement to results provided in accordance with GAAP because management believes these non-GAAP financial measures help investors' ability to analyze underlying trends in the Company's business, evaluate its performance relative to other companies in its industry and provide useful information to both management and investors by excluding certain items that may not be indicative of the Company's core operating results. Additionally, other companies may calculate Adjusted EBITDA differently than we do, limiting its usefulness as a comparative measure.

The following table is a reconciliation of net income to Adjusted EBITDA:

	Year ended December 31,		
(in millions)	**2022**	**2021**	**2020**
Net income	$ 545.3	$ 460.6	$ 52.9
Depreciation	131.7	150.9	162.9
Amortization	48.2	52.5	60.0
Interest expense, net	102.9	97.2	112.4
Income tax expense	210.9	124.6	6.1
EBITDA	1,039.0	885.8	394.3
Other operating expenses, net	32.8	107.5	89.4
Other (income) expense, net	(26.5)	(110.4)	61.0
Impairment charges	0.6	3.0	40.2
(Gain) loss on sale of business	—	(88.2)	50.6
Brazil VAT charge	—	—	0.3
Adjusted EBITDA	$ 1,045.9	$ 797.7	$ 635.8

The following table is a reconciliation of gross profit (exclusive of depreciation):

	Year ended December 31,		
(in millions)	2022	2021	2020
Net sales	$ 11,475.3	$ 9,535.5	$ 8,265.0
Cost of goods sold (exclusive of depreciation)	8,704.1	7,142.3	6,262.8
Gross profit (exclusive of depreciation)	$ 2,771.2	$ 2,393.2	$ 2,002.2

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

Financial Risk Management Objectives and Policies

The principal risks arising from our financial instruments are interest rate and foreign currency risk. We use derivative financial instruments to reduce exposure to fluctuations in foreign exchange rates and interest rates in certain limited circumstances described below. We follow a strict policy that prohibits trading in financial instruments other than to acquire and manage these hedging positions. We do not hold or issue derivative or other financial instruments for speculative purposes, or to hedge translation risk.

Interest Rate Risk

Our exposure to market risk for changes in interest rates relates primarily to our long-term debt obligations. Under our hedging policy, we seek to maintain an appropriate amount of fixed-rate debt obligations, either directly or effectively through interest rate derivative contracts that fix the interest rate payable on all or a portion of our floating rate debt obligations. We assess the anticipated mix of fixed versus floating amount of debt once a year, in connection with our annual budgeting process, with the purpose of hedging variability of interest expense and interest payments on our variable rate bank debt and maintaining a mix of both fixed and floating rate debt. As of December 31, 2022, approximately 68% of our debt was fixed rate after consideration of interest rate swap contracts. An increase in interest rates of 100 basis points on our floating rate debt as of December 31, 2022 would increase our annual interest expense by approximately $7.9 million, including the impact of derivatives.

Foreign Currency Risk

Because we conduct our business on an international basis in multiple currencies, we may be adversely affected by foreign exchange rate fluctuations. Although we report financial results in US dollars, a substantial portion of our net sales and expenses are denominated in currencies other than the US dollar, particularly the euro, Canadian dollar and European currencies other than the euro, including the British pound sterling. Fluctuations in exchange rates could significantly affect our reported results from period to period as we translate results in local currencies into US dollars. We have not used derivative instruments to hedge the translation risk related to earnings of foreign subsidiaries.

Additionally, our investments in EMEA, Canada and LATAM are subject to foreign currency risk. Currency fluctuations result in non-cash gains and losses that do not impact income before income taxes, but instead are recorded as accumulated other comprehensive loss in equity in our consolidated balance sheet. We do not hedge our investments in non-US entities because these investments are viewed as long-term in nature.

The majority of our foreign currency risk arising on cash, accounts receivable, accounts payable and loan balances denominated in currencies other than those in which we record the financial results for a business operation relates to exposures to the US dollar, euro or British pound sterling. The approximate change in our income before income taxes resulting from a 10% increase in the value of the US dollar, euro and British pound sterling with all other variables held constant, including the impact of foreign currency derivative instruments and based on our balances as of December 31, 2022, would be an increase of $2.1 million, an increase of $2.2 million and a decrease of $1.2 million, respectively.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

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Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of Univar Solutions Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Univar Solutions Inc. (the Company) as of December 31, 2022 and 2021, the related consolidated statements of operations, comprehensive income, changes in stockholders' equity and cash flows for each of the three years in the period ended December 31, 2022, and the related notes and financial statement schedule listed in the Index at Item 15(a) (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2022 and 2021, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2022, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2022, based on criteria established in Internal Control — Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework), and our report dated February 22, 2023 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective, or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Environmental Liabilities

Description of the Matter

At December 31, 2022, the Company's environmental liability balance was $90.9 million. As discussed in Note 18 of the consolidated financial statements, the Company is subject to various federal, state and local environmental laws and regulations that require environmental assessment or remediation efforts (collectively "environmental remediation work") at approximately 128 locations. In determining the appropriate level of environmental liabilities, the Company considers several factors such as information obtained from investigatory studies, scope of the remediation and related changes, the interpretation, application and enforcement of laws and regulations, changes in the costs of remediation programs, the development of alternative cleanup technologies and methods, and the relative level of the Company's involvement at various sites for which the Company is allegedly associated.

Auditing management's accrual for environmental liabilities was especially challenging because it involves judgmental underlying assumptions, including remediation methods, remediation time horizon and remediation cost estimates. These assumptions have a significant effect on the accrual for environmental liabilities.

How We Addressed the Matter in our Audit

We tested management's controls that address the risks of material misstatement relating to the measurement and valuation of the environmental liabilities. For example, we tested controls over management's review of the environmental liability calculations and the significant assumptions and data inputs used by management.

To test the accrual for environmental liabilities, we involved our specialist to assist us in evaluating the reasonableness of the Company's calculation and underlying assumptions. We performed audit procedures that included, among others, assessing key methodologies and testing the significant assumptions and the underlying data used by management. For example, we tested the site's current remediation status and remediation strategy, which included an analysis of the site's remediation timeline, regulatory requirements, remediation actions and related technologies.

/s/ Ernst & Young LLP
We have served as the Company's auditor since 2010.
Chicago, Illinois
February 22, 2023

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of Univar Solutions Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Univar Solutions Inc.'s internal control over financial reporting as of December 31, 2022, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Univar Solutions Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2022, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2022 and 2021, the related consolidated statements of operations, comprehensive income, changes in stockholders' equity and cash flows for each of the three years in the period ended December 31, 2022, and the related notes and financial statement schedule listed in the Index at Item 15(a) and our report dated February 22, 2023 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on Internal Control over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP
Chicago, Illinois
February 22, 2023

Univar Solutions Inc.
Consolidated Statements of Operations

(in millions, except per share data)	Year ended December 31, 2022	2021	2020
Net sales	$ 11,475.3	$ 9,535.5	$ 8,265.0
Cost of goods sold (exclusive of depreciation)	8,704.1	7,142.3	6,262.8
Operating expenses:			
Outbound freight and handling	476.8	403.7	344.4
Warehousing, selling and administrative	1,248.5	1,191.8	1,022.3
Other operating expenses, net	32.8	107.5	89.4
Depreciation	131.7	150.9	162.9
Amortization	48.2	52.5	60.0
Impairment charges	0.6	3.0	40.2
Total operating expenses	1,938.6	1,909.4	1,719.2
Operating income	832.6	483.8	283.0
Other (expense) income:			
Interest income	4.3	4.3	2.1
Interest expense	(107.2)	(101.5)	(114.5)
Gain (loss) on sale of business	—	88.2	(50.6)
Other income (expense), net	26.5	110.4	(61.0)
Total other (expense) income	(76.4)	101.4	(224.0)
Income before income taxes	756.2	585.2	59.0
Income tax expense	210.9	124.6	6.1
Net income	$ 545.3	$ 460.6	$ 52.9
Income per common share:			
Basic	$ 3.29	$ 2.71	$ 0.31
Diluted	$ 3.26	$ 2.69	$ 0.31
Weighted average common shares outstanding:			
Basic	165.8	170.2	169.0
Diluted	167.4	171.4	169.8

The accompanying notes are an integral part of these consolidated financial statements.

Univar Solutions Inc.
Consolidated Statements of Comprehensive Income

(in millions)	Year ended December 31, 2022	2021	2020
Net income	$ 545.3	$ 460.6	$ 52.9
Other comprehensive income (loss), net of tax:			
Foreign currency translation, net of tax of $0.1, $— and $(4.7) for the years ended December 31, 2022, 2021 and 2020, respectively	(64.4)	5.0	(10.7)
Pension and other postretirement benefits adjustment, net of tax of $0.1, $0.6 and $(4.7) for the years ended December 31, 2022, 2021 and 2020, respectively	(0.6)	(2.5)	20.2
Derivative financial instruments, net of tax of $(28.5), $(7.8) and $7.9 for the years ended December 31, 2022, 2021 and 2020, respectively	83.0	21.9	(17.3)
Total other comprehensive income (loss), net of tax	18.0	24.4	(7.8)
Comprehensive income	$ 563.3	$ 485.0	$ 45.1

The accompanying notes are an integral part of these consolidated financial statements.

Univar Solutions Inc.
Consolidated Balance Sheets

(in millions, except share and per share data)	December 31, 2022	December 31, 2021
Assets		
Current assets:		
Cash and cash equivalents	$ 385.3	$ 251.5
Trade accounts receivable, net of allowance for doubtful accounts of $13.1 and $15.8 at December 31, 2022 and 2021, respectively.	1,489.9	1,539.5
Inventories	1,137.8	932.2
Prepaid expenses and other current assets	217.8	169.1
Total current assets	3,230.8	2,892.3
Property, plant and equipment, net	1,055.0	1,031.0
Goodwill	2,288.2	2,310.4
Intangible assets, net	167.0	211.7
Deferred tax assets	20.7	29.4
Other assets	384.0	303.0
Total assets	$ 7,145.7	$ 6,777.8
Liabilities and stockholders' equity		
Current liabilities:		
Trade accounts payable	$ 982.5	$ 1,009.3
Current portion of long-term debt	38.9	41.5
Accrued compensation	204.7	196.4
Other accrued expenses	401.3	420.4
Total current liabilities	1,627.4	1,667.6
Long-term debt	2,426.9	2,223.5
Pension and other postretirement benefit liabilities	135.2	211.7
Deferred tax liabilities	106.2	56.1
Other long-term liabilities	355.8	326.4
Total liabilities	4,651.5	4,485.3
Commitments and contingencies		
Stockholders' equity:		
Preferred stock, $0.01 par value, 200,000,000 shares authorized, none issued	—	—
Common stock, $0.01 par value, 2,000,000,000 shares authorized, 173,237,533 and 171,199,938 shares issued at December 31, 2022 and 2021, respectively	1.7	1.7
Additional paid-in capital	3,046.0	3,048.5
Treasury stock at cost, 15,254,566 and 1,832,385 shares at December 31, 2022 and 2021, respectively	(409.1)	(50.0)
Retained earnings (accumulated deficit)	200.3	(345.0)
Accumulated other comprehensive loss	(344.7)	(362.7)
Total stockholders' equity	2,494.2	2,292.5
Total liabilities and stockholders' equity	$ 7,145.7	$ 6,777.8

The accompanying notes are an integral part of these consolidated financial statements.

Univar Solutions Inc.
Consolidated Statements of Changes in Stockholders' Equity

(in millions)	Common stock outstanding (shares)	Common stock	Additional paid-in capital	Treasury stock	Retained earnings (accumulated deficit)	Accumulated other comprehensive loss	Total
Balance as of January 1, 2020	168.7	$ 1.7	$ 2,968.9	$ —	$ (858.5)	$ (379.3)	$ 1,732.8
Net income	—	—	—	—	52.9	—	52.9
Other comprehensive loss, net of tax	—	—	—	—	—	(7.8)	(7.8)
Restricted stock units vested, net of tax withholdings	0.4	—	(2.9)	—	—	—	(2.9)
Stock option exercises	0.1	—	1.1	—	—	—	1.1
Employee stock purchase plan	0.1	—	1.5	—	—	—	1.5
Stock-based compensation expense	—	—	14.5	—	—	—	14.5
Other	—	—	0.2	—	—	—	0.2
Balance as of December 31, 2020	169.3	$ 1.7	$ 2,983.3	$ —	$ (805.6)	$ (387.1)	$ 1,792.3
Net income	—	—	—	—	460.6	—	460.6
Other comprehensive income, net of tax	—	—	—	—	—	24.4	24.4
Exercise of warrants	1.0	—	26.8	—	—	—	26.8
Restricted stock units vested, net of tax withholdings	0.3	—	(2.6)	—	—	—	(2.6)
Stock option exercises	0.6	—	13.4	—	—	—	13.4
Employee stock purchase plan	—	—	1.5	—	—	—	1.5
Stock-based compensation expense	—	—	25.4	—	—	—	25.4
Purchase of treasury stock	(1.8)	—	—	(50.0)	—	—	(50.0)
Other	—	—	0.7	—	—	—	0.7
Balance as of December 31, 2021	169.4	$ 1.7	$ 3,048.5	$ (50.0)	$ (345.0)	$ (362.7)	$ 2,292.5
Net income	—	—	—	—	545.3	—	545.3
Other comprehensive income, net of tax	—	—	—	—	—	18.0	18.0
Restricted stock units vested, net of tax withholdings	0.9	—	(11.7)	—	—	—	(11.7)
Stock option exercises	1.0	—	24.1	—	—	—	24.1
Employee stock purchase plan	0.1	—	1.6	—	—	—	1.6
Stock-based compensation expense	—	—	34.3	—	—	—	34.3
Purchase of treasury stock	(13.4)	—	(50.0)	(359.1)	—	—	(409.1)
Other	—	—	(0.8)	—	—	—	(0.8)
Balance as of December 31, 2022	158.0	$ 1.7	$ 3,046.0	$ (409.1)	$ 200.3	$ (344.7)	$ 2,494.2

The accompanying notes are an integral part of these consolidated financial statements.

Univar Solutions Inc.
Consolidated Statements of Cash Flows

(in millions)	Year ended December 31, 2022	2021	2020
Operating activities:			
Net income	$ 545.3	$ 460.6	$ 52.9
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation and amortization	179.9	203.4	222.9
Impairment charges	0.6	3.0	40.2
Amortization of deferred financing fees and debt discount	5.7	6.2	6.5
(Gain) loss on sale of business	—	(88.2)	50.6
Gain on sale of property, plant and equipment	(2.8)	(10.1)	(23.7)
Pension mark to market (gain) loss	(17.7)	(75.9)	52.8
Deferred income taxes	29.0	6.4	(32.4)
Stock-based compensation expense	34.3	25.4	14.5
Fair value adjustment for warrants	—	(33.8)	0.8
Other	0.8	3.4	5.4
Changes in operating assets and liabilities:			
Trade accounts receivable, net	22.9	(328.6)	(66.0)
Inventories	(220.5)	(270.2)	126.0
Prepaid expenses and other current assets	(12.7)	(35.0)	1.2
Trade accounts payable	(6.8)	255.1	(139.3)
Other, net	(11.6)	168.6	(85.5)
Net cash provided by operating activities	546.4	290.3	226.9
Investing activities:			
Purchases of property, plant and equipment	(153.8)	(110.9)	(111.3)
Purchases of businesses, net of cash acquired	(16.7)	(28.7)	(4.6)
Proceeds from sale of property, plant and equipment	7.8	29.0	46.5
Proceeds from sale of business	—	136.5	37.3
Other	1.0	(2.3)	(9.2)
Net cash (used) provided by investing activities	(161.7)	23.6	(41.3)
Financing activities:			
Proceeds from the issuance of long-term debt, net	199.6	995.0	—
Payments on long-term debt and finance lease obligations	(87.7)	(1,440.5)	(205.3)
Proceeds under revolving credit facilities	2,238.7	2,732.4	2,150.0
Payments under revolving credit facilities	(2,186.9)	(2,700.0)	(2,084.5)
Debt issuance costs	(2.1)	(1.0)	—
Taxes paid related to net share settlements of stock-based compensation awards	(7.8)	(2.6)	(2.9)
Purchases of treasury stock	(409.1)	(50.0)	—
Stock option exercises	23.9	13.4	1.1
Proceeds from the exercise of warrants	—	27.1	—
Other	(1.3)	1.6	1.6
Net cash used by financing activities	(232.7)	(424.6)	(140.0)
Effect of exchange rate changes on cash and cash equivalents	(18.2)	(24.4)	10.7
Net increase (decrease) in cash and cash equivalents	133.8	(135.1)	56.3
Cash and cash equivalents at beginning of period	251.5	386.6	330.3
Cash and cash equivalents at end of period	$ 385.3	$ 251.5	$ 386.6
Supplemental disclosure of cash flow information:			
Cash paid during the period for:			
Income taxes	$ 189.7	$ 98.0	$ 51.3
Interest, net of capitalized interest	88.7	88.0	104.7
Non-cash activities:			
Other liabilities related to the purchase of businesses	$ 1.4	$ 23.0	$ —
Additions of property, plant and equipment included in trade accounts payable and other accrued expenses	8.5	14.8	5.5
Additions of property, plant and equipment under a finance lease obligation	34.5	31.8	61.4
Additions of assets under an operating lease obligation	96.0	63.1	62.1

The accompanying notes are an integral part of these consolidated financial statements.

Univar Solutions Inc.
Notes to Consolidated Financial Statements

1. Nature of operations

Headquartered in Downers Grove, Illinois, Univar Solutions Inc. (the "Company") is a leading global distributor of commodity and specialty chemicals and ingredients, as well as a provider of value-added services, to customers across a wide range of industries. The Company's four reportable segments, defined as USA, EMEA, Canada and LATAM, represent the geographic areas under which the Company operates and manages its business. LATAM includes certain developing businesses in Latin America and the Asia-Pacific region.

2. Significant accounting policies

Basis of consolidation and presentation

The consolidated financial statements include the financial statements of the Company and its majority-owned subsidiaries. Subsidiaries are consolidated if the Company has a controlling financial interest, which may exist based on ownership of a majority of the voting interest, or based on the Company's determination that it is the primary beneficiary of a variable interest entity. All intercompany balances and transactions are eliminated in consolidation. Unless otherwise indicated, all financial data presented in these consolidated financial statements are expressed in US dollars.

Proceeds and payments under the revolving credit facilities for the years ended December 31, 2021 and 2020, which were previously reported net in the consolidated statements of cash flows, are now presented separately to conform to the current year presentation. Additionally, certain other immaterial amounts in the prior years' consolidated financial statements and notes have been reclassified to conform to the current year presentation.

Use of estimates

The preparation of consolidated financial statements in conformity with US GAAP requires management to make estimates and assumptions affecting the amounts reported and disclosed in the financial statements and accompanying notes. Actual results could differ materially from these estimates.

Cash and cash equivalents

Cash and cash equivalents include highly-liquid investments with an original maturity of three months or less that are readily convertible into known amounts of cash.

Certain of the Company's subsidiaries participate in a multi-currency, notional cash pooling arrangement with a third-party bank provider to manage global liquidity requirements (the "Notional Cash Pool"). Under the Notional Cash Pool, cash deposited by participating subsidiaries is pledged as security against the overdraft balances of other participating subsidiaries, providing legal rights of offset. As a result, the balances are presented on a net basis within cash and cash equivalents in the consolidated balance sheets. As of December 31, 2022, the net cash position of the Notional Cash Pool was $52.9 million, which consisted of a gross cash balance of $69.8 million less a bank overdraft balance of $16.9 million. As of December 31, 2021, the net cash position of the Notional Cash Pool was $43.2 million, which consisted of a gross cash balance of $146.0 million less a bank overdraft balance of $102.8 million.

Trade accounts receivable, net and allowance for doubtful accounts

Trade accounts receivable are stated at the invoiced amount, net of an allowance for doubtful accounts. The allowance for doubtful accounts reflects the Company's current estimate of credit losses expected to be incurred over the life of the trade accounts receivable. Collectability of the trade accounts receivable balance is assessed on an ongoing basis and determined based on the delinquency of customer accounts, the financial condition of individual customers, past collections experience and future economic expectations.

Changes in the allowance for doubtful accounts were as follows:

(in millions)	2022	2021	2020
Balance as of January 1	$ 15.8	$ 17.2	$ 12.9
Provision for credit losses	5.1	6.9	9.3
Write-offs	(7.7)	(8.1)	(4.4)
Recoveries	—	—	0.1
Dispositions	—	—	(0.5)
Foreign exchange	(0.1)	(0.2)	(0.2)
Balance as of December 31	$ 13.1	$ 15.8	$ 17.2

Inventories

Inventories consist primarily of products purchased for resale and are stated at the lower of cost or net realizable value. Inventory cost is determined based on the weighted average cost method and includes purchase price from producers net of rebates received, inbound freight and handling, and direct labor and other costs incurred to blend and repackage product, but excludes depreciation expense.

Property, plant and equipment, net

Property, plant and equipment are carried at historical cost, net of accumulated depreciation.Depreciation is recorded on a straight-line basis over the estimated useful life of each asset as follows:

Buildings	10-50 years
Main components of tank farms	5-40 years
Containers	2-15 years
Machinery and equipment	5-20 years
Furniture, fixtures and others	5-20 years
Information technology	3-10 years

The Company evaluates the useful life and carrying value of property, plant and equipment for impairment if an event occurs or circumstances change that would indicate the carrying value may not be recoverable. If the carrying amount of the asset group is not recoverable on an undiscounted cash flow basis, an impairment loss is recognized to the extent the asset group's carrying amount exceeds its estimated fair value.

Goodwill and intangible assets

Goodwill represents the excess of the aggregate purchase price over the fair value of the net assets acquired in business combinations. Goodwill is tested for impairment annually as of October 1, or between annual tests if an event occurs or circumstances change that would more likely than not reduce the fair value of a reporting unit below its carrying amount. The Company’s reporting units are USA, EMEA, Canada, Latin America and Asia-Pacific.

For each of the reporting units, the Company has the option to perform either the qualitative or quantitative test. In the event a reporting unit fails the qualitative test, it is required to perform the quantitative test. The quantitative impairment test considers both the income approach and the market approach to estimate a reporting unit’s fair value. Significant assumptions include forecasted earnings before interest, taxes, depreciation and amortization ("EBITDA"), market segment growth rates and discount rates based on a reporting unit's weighted average cost of capital.

If the fair value of a reporting unit is less than its carrying value, the Company will recognize an impairment for the lesser of the amount by which the reporting unit's carrying amount exceeds its fair value or the reporting unit’s goodwill carrying value. As of October 1, 2022, the Company performed its annual impairment assessment using the qualitative test and concluded that fair value exceeded carrying value for all reporting units.

The Company's intangible assets have finite lives and are amortized over their respective useful lives of2 to 20 years using the estimated pattern of economic benefit. Intangible assets are tested for impairment if an event occurs or circumstances change that indicates the carrying value may not be recoverable.

Short-term financing

Short-term financing includes bank overdrafts and short-term lines of credit.

Income taxes

Deferred tax assets and liabilities are determined based on differences between financial reporting and tax bases of assets and liabilities and are measured using enacted tax rates and laws that are expected to be in effect when the differences reverse. Deferred tax assets are also recognized for the estimated future effects of tax loss carryforwards. The effect on deferred taxes of changes in tax rates is recognized in the period in which the revised tax rate is enacted. Income tax effects are released from accumulated other comprehensive loss in the period the underlying item expires.

The Company records valuation allowances to reduce deferred tax assets to the extent it believes it is more likely than not that such assets will not be realized. In making such determinations, the Company considers all available positive and negative evidence, including scheduled reversals of deferred tax liabilities, forecasted and appropriate character of future taxable income, its experience with operating loss and tax credit carryforwards not expiring unused, tax planning strategies and the ability to carry back losses to prior years.

The Company is subject to global intangible low-taxed income ("GILTI"), which is a tax on foreign income in excess of a deemed return on tangible assets of foreign corporations. The Company treats taxes due on future US inclusions in taxable income related to GILTI as a current-period expense when incurred.

The Company accounts for unrecognized tax benefits through an assessment of whether a position is more likely than not to be sustained upon examination by taxing authorities based on its technical merits. The Company records liabilities for uncertain tax positions taken or expected to be taken in a tax return. The Company recognizes interest and penalties related to unrecognized tax benefits within interest expense and warehousing, selling and administrative expenses, respectively, in the consolidated statements of operations. Accrued interest and penalties are included in other accrued expenses and other long-term liabilities in the consolidated balance sheets.

On August 16, 2022, the Inflation Reduction Act ("IRA") was enacted into US law. Effective for tax years beginning after December 31, 2022, the IRA imposes a 15% corporate minimum tax, a 1% excise tax on share repurchases, and creates and extends certain tax-related energy incentives. Management does not expect the tax-related provisions of the IRA to have a material impact on the Company's consolidated financial statements.

Defined benefit pension plans

The Company sponsors several defined benefit pension plans and recognizes actuarial gains or losses, known as "mark to market" adjustments, as of the December 31 measurement date. The mark to market adjustments primarily include gains and losses resulting from changes in discount rates and the difference between the expected and actual rate of return on plan assets. Settlement gains and losses are recognized in the period in which the settlement occurs. The fair value of plan assets is used to calculate the expected return on plan assets component of net periodic benefit cost.

Leases

At the commencement date of a lease, the Company recognizes a liability to make lease payments and an asset representing the right to use the underlying asset during the lease term. The Company includes options to extend or terminate a lease within the lease term when it is reasonably certain the option will be exercised. The lease liability is measured at the present value of fixed lease payments over the lease term. The lease liability includes payments allocated to lease components, while payments allocated to non-lease components are expensed as incurred for all asset classes. When a contract excludes an implicit rate, the Company utilizes an incremental borrowing rate based on information available at the lease commencement date, including lease term and geographic region. The initial measurement of the right-of-use asset includes the initial measurement of the lease liability, fixed lease payments made in advance of the lease commencement date and initial direct costs incurred by the Company and excludes lease incentives. Variable lease payments, such as payments based on an index rate or usage, are expensed as incurred and excluded from lease liabilities and right-of-use assets.

Leases with an initial term of 12 months or less are classified as short-term leases and are not recorded on the consolidated balance sheets. The lease expense for short-term leases is recognized on a straight-line basis over the lease term.

Legal costs

Legal costs are expensed as incurred.

Environmental liabilities

Environmental liabilities are recognized for probable and reasonably estimable losses associated with environmental remediation. Incremental direct costs of the investigation, remediation effort and post-remediation monitoring are included in the estimated environmental liabilities. The Company discounts environmental liabilities if the liability and the respective payment streams are fixed or reliably determinable. Expected cash outflows related to environmental remediation for the next 12 months and amounts for which the timing is uncertain are reported as current within other accrued expenses in the consolidated balance sheets. The long-term portion of environmental liabilities is reported within other long-term liabilities in

the consolidated balance sheets. Environmental liabilities are not reduced for potential insurance recoveries. Insurance recoveries are recorded when it is probable that a recovery will be realized and are included within prepaid expenses and other current assets or other assets in the consolidated balance sheets. Environmental remediation expenses and insurance recoveries are included within warehousing, selling and administrative expenses in the consolidated statements of operations, unless associated with disposed operations, in which case such expenses are included in other operating expenses, net.

Revenue recognition

Revenue is measured as the amount of consideration the Company expects to receive in exchange for transferring a good or providing a service. Since the term between invoicing and payment is less than a year, revenue is not adjusted for the effects of a significant financing component. Revenue for bill-and-hold arrangements is recognized if the Company has a substantive customer request, the materials are properly segregated and designated as belonging to the customer, materials are ready to be transferred to the customer and the Company is unable to direct the materials to another customer.

Chemical distribution

Revenue is recognized when performance obligations under the terms of the contract are satisfied, which generally occurs at a point in time upon completion of the shipping process as indicated by the terms of the contract. Net sales include product sales and billings for freight and handling charges, net of discounts, expected returns, customer price and volume incentives and sales or other revenue-based taxes. The Company estimates customer price and volume incentives and expected customer returns based on historical experience.

Services

The Company recognizes revenue from services as they are performed and economic value is transferred to customers. Services provided to customers are primarily related to waste management and warehousing.

Crop sciences

The Company recognized revenue when control of products transferred to customers. The amount of consideration recorded varied due to price movements and rights granted to customers to return product. Customer payment terms often extended through a growing season, which was up to six months.

Beginning in 2021, crop sciences is no longer a revenue stream due to actions taken by the Company within the Canada segment in 2020.

Foreign currency

Assets and liabilities of foreign subsidiaries are translated into US dollars at period-end exchange rates. Income and expense accounts of foreign subsidiaries are translated into US dollars at the average exchange rates for the period. The net exchange gains and losses arising from translation are reflected as a component of foreign currency translation within accumulated other comprehensive income (loss) ("AOCI").

Transaction gains and losses are recognized in other income (expense), net in the consolidated statements of operations. Transaction gains and losses relating to intercompany borrowings that are an investment in a foreign subsidiary are reflected as a component of foreign currency translation within AOCI.

Stock-based compensation

The Company measures the total amount of employee stock-based compensation expense based on the grant date fair value of each award. Expense is recognized for each separately vesting tranche on a straight-line basis over the requisite service period, which is the shorter of the service period of the award or the period until the employees' retirement eligibility date. The Company recognizes forfeitures when incurred.

Fair value

Certain assets and liabilities are required to be recorded at fair value. The estimated fair values of those assets and liabilities have been determined using market information and valuation methodologies. Observable inputs reflect market data obtained from independent sources, while unobservable inputs reflect the Company's market assumptions. A financial instrument's categorization within the fair value hierarchy is based on the lowest level of input that is significant to the fair value measurement. There are three levels of inputs that may be used to measure fair value and a net asset value ("NAV") practical expedient:

Level 1	Quoted prices for *identical* instruments in active markets.
Level 2	Quoted prices for *similar* instruments in active markets; quoted prices for identical or similar instruments in markets that are not active; and model-derived valuation in which all significant inputs and significant value drivers are observable in active markets.
Level 3	Valuations derived from valuation techniques in which one or more significant inputs or significant value drivers are *unobservable*.
NAV	As a practical expedient, certain investments are measured based on the NAV per share (or equivalent) due to the absence of a readily available market price, and are excluded from the fair value hierarchy.

Derivatives

The Company uses derivative financial instruments to manage risks associated with foreign currency and interest rate fluctuations. The Company does not use derivative instruments for speculative trading purposes. The fair value of forward currency contracts is calculated by reference to current forward exchange rates for contracts with similar maturity profiles. The fair value of interest rate and cross currency swaps is determined by estimating the net present value of amounts to be paid under the agreement offset by the net present value of the expected cash inflows based on market rates and associated yield curves. Based on these valuation methodologies, these derivative contracts are classified as Level 2 in the fair value hierarchy. Changes in the fair value of derivative financial instruments are recognized in the consolidated statements of operations within interest expense or other income (expense), net. Cash flows associated with derivative financial instruments are recognized in the operating activities section of the consolidated statements of cash flows.

For derivatives designated as cash flow hedges, changes in the fair value of the derivative are recorded to AOCI and are reclassified to earnings when the underlying forecasted transaction affects earnings. For contracts designated as cash flow hedges, the Company reassesses the probability of the underlying forecasted transactions occurring on a quarterly basis. For derivatives not designated as hedging instruments, all changes in fair value are recorded to earnings in the current period.

Earnings per share

Basic earnings per share is based on the weighted average number of common shares outstanding during each period. Diluted earnings per share is based on the weighted average number of common shares and dilutive common share equivalents outstanding during each period. The Company reflects common share equivalents relating to stock options, non-vested restricted stock and non-vested restricted stock units in its computation of diluted weighted average shares outstanding, unless the effect of inclusion is anti-dilutive. The effect of dilutive securities is calculated using the treasury stock method.

Recently adopted accounting pronouncements

In November 2021, the Financial Accounting Standards Board ("FASB") issued Accounting Standards Update ("ASU") 2021-10 "Government Assistance" (Topic 832) - "Disclosures by Business Entities about Government Assistance" to increase the transparency of disclosures for government assistance and grants. The ASU requires annual disclosures pertaining to the types of received government assistance, accounting for the transactions and the related impacts on the reported financial results. The Company adopted this guidance on a prospective basis in the fourth quarter of 2022, which did not impact the consolidated financial statements and disclosures as the Company has not received significant government assistance.

In December 2022, the FASB issued ASU 2022-06 "Reference Rate Reform" (Topic 848) "Deferral of the Sunset Date of Topic 848", which deferred the sunset date of Topic 848 from December 31, 2022 to December 31, 2024. Topic 848 provides optional practical expedients and elections when accounting for contracts, hedging relationships and other transactions affected by the discontinuation of reference rates such as LIBOR, if certain criteria are met. The standard was effective upon issuance and the Company may apply the optional practical expedients and elections in Topic 848 prospectively through December 31, 2024. The Company adopted the standard in the fourth quarter of 2022, which did not impact the consolidated financial statements.

Accounting pronouncements issued and not yet adopted

In October 2021, the FASB issued ASU 2021-08 "Business Combinations" (Topic 805) – "Accounting for Contract Assets and Contract Liabilities from Contracts with Customers." This ASU requires an entity to recognize and measure contract assets and contract liabilities acquired in a business combination in accordance with Topic 606 (Revenue from Contracts with Customers). The standard is effective for public business entities for fiscal years beginning after December 15, 2022, including interim periods within those fiscal years. The Company will adopt this guidance on a prospective basis effective January 1, 2023, which is not expected to have a material impact on the consolidated financial statements.

In September 2022, the FASB issued ASU 2022-04 "Liabilities - Supplier Finance Programs" (Subtopic 405-50) - "Disclosure of Supplier Finance Program Obligations" to enhance the transparency of disclosure regarding supplier finance programs. The ASU requires a buyer in a supplier finance program to disclose information about the program's nature, activity during the period, changes from period to period and potential magnitude. The standard is effective for fiscal years beginning after December 15, 2022, including interim periods within those fiscal years, except for the disclosure of rollforward information, which is effective for fiscal years beginning after December 15, 2023. Retrospective application of the guidance is required for all disclosures except rollforward information, which requires prospective application. The Company has not yet determined the impact of the standard on its disclosures.

3. Business combinations

The Company makes acquisitions of certain businesses from time to time that are aligned with its strategic intent with respect to, among other factors, growth markets and adjacent product lines or technologies. Goodwill resulting from business combinations is largely attributable to the existing workforce of the acquired businesses and synergies expected to arise after the Company's acquisition of these businesses.

Vicom Distribución Productos Quimicos, S.L.

On July 29, 2022, the Company acquired all of the outstanding equity interests in Vicom, a leading regional specialty chemical distributor in Spain and Portugal, for cash consideration of $14.5 million. The transaction was accounted for under the acquisition method of accounting and the results of operations are included in the EMEA segment. The preliminary purchase price allocation included $3.4 million in identified intangible assets related to customer relationships and $5.6 million in goodwill, which is included in the EMEA segment and is not deductible for income tax purposes.

Sweetmix Distribuidora de Materias Primas Industriais Ltda

On December 1, 2021, the Company acquired all of the outstanding equity interests in Sweetmix, a food ingredients and CASE specialty chemical distribution company in Brazil. The purchase price, including measurement period adjustments, was $53.0 million, $28.7 million of which was paid in cash (net of cash acquired of $1.2 million) upon closing, with the remaining $23.1 million to be paid over the next five years. The transaction was accounted for under the acquisition method of accounting and the results of operations are included in the LATAM segment.

During the year ended December 31, 2022, the Company updated the purchase price allocation to reflect intangible asset fair value adjustments, purchase price adjustments and the deferred tax impacts of the recognized adjustments. The final purchase price allocation and measurement period adjustments are presented in the table below:

(in millions)	Initial Purchase Price Allocation	Measurement Period Adjustments	Final Purchase Price Allocation
Cash and cash equivalents	$ 1.2	$ —	$ 1.2
Trade accounts receivable, net	15.6	—	15.6
Inventories	8.5	—	8.5
Prepaid expenses and other current assets	2.6	—	2.6
Goodwill	33.8	(0.8)	33.0
Intangible assets, net	13.3	1.7	15.0
Trade accounts payable	(16.6)	—	(16.6)
Deferred tax liabilities	(4.5)	(0.8)	(5.3)
Other assets and liabilities, net	(1.0)	—	(1.0)
Purchase consideration	52.9	0.1	53.0
Less: Cash and cash equivalents	1.2	—	1.2
Purchase consideration, net of cash	$ 51.7	$ 0.1	$ 51.8

The goodwill is included in the LATAM segment and is not deductible for income tax purposes. The identified intangible assets relate to customer relationships that will be amortized over a period of eight years.

Zhuhai Techi Chem Silicone Industry Corporation

On December 18, 2020, the Company completed the acquisition of the specialty silicone solutions business of Techi Chem, a leading distributor of specialty silicone solutions used primarily for the CASE market within the China marketplace. The purchase price of $6.8 million was comprised of $4.6 million paid in cash and $2.2 million of contingent consideration. The transaction was accounted for under the acquisition method of accounting and the results of operations are included in the LATAM segment.

4. Dispositions

The dispositions below did not meet the criteria to be classified as a discontinued operation in the Company's consolidated financial statements since the dispositions did not represent a strategic shift that had, or will have, a major effect on the Company's operations and financial results.

Distrupol Business

On April 1, 2021, the Company completed the sale of its Distrupol business within the EMEA segment for total cash proceeds of $136.7 million. In 2021, the Company recorded an $87.6 million pre-tax gain on sale of business in the consolidated statements of operations, net of a release of cumulative foreign currency translation losses of $18.1 million. The sale of the Distrupol business was exempt from tax under local country subsidiary participation exemptions. The impact of the sale on US income taxes was minimal. The Company recorded income before income taxes attributable to the Distrupol business of $3.9 million and $10.3 million for the years ended December 31, 2021 and 2020, respectively.

Canadian Agriculture Services Business

On November 30, 2020, the Company completed the sale of its Canadian Agriculture services business within the Canada segment for total net cash proceeds of $39.3 million after closing transaction-related expenses. In 2020, the Company recorded a $31.5 million pre-tax loss on sale of business in the consolidated statements of operations. In 2021, the Company recognized a favorable adjustment of $0.7 million, decreasing the loss on sale recorded in 2020. The Company recorded income before income taxes attributable to the Canadian Agriculture services business of $2.8 million for the year ended December 31, 2020.

Industrial Spill and Emergency Response Businesses

On September 1, 2020, the Company completed the sale of its industrial spill and emergency response businesses to EnviroServe Inc. for total net cash proceeds of $6.2 million after closing transaction-related expenses. In 2020, the Company recorded a $10.5 million pre-tax loss on sale of business in the consolidated statements of operations. The Company recorded a loss before income taxes attributable to these businesses of $26.9 million for the year ended December 31, 2020.

Environmental Sciences Business

On December 31, 2019, the Company completed the sale of the Environmental Sciences business to AEA Investors LP for total net cash proceeds of $174.0 million plus a $5.0 million ($2.4 million present value) subordinated note receivable. In 2020, the Company recorded a net working capital adjustment of $8.2 million, reducing the proceeds and the gain on sale initially recorded in 2019.

5. Impairment charges

The Company reviews long-lived assets for impairment whenever events or changes in circumstances indicate an asset's carrying amount may not be recoverable. Significant estimates include forecasted EBITDA, working capital, capital expenditures and discount rates. As the inputs for testing recoverability and determining fair value of the asset groups are largely based on management's judgments and are not generally observable in active markets, the Company considers such inputs to be Level 3 measurements in the fair value hierarchy.

Year ended December 31, 2021

Throughout 2021, the Company announced the closure of certain operating facilities in the USA and Canada segments, which resulted in impairment charges related to property, plant and equipment, net of $3.0 million within the consolidated statements of operations.

Year ended December 31, 2020

During 2020, the Company determined there was a more likely than not expectation that the industrial spill and emergency response businesses within the USA segment would be sold. The Company determined this to be an impairment triggering event, requiring the assessment of the recoverability of these long-lived asset groups. The Company tested the recoverability and determined the assets to be impaired. As a result, the Company recorded a non-cash, pre-tax impairment charge of

$15.5 million, consisting of $12.8 million of intangible assets, net and $2.7 million of property, plant and equipment, net, within the consolidated statements of operations.

During 2020, the Company decided to cease further investment in, and seek to exit a contract related to certain technology assets, consisting of capitalized software and hardware components. This event represented an impairment triggering event requiring an impairment analysis within the Other segment. As a result, the Company recorded a pre-tax impairment charge of $21.7 million, inclusive of non-cash and cash components of $19.7 million and $2.0 million, respectively, relating to property, plant and equipment, net, within the consolidated statements of operations.

Additionally, the Company recorded impairment charges of $3.0 million related to property, plant and equipment, net within the consolidated statements of operations during the year ended December 31, 2020. The impairment charges were in connection with the announced closure of certain operating facilities in the USA segment and the wind down of the Canadian Agriculture wholesale distribution business.

6. Goodwill and intangible assets, net

Goodwill

The following table is a summary of goodwill activity by segment:

(in millions)	USA	EMEA	Canada	LATAM	Total
Balance as of January 1, 2021	$ 1,805.0	$ 8.7	$ 428.1	$ 28.6	$ 2,270.4
Acquisitions [1]	—	—	—	33.8	33.8
Purchase price adjustments [2]	—	—	—	(0.7)	(0.7)
Dispositions and other adjustments	7.6	(1.1)	—	(2.4)	4.1
Foreign currency translation	—	(0.2)	3.3	(0.3)	2.8
Balance as of December 31, 2021	1,812.6	7.4	431.4	59.0	2,310.4
Acquisitions [1]	—	5.6	—	—	5.6
Purchase price adjustments [2]	—	—	—	(0.8)	(0.8)
Foreign currency translation	—	(0.4)	(28.9)	2.3	(27.0)
Balance as of December 31, 2022	$ 1,812.6	$ 12.6	$ 402.5	$ 60.5	$ 2,288.2

(1) Acquisitions in 2021 and 2022 related to Sweetmix and Vicom, respectively.
(2) Purchase price adjustments in 2021 and 2022 related to the Techi Chem and Sweetmix acquisitions, respectively.

Accumulated goodwill impairment losses by segment were as follows:

(in millions)	USA	EMEA	Canada	LATAM	Total
Balance as of January 1, 2021	$ —	$ 216.7	$ 1.4	$ 45.3	$ 263.4
Balance as of December 31, 2021	—	170.9	1.4	43.5	215.8
Balance as of December 31, 2022	—	158.5	1.3	45.3	205.1

The year-over-year changes primarily relate to dispositions and foreign currency translation adjustments.

Intangible assets, net

The gross carrying amounts and accumulated amortization of the Company's intangible assets were as follows:

	December 31, 2022			December 31, 2021		
(in millions)	Gross	Accumulated amortization	Net	Gross	Accumulated amortization	Net
Customer relationships	$ 929.6	$ (765.2)	$ 164.4	$ 940.1	$ (732.8)	$ 207.3
Other	167.0	(164.4)	2.6	168.9	(164.5)	4.4
Total intangible assets	$ 1,096.6	$ (929.6)	$ 167.0	$ 1,109.0	$ (897.3)	$ 211.7

Other intangible assets consist of intellectual property trademarks, trade names, producer relationships and contracts, non-compete agreements and exclusive distribution rights.

The estimated annual amortization expense in each of the next five years is as follows:

(in millions)	
2023	$ 42.4
2024	33.0
2025	29.1
2026	24.0
2027	19.1

7. Revenue

The Company disaggregates revenues from contracts with customers by both geographic reportable segments and revenue contract types. Geographic reportable segmentation is pertinent to understanding the Company's revenues, as it aligns with how the Company reviews the financial performance of its operations. Revenue contract types are differentiated by the type of good or service the Company offers customers, since the contractual terms necessary for revenue recognition are unique to each of the identified revenue contract types.

The following table disaggregates external customer net sales by major category:

	Year ended December 31,		
(in millions)	2022	2021	2020
USA			
Chemical distribution	$ 7,247.3	$ 5,743.2	$ 4,698.1
Services	286.2	280.8	308.1
Total external customer net sales	$ 7,533.5	$ 6,024.0	$ 5,006.2
EMEA			
Chemical distribution	$ 2,064.6	$ 1,970.8	$ 1,695.8
Services	0.1	0.3	1.3
Total external customer net sales	$ 2,064.7	$ 1,971.1	$ 1,697.1
Canada			
Chemical distribution	$ 1,120.5	$ 919.2	$ 749.7
Crop sciences	—	—	315.1
Services	—	10.8	45.9
Total external customer net sales	$ 1,120.5	$ 930.0	$ 1,110.7
LATAM			
Chemical distribution	$ 739.5	$ 598.5	$ 441.5
Services	17.1	11.9	9.5
Total external customer net sales	$ 756.6	$ 610.4	$ 451.0
Consolidated			
Chemical distribution	$ 11,171.9	$ 9,231.7	$ 7,585.1
Crop sciences	—	—	315.1
Services	303.4	303.8	364.8
Total external customer net sales	$ 11,475.3	$ 9,535.5	$ 8,265.0

Deferred revenue

Deferred revenues are recognized as contract liabilities when customers provide the Company with consideration prior to the Company satisfying its performance obligations and are recognized in revenue when the performance obligations are met. Deferred revenues relate to revenues that are expected to be recognized within one year and are recorded within other accrued expenses in the consolidated balance sheets. Deferred revenues as of December 31, 2022 and 2021 were $15.3 million and $17.6 million, respectively.

Revenue recognized for the years ended December 31, 2022 and 2021 from amounts included in contract liabilities at the beginning of the respective periods was $16.8 million and $5.3 million, respectively.

8. Supplemental financial information

Other operating expenses, net

Other operating expenses, net consisted of the following:

	Year ended December 31,		
(in millions)	2022	2021	2020
Acquisition and integration related expenses	$ 1.7	$ 55.8	$ 62.4
Stock-based compensation expense	34.3	25.4	14.5
Restructuring charges	—	0.2	13.9
Other employee severance costs	—	8.7	14.8
Other facility closure costs	—	1.2	2.7
Multi-employer pension plan exit liability	—	31.2	—
Gain on sale of property, plant and equipment	(2.8)	(10.1)	(23.7)
Other	(0.4)	(4.9)	4.8
Total other operating expenses, net	$ 32.8	$ 107.5	$ 89.4

Other income (expense), net

Other income (expense), net consisted of the following:

	Year ended December 31,		
(in millions)	2022	2021	2020
Pension mark to market gain (loss)	$ 17.7	$ 75.9	$ (52.8)
Pension settlement gain	5.2	0.3	0.6
Non-operating retirement benefits	10.9	18.5	8.5
Foreign currency loss, net	(13.4)	(15.9)	(6.8)
Undesignated derivative instruments	6.9	5.8	(4.8)
Debt refinancing costs	—	(7.0)	(0.1)
Fair value adjustment for warrants	—	33.8	(0.8)
Other	(0.8)	(1.0)	(4.8)
Total other income (expense), net	$ 26.5	$ 110.4	$ (61.0)

Property, plant and equipment, net

Property, plant and equipment, net consisted of the following:

	December 31,	
(in millions)	2022	2021
Land and buildings	$ 823.2	$ 820.3
Tank farms	320.9	316.7
Machinery, equipment and other	1,055.9	1,015.3
Less accumulated depreciation	(1,248.9)	(1,207.8)
Subtotal	951.1	944.5
Work in progress	103.9	86.5
Property, plant and equipment, net	$ 1,055.0	$ 1,031.0

9. Restructuring charges

Restructuring charges relate to the implementation of several regional strategic initiatives aimed at streamlining the Company's cost structure and improving its operations. These actions primarily resulted in workforce reductions and other facility rationalization costs, with underlying plans substantially completed by December 31, 2020.

The following table summarizes activity related to restructuring liabilities:

(in millions)	Employee Termination Costs	Facility Exit Costs	Other Exit Costs	Total
Restructuring liabilities as of January 1, 2020	$ 3.7	$ 1.9	$ 0.2	$ 5.8
Restructuring charges	11.7	—	2.2	13.9
Cash paid	(12.4)	(0.5)	—	(12.9)
Non-cash and other	0.2	—	0.1	0.3
Restructuring liabilities as of December 31, 2020	3.2	1.4	2.5	7.1
Restructuring charges	0.2	—	—	0.2
Cash paid	(4.3)	—	(2.3)	(6.6)
Non-cash and other	1.2	(1.3)	(0.1)	(0.2)
Restructuring liabilities as of December 31, 2021	$ 0.3	$ 0.1	$ 0.1	$ 0.5

No restructuring charges were recorded during the year ended December 31, 2022. Restructuring liabilities of $0.1 million and $0.5 million were classified in other accrued expenses in the consolidated balance sheets as of December 31, 2022 and 2021, respectively.

10. Earnings per share

The following table presents the basic and diluted earnings per share computations:

	Year ended December 31,		
(in millions, except per share data)	2022	2021	2020
Net income	$ 545.3	$ 460.6	$ 52.9
Weighted average common shares outstanding			
Basic	165.8	170.2	169.0
Effect of dilutive securities: stock compensation plans	1.6	1.2	0.8
Diluted	167.4	171.4	169.8
Income per common share			
Basic	$ 3.29	$ 2.71	$ 0.31
Diluted	$ 3.26	$ 2.69	$ 0.31

The common shares that were not included in the computation of diluted earnings per share because their inclusion would be anti-dilutive were as follows:

	Year ended December 31,		
(in millions)	2022	2021	2020
Stock options	0.6	2.0	4.3
Restricted stock	—	—	0.1
Warrants	—	—	7.6

On November 1, 2022, the Company entered into an accelerated share repurchase agreement ("ASR") with Goldman Sachs & Co. LLC ("Goldman"), which was accounted for as an open market repurchase of its common stock on the trade date and a forward contract indexed to its common stock. Refer to "Note 14: Share repurchase program and stock-based compensation" for additional information. The Company reflected the initial share delivery as an immediate reduction in common shares outstanding. The effect of the forward contract was excluded from the computation of diluted earnings per share for the year ended December 31, 2022 as its inclusion would have been anti-dilutive.

11. Debt

Short-term financing

The Company had no outstanding balance in short-term financing facilities as of December 31, 2022 and 2021.

The Company had $127.7 million and $141.9 million of outstanding letters of credit as of December 31, 2022 and 2021, respectively.

Long-term debt

Long-term debt consisted of the following:

	December 31,	
(in millions)	**2022**	**2021**
Senior Term Loan Facilities:		
Term B-5 Loan due 2026, variable interest rate of6.38% and 2.10% at December 31, 2022 and 2021, respectively	$ 388.0	$ 392.0
Term B-6 Loan due 2028, variable interest rate of6.13% and 2.10% at December 31, 2022 and 2021, respectively	985.0	995.0
Asset Backed Loan ("ABL") Facilities:		
Senior ABL Credit Facility due 2027, variable interest rate of5.59% at December 31, 2022	353.0	—
Senior ABL Term Loan due 2027, variable interest rate of6.17% at December 31, 2022	200.0	—
North American ABL Facility due 2024, variable interest rate of1.43% at December 31, 2021	—	297.9
Senior Unsecured Notes:		
Senior Unsecured Notes due 2027, fixed interest rate of5.13% at December 31, 2022 and 2021	454.0	500.0
Finance lease obligations	104.3	101.9
Total long-term debt before unamortized debt issuance costs and discount	2,484.3	2,286.8
Less: unamortized debt issuance costs and discount on debt	(18.5)	(21.8)
Total long-term debt	2,465.8	2,265.0
Less: current maturities	(38.9)	(41.5)
Total long-term debt, excluding current maturities	$ 2,426.9	$ 2,223.5

The weighted average interest rate on long-term debt, including the effect of designated and undesignated derivative instruments (refer to “Note 16: Derivatives” for more information), was 4.36% and 3.25% as of December 31, 2022 and 2021, respectively.

As of December 31, 2022, future contractual maturities of long-term debt, excluding finance lease obligations, were as follows:

(in millions)	
2023	$ 14.0
2024	14.0
2025	14.0
2026	486.0
2027	917.0
Thereafter	935.0
Total	$ 2,380.0

Refer to “Note 19: Leasing” for additional information regarding finance lease obligations.

Senior Term Loan Facilities

In 2019, the Company entered into the Fifth Amendment to its Credit Agreement dated July 1, 2015 ("Credit Agreement"), which provided a Term B-5 Loan facility in an aggregate initial principal amount of $400 million that matures on July 1, 2026 (“Term B-5 Loan”). The Term B-5 Loan is payable in quarterly installments of 0.25% of the aggregate initial principal amount. The interest rate applicable to the Term B-5 Loan is based on, at the Company's option, (i) a fluctuating rate of interest determined by reference to a base rate plus an applicable margin of 1.00% or (ii) a Eurocurrency rate plus an applicable margin of 2.00%. The Company can prepay the Term B-5 Loan in whole or part without penalty.

In 2020, using the proceeds from the sale of the Environmental Sciences business, the Company repaid $174.0 million of previously incurred term loan debt.

In 2021, to secure favorable interest rates and extend principal maturities, the Company entered into the Sixth Amendment to its Credit Agreement, which provided a Term B-6 Loan facility in an aggregate initial principal amount of $1.0 billion that matures on June 3, 2028 ("Term B-6 Loan"). The proceeds from the Term B-6 Loan and an incremental borrowing of $274.2 million under the North American ABL Facility were used to repay in full previously incurred term loan debt and satisfy related lending and refinancing fees. As a result of the issuance of the Term B-6 Loan and the repayment of the Term B-3 Loan, the Company recognized a loss on extinguishment of debt of $2.0 million and debt refinancing costs of $6.9 million during the second quarter of 2021, both of which are included in other income (expense), net in the consolidated statements of operations.

The Term B-6 Loan is payable in quarterly installments of 0.25% of the aggregate initial principal amount beginning in September 2021. The interest rate applicable to the Term B-6 Loan is based on, at the Company's option, (i) a fluctuating rate of interest determined by reference to a base rate plus an applicable margin of 1.00% or (ii) a Eurocurrency rate plus an applicable margin of 2.00% (in each case with a 0.25% step down based on the achievement of a specific leverage level). The Company can prepay the Term B-6 Loan in whole or part without penalty.

ABL Facilities

On October 27, 2022, the Company and certain of its US, Canadian and European subsidiaries entered into the Second Amended and Restated ABL Credit Agreement (the "Senior ABL Facility"). The Senior ABL Facility amends and restates in full the amended and restated ABL facility entered into by the Company on February 28, 2019 ("North American ABL Facility"). In connection with the entry into the Senior ABL Facility, the Company terminated its existing European ABL Credit Agreement and the Euro ABL Facility thereunder. The Senior ABL Facility, which matures on October 27, 2027, provides a five-year senior secured ABL credit facility in an aggregate principal amount of $1.6 billion ("Senior ABL Credit Facility") and a five-year senior secured term loan facility in an aggregate principal amount of $200 million ("Senior ABL Term Loan"). Borrowing availability is determined by a borrowing base consisting of eligible inventory, eligible accounts receivable and cash of the Company and certain of its subsidiaries. The Senior ABL Term Loan is payable in quarterly installments of 12.5% of the aggregate initial principal amount beginning in March 2026.

The interest rates applicable to the Senior ABL Facility will be based on, at the Company's option, (i) with respect to the Senior ABL Term Loan, a fluctuating rate of interest determined by reference to either a base rate plus an applicable margin of 0.75% or a term SOFR rate plus an applicable margin of 1.75%, in each case with one 0.25% step down based on achievement of a specific leverage level and (ii) with respect to the Senior ABL Credit Facility, a fluctuating rate of interest determined by reference to a base rate plus an applicable margin of 0.125%, 0.25% or 0.375% or a term SOFR rate plus an applicable margin of 1.125%, 1.25% or 1.375%, with the margin in each case based on average combined testing availability. The applicable margin will be adjusted after the completion of each full fiscal quarter based upon the pricing grid in the Senior ABL Facility. The Senior ABL Facility contains a number of customary affirmative and negative covenants and events of default.

During 2021, the Company made a series of prepayments on the Canadian ABL term loan using the proceeds from the sale of the Canadian Agriculture services business and cash on hand, which resulted in the full extinguishment of the loan.

Senior Unsecured Notes

In 2019, the Company issued $500.0 million in Senior Unsecured Notes due December 1, 2027 (“2027 Senior Notes”), with a fixed interest rate of 5.125%. The Company can prepay the 2027 Senior Notes in whole or part at a premium on or after December 1, 2022 and without a premium on or after December 1, 2024. The Company repurchased $46.0 million of the 2027 Senior Notes in the open market during the second quarter of 2022, resulting in a gain on extinguishment of debt of $1.5 million within other income (expense), net in the consolidated statements of operations.

Debt costs

As a result of actions within its debt portfolio, the Company recognized a gain on extinguishment of debt of $1.5 million and losses of $2.3 million and $1.8 million during the years ended December 31, 2022, 2021 and 2020, respectively. The Company recognized no debt refinancing costs during the year ended December 31, 2022 and recognized debt refinancing costs of $7.0 million and $0.1 million during the years ended December 31, 2021 and 2020, respectively.

Borrowing availability and assets pledged as collateral

Borrowing availability is determined based upon available qualifying collateral, as defined in the Senior ABL Facility. As of December 31, 2022, the Company's borrowing availability under the Senior ABL Facility was $675.2 million.

Unused line fees were as follows:

	December 31,	
	2022	2021
$1.6 billion Senior ABL Credit Facility	0.250 %	N/A
$1.525 billion North American ABL Facility	N/A	0.300 %
€200 million Euro ABL Facility	N/A	0.375 %

The Senior ABL Facility is secured by a first priority lien on the loan parties' accounts receivable, inventory and cash as well as a second priority lien on substantially all other assets of the US subsidiary loan parties and other assets of certain European subsidiary loan parties in each case subject to various limitations and exceptions. In addition, 65% of the shares of certain foreign subsidiaries have been pledged as security. The Senior Term Loan Facilities are secured by substantially all of the assets of the US operating and management subsidiaries and a second priority lien on such accounts receivable and inventory. The Company had assets pledged of approximately $3.1 billion as of December 31, 2022 and 2021.

Debt covenants

The Company is in compliance with all debt covenants. The Senior ABL Facility includes a springing fixed charge coverage ratio of1.0x, applicable only if availability is less than or equal to 10% of the borrowing capacity. As of December 31, 2022, the Company had availability in excess of the minimum borrowing capacity and therefore this financial covenant is not applicable.

Other information

The fair values of debt were based on current market quotes for similar borrowings and credit risk adjusted for liquidity, margins and amortization, as necessary, and are classified as Level 2 in the fair value hierarchy.

	December 31, 2022		December 31, 2021	
(in millions)	Carrying amount	Fair value	Carrying amount	Fair value
Total long-term debt	$ 2,465.8	$ 2,459.3	$ 2,265.0	$ 2,307.8

The Company is exposed to credit loss and loss of liquidity availability if the financial institutions or counterparties issuing the debt securities fail to perform. The Company minimizes exposure to these credit risks by dealing with a diversified group of investment grade financial institutions. The Company manages credit risk by monitoring the credit ratings and market indicators of credit risk of its lending counterparties, and does not anticipate counterparty non-performance.

12. Employee benefit plans

Defined benefit pension plans

The Company sponsors defined benefit plans that provide pension benefits for employees upon retirement in certain jurisdictions including the US, Canada, United Kingdom and several other European countries. The US, Canada and United Kingdom defined benefit pension plans are closed to new entrants.

United Kingdom buy-in contract

Continuing the Company’s efforts to reduce retirement plan risk while delivering promised benefits to plan participants, the Trustees of the Univar Pension Scheme (1978) (“the UK Plan”) executed a buy-in contract in December 2021. The buy-in arrangement is an insurance contract providing substantially all future benefit plan payments to all UK Plan participants. As such, this transaction significantly mitigates volatility by removing investment, longevity, interest rate and inflation risk. However, the primary benefit obligation remains with the Company.

The buy-in contract remains an asset of the UK Plan and is part of the plan's investment strategy. The UK Plan transferred substantially all of the Company’s UK pension plan assets to the insurer in exchange for an insurance contract at the effective date of the buy-in. The buy-in contract had a fair value of $241.5 million as of December 31, 2022 and is classified as a Level 3 investment. The projected benefit obligation of the UK Plan as of December 31, 2022 was $198.3 million, which was net of $19.4 million of prior service credits recorded in accumulated other comprehensive loss.

US pension settlement

In November 2022, the Company entered into an agreement with an insurance company to purchase an irrevocable group annuity contract. The Company transferred US qualified pension plan liabilities of $72.6 million and plan assets of $67.7 million, resulting in a non-cash pre-tax settlement gain of $4.9 million.

The Company expects to continue to consider future opportunities to prudently manage costs and risks of its global retirement plans. Future transactions could result in non-cash settlement or mark to market charges which may impact earnings and could be material to a given reporting period.

The following table summarizes the Company's defined benefit pension plans' benefit obligations, plan assets and funded status:

	Domestic		Foreign		Total	
	Year ended December 31,		Year ended December 31,		Year ended December 31,	
(in millions)	2022	2021	2022	2021	2022	2021
Change in projected benefit obligations:						
Projected benefit obligations at beginning of year	$ 731.6	$ 774.9	$ 578.7	$ 649.8	$ 1,310.3	$ 1,424.7
Service cost	—	—	1.3	1.8	1.3	1.8
Interest cost	20.5	19.3	10.9	10.0	31.4	29.3
Benefits paid	(37.1)	(36.0)	(24.1)	(32.7)	(61.2)	(68.7)
Plan amendments	—	—	—	(0.4)	—	(0.4)
Settlements	(72.6)	—	(2.9)	—	(75.5)	—
Actuarial gain[1]	(172.6)	(26.6)	(169.5)	(44.0)	(342.1)	(70.6)
Foreign exchange and other	—	—	(48.0)	(5.8)	(48.0)	(5.8)
Projected benefit obligations at end of year	$ 469.8	$ 731.6	$ 346.4	$ 578.7	$ 816.2	$ 1,310.3
Change in the fair value of plan assets:						
Plan assets at beginning of year	$ 562.9	$ 541.8	$ 625.1	$ 654.0	$ 1,188.0	$ 1,195.8
Actual return on plan assets	(117.0)	44.3	(166.3)	5.6	(283.3)	49.9
Contributions by employer	13.7	12.8	3.4	4.5	17.1	17.3
Benefits paid	(37.1)	(36.0)	(24.1)	(32.7)	(61.2)	(68.7)
Settlements	(67.7)	—	(2.6)	—	(70.3)	—
Foreign exchange and other	—	—	(56.8)	(6.3)	(56.8)	(6.3)
Plan assets at end of year	354.8	562.9	378.7	625.1	733.5	1,188.0
Funded status at end of year	$ (115.0)	$ (168.7)	$ 32.3	$ 46.4	$ (82.7)	$ (122.3)
Accumulated benefit obligation	$ 469.8	$ 731.6	$ 340.8	$ 568.3	$ 810.6	$ 1,299.9

(1) The actuarial gains for the years ended December 31, 2022 and 2021 were primarily due to changes in the discount rate assumptions utilized in measuring plan obligations.

Net amounts related to the Company's defined benefit pension plans recognized in the consolidated balance sheets consist of:

	Domestic		Foreign		Total	
	December 31,		December 31,		December 31,	
(in millions)	2022	2021	2022	2021	2022	2021
Overfunded net benefit obligation in other assets	$ —	$ —	$ 56.6	$ 93.3	$ 56.6	$ 93.3
Current portion of net benefit obligation in other accrued expenses	(3.5)	(3.5)	(1.9)	(2.0)	(5.4)	(5.5)
Long-term portion of net benefit obligation in pension and other postretirement benefit liabilities	(111.5)	(165.2)	(22.4)	(44.9)	(133.9)	(210.1)
Net (liability) asset recognized at end of year	$ (115.0)	$ (168.7)	$ 32.3	$ 46.4	$ (82.7)	$ (122.3)

The following table summarizes defined benefit pension plans with accumulated benefit obligations in excess of plan assets:

	Domestic		Foreign		Total	
	December 31,		December 31,		December 31,	
(in millions)	2022	2021	2022	2021	2022	2021
Accumulated benefit obligation	$ 469.8	$ 731.6	$ 30.1	$ 179.7	$ 499.9	$ 911.3
Fair value of plan assets	354.8	562.9	9.4	143.3	364.2	706.2

The following table summarizes defined benefit pension plans with projected benefit obligations in excess of plan assets:

	Domestic		Foreign		Total	
	December 31,		December 31,		December 31,	
(in millions)	2022	2021	2022	2021	2022	2021
Projected benefit obligation	$ 469.8	$ 731.6	$ 130.7	$ 190.2	$ 600.5	$ 921.8
Fair value of plan assets	354.8	562.9	106.5	143.3	461.3	706.2

The following table summarizes the components of net periodic benefit cost (income) recognized in the consolidated statements of operations related to defined benefit pension plans:

	Domestic			Foreign			Total		
	Year ended December 31,			Year ended December 31,			Year ended December 31,		
(in millions)	2022	2021	2020	2022	2021	2020	2022	2021	2020
Service cost [1]	$ —	$ —	$ —	$ 1.3	$ 1.8	$ 1.9	$ 1.3	$ 1.8	$ 1.9
Interest cost [2]	20.5	19.3	23.2	10.9	10.0	11.7	31.4	29.3	34.9
Expected return on plan assets [2]	(30.3)	(29.2)	(28.5)	(11.3)	(15.1)	(14.8)	(41.6)	(44.3)	(43.3)
Amortization of prior service credit [2]	—	—	—	(0.7)	(3.5)	(0.1)	(0.7)	(3.5)	(0.1)
Settlements [3]	(4.9)	—	—	(0.3)	(0.3)	(0.6)	(5.2)	(0.3)	(0.6)
Actuarial (gain) loss [4]	(25.2)	(41.7)	28.9	8.0	(34.2)	23.7	(17.2)	(75.9)	52.6
Net periodic benefit (income) cost	$ (39.9)	$ (51.6)	$ 23.6	$ 7.9	$ (41.3)	$ 21.8	$ (32.0)	$ (92.9)	$ 45.4

(1) Service cost is included in warehousing, selling and administrative expenses.
(2) These amounts are included in other income (expense), net, and represent non-operating retirement benefits.
(3) Settlements are included in other income (expense), net.
(4) Actuarial (gain) loss, or mark to market, includes measurement gains and losses resulting from changes since the prior measurement date in assumptions and plan experience, as well as the difference between the expected and actual return on plan assets. These amounts are recorded in other income (expense), net.

As of December 31, 2022 and 2021, amounts recorded in AOCI related to foreign defined benefit pension plans.The following table summarizes pre-tax amounts included in AOCI related to pension plan amendments:

	December 31,	
(in millions)	2022	2021
Net prior service credit	$ 18.9	$ 22.2

Actuarial assumptions

The significant weighted average actuarial assumptions used in determining the benefit obligations and net periodic benefit cost (income) for the Company's defined benefit plans were as follows:

	Domestic		Foreign	
	December 31,		December 31,	
	2022	2021	2022	2021
Actuarial assumptions used to determine benefit obligations at end of period:				
Discount rate	5.49 %	2.89 %	4.69 %	2.05 %
Expected annual rate of compensation increase	N/A	N/A	1.97 %	1.87 %

	Domestic			Foreign		
	Year ended December 31,			Year ended December 31,		
	2022	2021	2020	2022	2021	2020
Actuarial assumptions used to determine net periodic benefit cost (income) for the period:						
Discount rate	2.89 %	2.55 %	3.28 %	2.05 %	1.55 %	2.14 %
Expected rate of return on plan assets	6.00 %	6.00 %	6.50 %	2.13 %	2.46 %	2.75 %
Expected annual rate of compensation increase	N/A	N/A	N/A	1.87 %	2.91 %	2.85 %

Discount rates are used to measure benefit obligations and the interest cost component of net periodic benefit cost (income). The Company selects its discount rates based on the consideration of equivalent yields on high-quality fixed income investments at each measurement date. Discount rates are based on a benefit cash flow-matching approach and represent the rates at which the Company's benefit obligations could effectively be settled as of the measurement date.

For domestic defined benefit plans, the discount rates are based on a hypothetical bond portfolio approach. The hypothetical bond portfolio is constructed to comprise AA-rated corporate bonds whose cash flow from coupons and maturities match the expected future plan benefit payments.

The discount rates for the foreign defined benefit plans are based on a yield curve approach. For plans in countries with a sufficient corporate bond market, the expected future benefit payments are matched with a yield curve derived from AA-rated corporate bonds, subject to minimum amounts outstanding and meeting other selection criteria. For plans in countries without a sufficient corporate bond market, the yield curve is constructed based on prevailing government yields and an estimated credit spread to reflect a corporate risk premium. The discount rate to value UK Plan liabilities continues to be set with reference to the yields on high quality corporate bonds.

The expected long-term rate of return on plan assets reflects management's expectations on long-term average rates of return on funds invested to provide for benefits included in the benefit obligations. The long-term rate of return assumptions are based on the outlook for equity and fixed income returns, with consideration of historical returns, asset allocations, investment strategies and premiums for active management when appropriate. Assumptions reflect the expected rates of return at the beginning of the year.

Plan asset management

Plan assets for defined benefit plans are invested in global equity, debt securities or insurance contracts through professional investment managers with the objective to achieve targeted risk adjusted returns and to maintain liquidity sufficient to fund current benefit payments. Each funded defined benefit plan has an investment policy that is administered by plan trustees with the objective of meeting targeted asset allocations based on the circumstances of that particular plan.

The investment strategy followed by the Company varies by country depending on the circumstances of the underlying plan. Less mature plan benefit obligations are funded by using more equity securities as they are expected to achieve long-term growth while exceeding inflation. More mature plan benefit obligations are funded using a higher allocation of fixed income securities as they are expected to produce current income with limited volatility.

The Company has adopted a dynamic investment strategy whereby as the plan funded status improves, the investment strategy is migrated to more liability matching assets, and return seeking assets are reduced. Risk management practices include the use of multiple asset classes for diversification purposes. Specific guidelines for each asset class and investment manager are implemented and monitored. As noted above, substantially all of the UK Plan assets were within a buy-in contract as of December 31, 2022 and 2021, subsequent to the transaction executed in December 2021.

The weighted average target asset allocation for defined benefit pension plans as of December 31, 2022 was as follows:

	Domestic	Foreign
Asset category:		
Equity securities	45.0 %	11.0 %
Debt securities	40.0 %	16.0 %
Other	15.0 %	73.0 %
Total	100.0 %	100.0 %

Plan asset valuation methodologies

Plan asset	Description of methodology
Cash	This represents cash at banks at fair value, reflected within Level 1 of the fair value hierarchy.
Investment funds	Fair values are based on the NAV of the units held at year end. The NAVs are based on the fair value of the underlying assets of the funds, less their liabilities, divided by the number of units outstanding at the valuation date. The funds are traded on private markets that are not active, with unit prices based on observable (Level 2) or unobservable (Level 3) market data based on the fund's underlying assets.
Buy-in contract	This represents the UK Plan buy-in contract assets. Fair values are based on initial pricing adjusted for changes in government and corporate bond yields, inflation, demographic assumptions and changes in participant data, and as such are considered Level 3 based on the significant unobservable inputs used in deriving the assets' fair values.
Insurance contracts	Fair values are based on the present value of the accrued benefit, and as such are considered Level 3 based on the significant unobservable inputs used in deriving the asset's fair values.
Investments measured at NAV	Investment in a real estate fund, which invests in real estate assets. The investment in properties by the real estate fund is carried at fair value, which is estimated based on the price that would be received to sell an asset in an orderly transaction between marketplace participants at the measurement date. The investment can be withdrawn quarterly, after a one hundred days notice period.

Fair value of plan assets

The following table presents domestic plan assets categorized within the three-level fair value hierarchy:

	December 31, 2022				December 31, 2021			
(in millions)	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
Cash	$ 2.8	$ —	$ —	$ 2.8	$ 4.3	$ —	$ —	$ 4.3
Investment funds [1]	—	292.5	27.6	320.1	—	503.0	27.6	530.6
Subtotal	$ 2.8	$ 292.5	$ 27.6	322.9	$ 4.3	$ 503.0	$ 27.6	534.9
Investments measured at NAV				31.9				28.0
Total				$ 354.8				$ 562.9

(1) This category includes investments of 24.7% and 21.8% in US equities, 17.9% and 17.9% in non-US equities, 25.9% and 41.9% in US corporate bonds, 0.1% and 2.2% in non-US corporate bonds, 19.9% and 0.0% in US government bonds, 2.9% and 0.0% in non-US government bonds and 8.6% and 16.2% in other investments as of December 31, 2022 and 2021, respectively.

The following table presents foreign plan assets categorized within the three-level fair value hierarchy:

	December 31, 2022				December 31, 2021			
(in millions)	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
Cash	$ 1.9	$ —	$ —	$ 1.9	$ 18.8	$ —	$ —	$ 18.8
Investments:								
Investment funds [1]	—	118.9	—	118.9	—	147.4	—	147.4
Buy-in contract	—	—	241.5	241.5	—	—	436.1	436.1
Insurance contracts	—	—	16.4	16.4	—	—	22.8	22.8
Total investments	—	118.9	257.9	376.8	—	147.4	458.9	606.3
Total	$ 1.9	$ 118.9	$ 257.9	$ 378.7	$ 18.8	$ 147.4	$ 458.9	$ 625.1

(1) This category includes investments of 15.0% and 0.6% in US equities, 23.6% and 45.4% in non-US equities, 38.7% and 39.7% in non-US corporate bonds, 13.7% and 5.4% in non-US government bonds and 9.0% and 8.9% in other investments as of December 31, 2022 and 2021, respectively.

Changes in the plan assets valued using significant unobservable inputs (Level 3) were as follows:

	Domestic	
	Investment funds	
(in millions)	2022	2021
Balance as of January 1	$ 27.6	$ —
Actual return on plan assets still held at year end	—	—
Purchases, sales and settlements, net	—	—
Transfer from Level 2 classification	—	27.6
Balance as of December 31	$ 27.6	$ 27.6

	Foreign	
	Buy-in and Insurance contracts	
(in millions)	2022	2021
Balance as of January 1	$ 458.9	$ 26.6
Actual return on plan assets still held at year end	(142.3)	(34.1)
Purchases, sales and settlements, net	(13.9)	475.4
Foreign exchange	(44.8)	(9.0)
Balance as of December 31	$ 257.9	$ 458.9

Contributions

The Company expects to contribute approximately $13.5 million and $2.5 million to its domestic and foreign defined benefit pension plan funds in 2023, respectively, including direct payments to plan participants in unfunded plans.

Benefit payments

Estimated future benefit payments are as follows:

	Defined benefit pension plans		
(in millions)	Domestic	Foreign	Total
2023	$ 33.4	$ 22.4	$ 55.8
2024	34.0	22.8	56.8
2025	34.8	23.8	58.6
2026	35.3	23.8	59.1
2027	35.7	24.0	59.7
2028 through 2032	178.9	126.1	305.0

Defined contribution plans

The Company provides defined contribution plans and had contribution expense of $52.9 million, $51.4 million and $43.8 million for the years ended December 31, 2022, 2021 and 2020, respectively.

Multi-employer pension plans

The Company contributes to certain multi-employer pension plans based on obligations arising from collective bargaining agreements. The risks of participating in these multi-employer plans are different from single-employer plans in the following aspects:

- Assets contributed to the multi-employer plan by the Company may be used to provide benefits to employees of other participating employers.
- If the Company stops contributing to the plan, the unfunded obligations of the plan may be borne by the remaining participating employers. Similarly, the Company could be liable for underfunded obligations of other, departed employers.
- If the Company chooses to stop participating in some of its multi-employer plans, it may be required to pay those plans an amount based on the underfunded status of the plan, referred to as a withdrawal liability. A withdrawal liability will be recorded when it is probable that a liability exists and the liability can be reasonably estimated.

The Company's participation in these plans for the annual period ended December 31, 2022 is outlined in the table below. The Pension Protection Act ("PPA") zone status is the most recently available and is certified by the plan's actuary. Among other factors, plans in the "red zone" are less than 65 percent funded, plans in the "yellow zone" are less than 80 percent funded and plans in the "green zone" are at least 80 percent funded. The "FIP/RP status pending/implemented" column indicates plans for which a financial improvement plan ("FIP") or a rehabilitation plan ("RP") is either pending or has been implemented. In addition to regular plan contributions, the Company may be subject to a surcharge if the plan is in the red zone. The "Surcharge imposed" column indicates whether a surcharge has been imposed on contributions to the plan. The last column lists the expiration date(s) of the collective bargaining agreement(s) ("CBA") to which the plans are subject. Contributions may be required in future periods by the CBAs, statutory obligations or other contractual obligations.

If the Company were to cease making contributions to these plans atone or more or all locations, it could trigger a withdrawal liability that could be material to the Company's results of operations and cash flows. Calculating any such withdrawal liability depends on a number of factors that are out of the Company's control and subject to change. A withdrawal liability represents an employer's proportional share of the multi-employer plan's unfunded vested benefits ("UVBs"). UVBs equal the value of non-forfeitable benefits owed by the plan, less the value of the plan's assets. The value of assets and liabilities are determined using actuarial assumptions that reflect the actuary's best estimate of anticipated UVBs for that employer.

A significant portion of the Company's exposure historically resided with the eight facilities that participated in the Central States, Southeast and Southwest Areas Pension Plan ("Central States Pension Fund"). During the year ended December 31, 2021, the Company recognized its best estimate of a withdrawal liability of $31.2 million related to triggering events at alleight sites, culminating in the Company ceasing to participate in the Central States Pension Fund. Upon an agreed final funding assessment with the Central States Pension Fund, the Company will recognize any differences between the estimated and actual withdrawal liability. As of December 31, 2022, this balance is unchanged. The Company estimates its cash obligation to be approximately $1.9 million annually for each of the next 20 years. The net present value of the withdrawal liability was determined using a risk-free interest rate. Amounts associated with the withdrawal liability are included in other operating expenses, net in the consolidated statements of operations and other accrued expenses and other long-term liabilities in the consolidated balance sheets.

(in millions)	EIN/Pension plan number	PPA zone status		FIP/RP status pending/implemented	Contributions [a] Year ended December 31,			Surcharge imposed	Number of and expiration dates of collective bargaining agreement(s)
		2022	2021		2022	2021	2020		
Western Conference of Teamsters Pension Plan	91-6145047/001	Green as of January 1, 2022	Green as of January 1, 2021	No	$ 1.4	$ 1.4	$ 1.5	No	(b)
Central States, Southeast and Southwest Areas Pension Plan	36-6044243/001	Red as of January 1, 2021	Red as of January 1, 2020	Implemented	—	0.7	1.1	No	(c)
Total contributions					$ 1.4	$ 2.1	$ 2.6		

(a) Plan contributions by the Company did not represent more than five percent of total contributions to the plans as indicated in the plans' most recently available annual report.
(b) As of December 31, 2022, the Company had six agreements expiring through July 31, 2025.
(c) The Company either terminated the CBA or the Union voted to amend the CBA covered by this fund. As a result, the Company has withdrawn from the fund and recognized expense for its estimated withdrawal liability.

13. Income taxes

For financial reporting purposes, income before income taxes includes the following components:

(in millions)	Year ended December 31, 2022	2021	2020
Income before income taxes			
US	$ 506.9	$ 203.1	$ (47.7)
Foreign	249.3	382.1	106.7
Total income before income taxes	$ 756.2	$ 585.2	$ 59.0

The expense for income taxes is summarized as follows:

(in millions)	Year ended December 31, 2022	2021	2020
Current:			
Federal	$ 87.4	$ 44.4	$ (3.4)
State	21.7	15.2	(1.5)
Foreign	72.8	58.6	43.4
Total current	181.9	118.2	38.5
Deferred:			
Federal	23.3	0.3	(20.2)
State	5.4	(0.7)	(2.7)
Foreign	0.3	6.8	(9.5)
Total deferred	29.0	6.4	(32.4)
Total income tax expense	$ 210.9	$ 124.6	$ 6.1

Differences between actual provisions for income taxes and provisions for income taxes at the US federal statutory rate of 21.0% were as follows:

(in millions)	Year ended December 31, 2022	2021	2020
US federal statutory income tax expense applied to income before income taxes	$ 158.8	$ 122.8	$ 12.5
State income taxes, net of federal benefit	22.7	10.8	(4.6)
Foreign tax rate differential	11.0	10.5	5.7
Distributions from foreign subsidiaries	0.9	—	(9.9)
Global intangible low-taxed income	2.8	31.5	12.9
Disposition of business	0.1	(20.6)	(5.0)
Change in valuation allowance, net	5.7	(11.3)	(69.3)
Foreign tax credit	(3.2)	(29.8)	58.8
Non-deductible expenses	8.2	5.8	2.6
Warrants	—	(7.1)	0.2
Change in statutory income tax rates	(0.9)	5.1	1.5
Other	4.8	6.9	0.7
Total income tax expense	$ 210.9	$ 124.6	$ 6.1
Effective income tax rate	27.9 %	21.3 %	10.3 %

The consolidated deferred tax assets and liabilities are detailed as follows:

(in millions)	December 31, 2022	December 31, 2021
Deferred tax assets:		
Net operating loss carryforwards ("NOLs")	$ 17.6	$ 22.8
Environmental liabilities	22.7	22.1
Interest	—	5.9
Tax credit and capital loss carryforwards	4.9	4.9
Pension	33.6	41.5
Compensation	18.2	36.8
Inventory	13.3	12.5
Lease liabilities	50.3	42.0
Other temporary differences	15.8	22.1
Gross deferred tax assets	176.4	210.6
Valuation allowance	(12.7)	(8.6)
Deferred tax assets, net of valuation allowance	163.7	202.0
Deferred tax liabilities:		
Property, plant and equipment, net	(96.7)	(102.5)
Intangible assets	(77.4)	(83.2)
Right-of-use lease assets	(48.7)	(39.8)
Interest	(20.7)	—
Other temporary differences	(5.7)	(3.2)
Deferred tax liabilities	(249.2)	(228.7)
Net deferred tax liability	$ (85.5)	$ (26.7)

As of December 31, 2022, the Company has $17.6 million (tax effected) of NOLs, of which $0.7 million will expire in the period 2023 through 2027, $3.8 million will expire in the period between 2028 and 2041 and the remaining $13.1 million has no expiration. Additionally, the Company has approximately $4.9 million of foreign tax credits and capital loss carryforwards. The Company does not expect future earnings of the appropriate character of taxable income to allow it to utilize certain of these tax attributes in future years. Therefore, the Company maintains a valuation allowance of $12.7 million on these tax attributes and other deferred tax assets.

Foreign Tax Effects

The Company earns a significant amount of its operating income outside of the US. As of December 31, 2022, the Company is indefinitely reinvested with respect to its US directly-owned subsidiary earnings. Therefore, the Company has not recognized a deferred tax liability on its investment in foreign subsidiaries. The Company is subject to US income tax on substantially all foreign earnings under the GILTI provisions of the 2017 Tax Cuts and Jobs Act, while a significant portion of remaining foreign earnings are eligible for the dividends received deduction. As a result, a portion of any future repatriation of $628.8 million of undistributed earnings may be subject to US income tax, as well as state and local income taxes, withholding taxes and currency translation gains or losses. It is impracticable to calculate the exact amount. Additionally, gains and losses on any future taxable dispositions of US-owned foreign affiliates continue to be subject to US income tax.

Unrecognized Tax Benefits

The changes in unrecognized tax benefits included in other long-term liabilities, excluding interest and penalties, were as follows:

(in millions)	2022	2021	2020
Balance as of January 1	$ 40.1	$ 35.0	$ 1.1
Increase for tax positions of current year	—	19.4	16.4
Increase for tax positions of prior years	0.4	0.9	17.8
Decrease for tax positions of prior years	—	(14.4)	—
Reductions due to the statute of limitations expiration	(0.1)	(0.7)	(0.3)
Reductions due to settlement with taxing authorities	(0.3)	—	—
Foreign exchange	—	(0.1)	—
Balance as of December 31	$ 40.1	$ 40.1	$ 35.0

As of December 31, 2022 and 2021, there were $40.1 million of unrecognized tax benefits that if recognized would affect the annual effective tax rate. The Company files income tax returns in the US and various state and foreign jurisdictions. Generally, tax years are open for review by taxing authorities for a period of three years. As of December 31, 2022, the Company has limited audit activity for tax years back to 2008 through 2010, as well as for the periods 2012 through 2020. The Company continues to believe its positions are supportable; however, due to uncertainties in any tax audit outcome, the Company's estimates of the ultimate settlement of uncertain tax positions may change and the actual tax benefits may differ from the estimates.

The Company recognized $2.4 million, $0.1 million and $0.6 million of interest and/or penalties related to income tax matters in interest expense related to unrecognized tax benefits in the consolidated statements of operations for the years ended December 31, 2022, 2021 and 2020, respectively. The Company had $4.3 million and $2.0 million of interest and penalties reflected within other long-term liabilities on the consolidated balance sheets as of December 31, 2022 and 2021, respectively.

14. Share repurchase program and stock-based compensation

Share repurchase program

On November 1, 2021, the Company announced that its Board of Directors had authorized a share repurchase program of up to $500.0 million of its outstanding common stock, which expires on October 27, 2026. On November 1, 2022, the Company announced that its Board of Directors had approved an increase in the amount of authorized repurchases under the program of $1.0 billion, resulting in a total authorized repurchase amount of $1.5 billion. The program does not require the repurchase of any minimum number of shares and can be suspended, modified or discontinued at any time at the Company's discretion. Under the share repurchase program, the Company may purchase shares from time to time at the discretion of management through open market purchases, privately negotiated transactions, block trades, accelerated or other structured share repurchase programs, or other means.

On November 1, 2022, the Company entered into an ASR with Goldman to repurchase $200.0 million of its common stock. On November 3, 2022, the Company paid $200.0 million to Goldman and received an initial delivery of approximately 5.8 million shares of its common stock, which represented 75% of the notional value of the ASR divided by the closing price of the Company's common stock on November 1, 2022. The final number of shares repurchased under the ASR was based on the daily volume-weighted average prices for Rule 10b-18 eligible transactions in the Company's common stock during the term of the ASR, less a discount and subject to adjustment pursuant to the terms of the ASR. In February 2023, the final settlement was determined to be an additional delivery of approximately 0.5 million shares of the Company's common stock from Goldman.

As of December 31, 2022, the aggregate purchase price of the ASR was recorded as a reduction to stockholders' equity, consisting of a $150.0 million increase in treasury stock to reflect the value of the initial share delivery and a $50.0 million decrease in additional paid-in capital pending final settlement of the ASR. The amount recorded in additional paid-in capital will be reclassified to treasury stock in the first quarter of 2023 in connection with the final settlement of the ASR.

In addition to the ASR, the Company repurchased on the open market approximately 7.7 million shares for $209.1 million during the year ended December 31, 2022 and approximately 1.8 million shares for $50.0 million during the year ended December 31, 2021. The Company's remaining stock repurchase authorization under the program was approximately $1,040.9 million as of December 31, 2022.

Stock-based compensation

In May 2020, the Company replaced the Univar Inc. 2017 Omnibus Equity Incentive Plan (the "2017 Plan") with the Univar Solutions Inc. 2020 Omnibus Incentive Plan (the "2020 Plan"). The 2017 Plan had no further awards granted and any available reserves under the 2017 Plan were terminated and not transferred to the 2020 Plan. There were no changes to the outstanding awards related to the 2017 Plan, the Univar Inc. 2015 Stock Incentive Plan and Univar Inc. 2011 Stock Incentive Plan (together with the 2017 Plan and the 2020 Plan, the "Plans").

The 2020 Plan allows the Company to issue awards to employees, consultants and directors of the Company and its subsidiaries. Awards may be made in the form of stock options, stock purchase rights, restricted stock, restricted stock units, performance shares, performance units, stock appreciation rights, dividend equivalents, deferred share units or other stock-based awards. As of December 31, 2022, there were 8.6 million shares authorized under the Plans. The Company generally issues authorized but previously unissued shares to satisfy stock option exercises and stock award vestings.

For the years ended December 31, 2022, 2021 and 2020, the Company recognized total stock-based compensation expense within other operating expenses, net of $34.3 million, $25.4 million and $14.5 million, and a net tax benefit relating to stock-based compensation expense of $4.2 million, $2.2 million and $2.0 million, respectively.

Stock options

Stock options expire ten years after the grant date and generally become exercisable over a three-year period or less, based on continued employment, with annual vesting. The exercise price of a stock option is determined based upon the fair value of the common stock at the time of each grant. Participants have no stockholder rights until the time of exercise.

The following table reflects stock option activity under the Plans:

	Number of stock options	Weighted-average exercise price	Weighted-average remaining contractual term (in years)	Aggregate intrinsic value (in millions)
Outstanding at January 1, 2022	3,290,543	$ 24.22		
Granted	—	—		
Exercised	(1,031,893)	23.41		
Forfeited and expired	(52,290)	24.35		
Outstanding at December 31, 2022	2,206,360	$ 24.60	5.4	$ 15.9
Exercisable at December 31, 2022	1,880,396	$ 24.89	5.1	$ 13.0
Expected to vest after December 31, 2022	325,964	$ 22.94	7.1	$ 2.9

As of December 31, 2022, the Company had unrecognized stock-based compensation expense related to non-vested stock options of approximately $0.1 million, which will be recognized over a weighted-average period of 0.1 years.

	Year ended December 31,		
(in millions)	2022	2021	2020
Total intrinsic value of stock options exercised	$ 7.8	$ 2.3	$ 0.3

Restricted stock

Non-vested restricted stock relates to awards for members of the Company's Board of Directors which vest over 12 months. The grant date fair value of restricted stock is based on the market price of the common stock on that date. Non-vested shares of restricted stock may not be sold or transferred and are subject to forfeiture until vesting. Both vested and non-vested shares of restricted stock are included in the Company's shares outstanding.

The following table reflects restricted stock activity under the Plans:

	Number of restricted stock	Weighted-average grant date fair value
Non-vested at January 1, 2022	23,056	$ 24.29
Granted	22,191	29.75
Vested	(26,727)	24.70
Forfeited	—	—
Non-vested at December 31, 2022	18,520	$ 30.24

As of December 31, 2022, the Company had unrecognized stock-based compensation expense related to non-vested restricted stock awards of approximately $0.2 million, which will be recognized over a weighted-average period of 0.3 years. The weighted-average grant date fair value of granted restricted stock was $24.29 and $13.91 in 2021 and 2020, respectively.

Restricted stock units ("RSUs")

RSUs awarded to employees generally vest in three equal annual installments, subject to continued employment. Each RSU converts into one share of Univar Solutions common stock on the applicable vesting date. RSUs may not be sold, pledged or otherwise transferred until they vest and are subject to forfeiture. The grant date fair value is based on the market price of Univar Solutions stock on that date.

The following table reflects RSU activity under the Plans:

	Number of Restricted Stock Units	Weighted-average grant date fair value
Non-vested at January 1, 2022	1,553,491	$ 21.61
Granted	827,056	28.62
Vested	(805,585)	22.13
Forfeited	(118,560)	23.04
Non-vested at December 31, 2022	1,456,402	$ 25.18

As of December 31, 2022, the Company had unrecognized stock-based compensation expense related to non-vested RSU awards of approximately $12.8 million, which will be recognized over a weighted-average period of 1.0 years. The weighted-average grant date fair value of granted RSU awards was $21.66 and $21.29 in 2021 and 2020, respectively.

Performance-based restricted stock units ("PRSUs")

The Company awards performance-based shares to certain employees. These awards vest upon the passage of time and the achievement of performance conditions, which is generally over a three year period. The Company reviews progress toward the attainment of the performance conditions each quarter during the vesting period. When it is probable the minimum performance conditions for the award will be achieved, the Company begins recognizing the expense equal to the proportionate share of the total fair value. The total expense recognized over the duration of performance awards will equal the grant date fair value multiplied by the number of shares ultimately awarded based on the level of attainment of the performance conditions.

For grants with market conditions, the fair value is determined on the grant date by using a Monte Carlo simulation model, with a duration of three years. The PRSUs awarded may be subject to a downward or upward adjustment depending on the total stockholder return achieved by the Company during the particular performance period relative to the total stockholder return ranking among peer companies as specified in the respective PRSU award agreement. The total expense recognized over the duration of the award is not adjusted based on the level of attainment of the market conditions.

The following table reflects PRSU activity under the Plans:

	Number of Performance-Based Restricted Stock Units	Weighted-average grant date fair value
Non-vested at January 1, 2022	709,260	$ 21.77
Granted	243,750	27.94
Vested	(409,870)	22.81
Forfeited	(13,950)	21.46
Non-vested at December 31, 2022	529,190	$ 23.82

As of December 31, 2022, the Company had unrecognized stock-based compensation expense related to non-vested PRSU awards of approximately $7.7 million, which will be recognized over a weighted-average period of 1.5 years.

Fair value

(in millions)	Year ended December 31, 2022	2021	2020
Fair value of restricted stock, RSUs and PRSUs vested	$ 40.1	$ 9.6	$ 9.2

Employee stock purchase plan

The Univar Solutions Inc. Employee Stock Purchase Plan, or ESPP, authorizing the issuance of up to 2.0 million shares of the Company's common stock allows qualified participants to purchase the Company's common stock at 95% of its market price during the last day of two offering periods in each calendar year. The first offering period is January through June, and the second from July through December. As of December 31, 2022, the total number of shares issued under the plan for the two offering periods in 2022 was 60,942 shares.

Stock-based compensation fair value assumptions

The fair value of the Company's stock that is factored into the fair value of stock options and utilized for restricted stock, RSUs and PRSUs with internally developed performance conditions is based on the grant date closing price on the New York Stock Exchange.

The Company uses a Monte Carlo simulation to calculate the fair value of PRSUs with market conditions. The weighted average grant date fair value of PRSUs with market conditions was $27.94 and $20.36 for the years ended December 31, 2022 and 2021, respectively. The weighted-average assumptions under the Monte Carlo simulation model were as follows:

	Year ended December 31, 2022	2021
Risk-free interest rate [1]	1.6 %	0.3 %
Expected dividend yield	—	—
Expected volatility [2]	51.7 %	53.8 %

(1) The risk-free interest rate was based on the US Treasury yield for a period in years equal to the remaining performance period from date of grant.
(2) The expected volatility for each grant was determined based on the historical weekly stock price returns of the Company's common stock over a period equal to the remaining term of the performance period from the date of grant.

The Company uses the Black-Scholes-Merton option valuation model to calculate the fair value of stock options granted. No stock options were granted in 2021 or 2022. The weighted-average grant date fair value of stock options was $6.17 for the year ended December 31, 2020. The weighted-average assumptions used under the Black-Scholes-Merton option valuation model were as follows:

	Year ended December 31, 2020
Risk-free interest rate [1]	1.4 %
Expected dividend yield	—
Expected volatility [2]	24.4 %
Expected term (years) [3]	6.0

(1) The risk-free interest rate was based on the US Treasury yield for a term consistent with the expected term of the stock options at the time of grant.
(2) As the Company did not have sufficient historical volatility data, the expected volatility was based on the average historical data of a peer group of public companies over a period equal to the expected term of the stock options.
(3) As the Company did not have sufficient historical exercise data under the Plans, the expected term was based on the average of the vesting period of each tranche and the original contract term of 10 years.

15. Fair value measurements

The Company classifies its financial instruments according to the fair value hierarchy described in "Note 2: Significant accounting policies."

Items measured at fair value on a recurring basis

The following table is a reconciliation of recurring fair value measurements that use significant unobservable inputs (Level 3), which consisted of the warrant liability related to the Nexeo acquisition and Techi Chem acquisition contingent consideration liabilities:

	Warrant Liability		Contingent Consideration	
(in millions)	2021	2020	2021	2020
Fair value as of January 1	$ 33.8	$ 33.0	$ 2.2	$ —
Additions	—	—	—	2.2
Fair value adjustments	(33.8)	0.8	—	—
Foreign exchange	—	—	0.1	—
Transfers out of Level 3	—	—	(2.3)	—
Fair value as of December 31	$ —	$ 33.8	$ —	$ 2.2

In conjunction with the Nexeo acquisition in 2019, the Company assumed warrants that were converted into the right to receive, upon exercise, approximately 7.6 million shares of the Company's common stock. During 2021, a portion of the outstanding warrants were exercised at a price of $27.80, resulting in the issuance of 973,717 shares of common stock and the receipt of $27.1 million in cash proceeds. All remaining warrants expired on June 9, 2021, resulting in the write-off of the warrants' fair value within other income (expense), net in the consolidated statements of operations.

As the performance period for the Techi Chem contingent consideration ended in 2021, and inputs to the liability were no longer unobservable, the Company no longer considered the balance a Level 3 classification.

16. Derivatives

Foreign currency derivatives

The Company uses forward currency contracts to hedge earnings from the effects of foreign exchange rates relating to certain of the Company's intercompany and third-party receivables and payables denominated in foreign currencies. These derivative instruments are not formally designated as cash flow hedges by the Company and the terms of these instruments range from one to three months.

Interest rate swap contracts

The objective of the Company's designated interest rate swap contracts is to offset the variability of cash flows in LIBOR and SOFR indexed debt interest payments attributable to changes in the benchmark interest rates related to the Term B-6 Loan (previously the Term B-3 Loan) and a portion of debt outstanding under the Senior ABL Facility (previously the North American ABL Facility).

In June 2021, the Company executed two interest rate swap contracts, both effective June 30, 2023, to replace existing interest rate swap contracts with maturities occurring between June 2023 and June 2024. These interest rate swap contracts contain an initial aggregate notional value of $250.0 million from June 2023 to June 2024 that increases to an aggregate notional value of $500.0 million from June 2024 to May 2028.

The Company also uses undesignated interest rate swap contracts to manage interest rate variability.

Cross currency swap contracts

Cross currency swap contracts are used to effectively convert the Term B-5 Loan's principal amount of floating rate US dollar denominated debt, including interest payments, to fixed-rate Euro denominated debt. The cross currency swap contracts mature in November 2024 and approximately 95% of the contracts are designated as a cash flow hedge.

The Company also uses undesignated cross currency swap contracts to manage interest rate variability and mitigate foreign exchange exposure.

Notional amounts and fair value of derivative instruments

The following table presents the notional amounts of the Company's outstanding derivative instruments by type:

	December 31,	
(in millions)	2022	2021
Designated Derivatives:		
Interest rate swap contracts	$ 650.0	$ 650.0
Cross currency swap contracts	381.0	381.0
Undesignated Derivatives:		
Foreign currency derivatives	$ 149.2	$ 179.0
Interest rate swap contracts	100.0	100.0
Cross currency swap contracts	19.0	19.0

The following table presents the pre-tax gains (losses) recognized in accumulated other comprehensive loss related to designated derivative instruments:

	Amount of gain (loss) recognized in accumulated other comprehensive loss			Amount of gain (loss) to be reclassified to consolidated statement of operations within the next 12 months
	Year ended December 31,			
(in millions)	2022	2021	2020	
Effect of derivative instruments designated and qualifying as cash flow hedges:				
Interest rate swap contracts	$ 70.0	$ 4.7	$ (31.5)	$ 19.0
Cross currency swap contracts	71.7	35.7	(34.9)	19.7

The following table presents the pre-tax effects of derivative instruments on the consolidated statements of operations:

	Year ended December 31,					
	2022		2021		2020	
(in millions)	Interest expense	Other income (expense), net	Interest expense	Other income (expense), net	Interest expense	Other income (expense), net
Total amounts per Consolidated Statements of Operations	$ (107.2)	$ 26.5	$ (101.5)	$ 110.4	$ (114.5)	$ (61.0)
Effect of derivative instruments designated and qualifying as cash flow hedges:						
Interest rate swap contracts	$ (0.1)	$ —	$ (17.9)	$ —	$ (12.9)	$ —
Cross currency swap contracts	8.4	21.9	1.2	27.4	3.5	(31.8)
Effect of undesignated derivatives:						
Foreign currency derivatives	$ —	$ (2.3)	$ —	$ 2.8	$ —	$ 3.2
Interest rate swap contracts	—	5.6	—	1.2	—	(6.3)
Cross currency swap contracts	—	3.6	—	1.8	—	(1.7)

The following table presents the Company's gross assets and liabilities measured on a recurring basis and classified as Level 2 within the fair value hierarchy:

	Derivative Assets			Derivative Liabilities		
		December 31,			December 31,	
(in millions)	Balance Sheet Classification	2022	2021	Balance Sheet Classification	2022	2021
Designated Derivatives:						
Cross currency swap contracts	Prepaid expenses and other current assets	$ 19.7	$ 2.8	Other accrued expenses	$ —	$ —
Cross currency swap contracts	Other assets	35.4	—	Other long-term liabilities	—	12.7
Interest rate swap contracts	Prepaid expenses and other current assets	19.0	—	Other accrued expenses	—	8.1
Interest rate swap contracts	Other assets	35.2	—	Other long-term liabilities	—	7.8
Total designated derivatives		$ 109.3	$ 2.8		$ —	$ 28.6
Undesignated Derivatives:						
Foreign currency contracts	Prepaid expenses and other current assets	$ —	$ 1.8	Other accrued expenses	$ 0.9	$ 0.8
Cross currency swap contracts	Prepaid expenses and other current assets	1.0	0.1	Other accrued expenses	—	—
Cross currency swap contracts	Other assets	1.8	—	Other long-term liabilities	—	0.6
Interest rate swap contracts	Prepaid expenses and other current assets	2.7	—	Other accrued expenses	—	1.7
Interest rate swap contracts	Other assets	0.5	—	Other long-term liabilities	—	1.1
Total undesignated derivatives		$ 6.0	$ 1.9		$ 0.9	$ 4.2
Total derivatives		$ 115.3	$ 4.7		$ 0.9	$ 32.8

17. Accumulated other comprehensive loss

The following table presents the changes in accumulated other comprehensive loss by component, net of tax:

(in millions)	Cash flow hedges	Defined benefit pension	Currency translation	Total AOCI
Balance as of January 1, 2020	$ (15.4)	$ (1.0)	$ (362.9)	$ (379.3)
Other comprehensive (loss) income before reclassifications [1]	(45.6)	20.3	(10.7)	(36.0)
Amounts reclassified from accumulated other comprehensive loss	28.3	(0.1)	—	28.2
Net current period other comprehensive (loss) income	(17.3)	20.2	(10.7)	(7.8)
Balance as of December 31, 2020	$ (32.7)	$ 19.2	$ (373.6)	$ (387.1)
Other comprehensive income (loss) before reclassifications	29.8	0.4	(13.1)	17.1
Amounts reclassified from accumulated other comprehensive loss	(7.9)	(2.9)	—	(10.8)
Amounts reclassified related to dispositions [2]	—	—	18.1	18.1
Net current period other comprehensive income (loss)	21.9	(2.5)	5.0	24.4
Balance as of December 31, 2021	$ (10.8)	$ 16.7	$ (368.6)	$ (362.7)
Other comprehensive income (loss) before reclassifications	105.5	—	(64.4)	41.1
Amounts reclassified from accumulated other comprehensive loss	(22.5)	(0.6)	—	(23.1)
Net current period other comprehensive income (loss)	83.0	(0.6)	(64.4)	18.0
Balance as of December 31, 2022	$ 72.2	$ 16.1	$ (433.0)	$ (344.7)

(1) Defined benefit pension includes a $ 25.0 million pre-tax adjustment related to a plan amendment for the UK Plan.
(2) In conjunction with the sale of the Distrupol business, the Company released the associated cumulative foreign currency translation losses and included the release as part of the net gain on sale of business.

The following table is a summary of the amounts reclassified from accumulated other comprehensive loss to net income:

		Year ended December 31,		
(in millions)	Statement of Operations Classification	2022 [1]	2021 [1]	2020 [1]
Amortization of defined benefit pension items:				
Prior service credit	Other income (expense), net	$ (0.7)	$ (3.5)	$ (0.1)
Tax expense	Income tax expense	0.1	0.6	—
Net of tax		(0.6)	(2.9)	(0.1)
Cash flow hedges:				
Interest rate swap contracts	Interest expense	0.1	17.9	12.9
Cross-currency swap contracts	Interest expense and other income (expense), net	(30.3)	(28.6)	28.3
Tax expense (benefit)	Income tax expense	7.7	2.8	(12.9)
Net of tax		(22.5)	(7.9)	28.3
Total reclassifications for the period, net of tax		$ (23.1)	$ (10.8)	$ 28.2

(1) Amounts in parentheses represent income in the consolidated statements of operations.

18. Commitments and contingencies

Litigation

In the ordinary course of business, the Company is subject to pending or threatened claims, lawsuits, regulatory matters and administrative proceedings from time to time. Where appropriate the Company has recorded provisions in the consolidated financial statements for these matters. The liabilities for injuries to persons or property are in some instances covered by liability insurance, subject to various deductibles and self-insured retentions.

Other than as disclosed, the Company is not aware of any claims, lawsuits, regulatory matters or administrative proceedings, pending or threatened, that are likely to have a material effect on its overall financial position, results of operations, or cash flows. However, the Company cannot predict the outcome of any present or future claims or litigation or the potential for future claims or litigation and adverse developments could negatively impact earnings or cash flows in a particular future period.

Asbestos Claims

The Company is subject to liabilities from claims alleging personal injury from exposure to asbestos. The claims result primarily from an indemnification obligation related to Univar Solutions USA Inc.'s ("Univar") 1986 purchase of McKesson Chemical Company from McKesson Corporation ("McKesson"). Univar is pursuing insurance coverage for certain matters under McKesson's historical insurance coverage to partially offset the impact of any fees, settlements, or judgments that Univar is obligated to pay because of its obligation to McKesson. As of December 31, 2022, there were approximately 232 asbestos-related cases for which Univar has the obligation to defend and indemnify; however, this number tends to fluctuate up and down over time. Historically, the vast majority of these asbestos cases have been dismissed without payment or with a nominal payment. While the Company is unable to predict the outcome of these matters, it does not believe, based upon currently available facts, that the ultimate resolution of any of these matters will have a material effect on its overall financial position, results of operations or cash flows.

Unclaimed Property Audit

The Company and its subsidiaries were the subject of an unclaimed property audit request issued by the State of Delaware in 2015. On June 29, 2022, the State of Delaware notified the Company that it was closing its examination with a $0 finding. On September 13, 2022, the parties executed a confidential settlement agreement and release.

Canada Revenue Agency

In October 2022, the Company received notice from the Canada Revenue Agency ("CRA") proposing that certain historical financing transactions between one of the Company's Canadian subsidiaries (Univar Canada Ltd.) and one of the Company's US subsidiaries (Univar Holdco Canada LLC) should be recharacterized as equity and not debt for the 2015 and 2016 tax years. The CRA has proposed that certain deductions claimed by the Canadian entity should be denied, resulting in additional tax due, as well as penalties and interest on the unpaid tax. The proposed assessment against the Company, inclusive of interest and penalties of Canadian Dollar ("C$") 22.0 million, totals C$51.0 million.

It is possible that the CRA might take a similar position in relation to two additional tax years (2017 and 2018), but the Company has not received a proposal in relation to those years. The transactions that are being challenged by the CRA for 2015 and 2016 do not apply in periods after 2018.

The Company believes that the tax position previously taken was proper and it will defend itself as appropriate. The Company has not recorded any liabilities in its consolidated financial statements for this matter, as it believes it is more likely than not that the Company's position will be sustained.

Environmental

The Company is subject to various federal, state and local environmental laws and regulations that require environmental assessment or remediation efforts (collectively "environmental remediation work") and from time to time the Company becomes aware of compliance matters regarding possible or alleged violations of these laws or regulations. For example, over the years, the Company has been identified as a "potentially responsible party" ("PRP") under the Comprehensive Environmental Response, Compensation and Liability Act and/or similar state laws that impose liability for costs relating to environmental remediation work at various sites. As a PRP, the Company may be required to pay a share of the costs of investigation and cleanup of certain sites. The Company is currently engaged in environmental remediation work at approximately 128 locations, some that are now or were previously Company-owned/occupied and some that were never Company-owned/occupied ("non-owned sites").

The Company's environmental remediation work at some sites is being conducted pursuant to governmental proceedings or investigations. At other sites, the Company, with appropriate state or federal agency oversight and approval, is conducting the environmental remediation work voluntarily. The Company is currently undergoing remediation efforts or is in the process of active review of the need for potential remediation efforts at approximately 107 current or formerly Company-owned/occupied

sites. In addition, the Company may be liable as a PRP for a share of the clean-up of approximately21 non-owned sites. These non-owned sites are typically (a) locations of independent waste disposal or recycling operations with alleged or confirmed contaminated soil and/or groundwater to which the Company may have shipped waste products or drums for re-conditioning, or (b) contaminated non-owned sites near historical sites owned or operated by the Company or its predecessors from which contamination is alleged to have arisen.

In determining the appropriate level of environmental liabilities, the Company considers several factors such as information obtained from investigatory studies; the scope of remediation (including any changes over time); the interpretation, application and enforcement of laws and regulations; changes in the costs of remediation programs; the development of alternative cleanup technologies and methods; and the relative level of the Company's involvement at various sites for which the Company is allegedly associated. The level of annual expenditures for remedial, monitoring and investigatory activities will change in the future as major components of planned remediation activities are completed and the scope, timing and costs of existing activities are changed. Project lives, and therefore cash flows, may range from 2 to 30 years, depending on the specific site and type of remediation project.

Although the Company believes that its accruals are adequate for environmental contingencies, it is possible, due to the uncertainties noted above, that additional accruals could be required in the future that could have a material effect on the overall financial position, results of operations or cash flows in a particular period.

Changes in total environmental liabilities, which were measured on an undiscounted basis, were as follows:

(in millions)	2022	2021
Environmental liabilities as of January 1	$ 88.1	$ 79.6
Revised obligation estimates	25.9	29.0
Payments	(22.7)	(20.3)
Foreign exchange	(0.4)	(0.2)
Environmental liabilities as of December 31	$ 90.9	$ 88.1

		December 31,	
(in millions)	Balance Sheet Classification	2022	2021
Current environmental liabilities	Other accrued expenses	$ 36.5	$ 39.3
Long-term environmental liabilities	Other long-term liabilities	54.4	48.8

As of December 31, 2022, receivables for insurance recoveries of $6.7 million and $9.3 million were recorded within prepaid expenses and other current assets and other assets, respectively, in the consolidated balance sheets. No receivables for insurance recoveries were recorded as of December 31, 2021. Insurance recoveries of $16.7 million were recorded within warehousing, selling and administrative expenses in the consolidated statements of operations for the year ended December 31, 2022.

The Company manages estimated cash flows by project. These estimates are subject to change if there are modifications to the scope of the remediation plan or if other factors, either external or internal, change the timing of the remediation activities. The Company periodically reviews the status of all existing or potential environmental liabilities and adjusts its accruals based on all available, relevant information. Based on current estimates, the expected payments for environmental remediation for the next five years and thereafter at December 31, 2022 were as follows:

(in millions)	
2023	$ 36.5
2024	12.4
2025	6.9
2026	5.3
2027	4.5
Thereafter	25.3
Total	$ 90.9

19. Leasing

The Company leases certain warehouses and distribution centers, office space, transportation equipment and other machinery and equipment. The majority of the leases have remaining terms of 1 to 8 years. The lease agreements do not contain any material residual value guarantees or material restrictive covenants. The Company subleases certain facilities to third parties.

(in millions)	Balance Sheet Classification	December 31, 2022	December 31, 2021
Assets			
Operating lease assets	Other assets	$ 200.7	$ 164.3
Finance lease assets	Property, plant and equipment, net[1]	107.4	102.1
Total lease assets		$ 308.1	$ 266.4
Liabilities			
Current liabilities:			
Current portion of operating lease liabilities	Other accrued expenses	$ 43.0	$ 45.7
Current portion of finance lease liabilities	Current portion of long-term debt	24.9	27.5
Noncurrent liabilities:			
Operating lease liabilities	Other long-term liabilities	164.4	125.5
Finance lease liabilities	Long-term debt	79.4	74.4
Total lease liabilities		$ 311.7	$ 273.1

(1) Finance lease right-of-use assets are recorded net of accumulated depreciation of $ 61.6 million and $75.8 million as of December 31, 2022 and 2021, respectively.

Lease cost

(in millions)	Year ended December 31, 2022			Year ended December 31, 2021			Year ended December 31, 2020		
Statement of Operations Classification	Operating Leases	Finance Leases	Total	Operating Leases	Finance Leases	Total	Operating Leases	Finance Leases	Total
Cost of goods sold (exclusive of depreciation)	$ 22.6	$ —	$ 22.6	$ 22.6	$ —	$ 22.6	$ 18.8	$ —	$ 18.8
Outbound freight and handling	6.6	—	6.6	6.6	—	6.6	6.1	—	6.1
Warehousing, selling and administrative	27.6	—	27.6	28.3	—	28.3	32.7	—	32.7
Depreciation	—	26.6	26.6	—	26.4	26.4	—	24.9	24.9
Interest expense	—	3.8	3.8	—	3.7	3.7	—	3.4	3.4
Total gross lease component costs	$ 56.8	$ 30.4	87.2	$ 57.5	$ 30.1	87.6	$ 57.6	$ 28.3	85.9
Variable lease costs			1.2			1.7			0.9
Short-term lease costs			7.7			6.7			25.7
Total gross lease costs			96.1			96.0			112.5
Less: sublease income			0.9			2.1			2.5
Total net lease costs			$ 95.2			$ 93.9			$ 110.0

Maturity of lease liabilities

(in millions)	Operating Leases	Finance Leases	Total
2023	$ 53.2	$ 28.7	$ 81.9
2024	39.5	25.2	64.7
2025	30.6	23.2	53.8
2026	27.0	20.0	47.0
2027	22.5	11.5	34.0
2028 and after	93.1	6.3	99.4
Total lease payments	265.9	114.9	380.8
Less: interest	58.5	10.6	69.1
Present value of lease liabilities	$ 207.4	$ 104.3	$ 311.7

Lease term and discount rate

	December 31,	
(in millions)	2022	2021
Weighted-average remaining lease term (years)		
Operating leases	8.3	6.7
Finance leases	6.3	6.5
Weighted-average discount rate		
Operating leases	4.91 %	3.99 %
Finance leases	4.29 %	3.59 %

Other information

	Year ended December 31,		
(in millions)	2022	2021	2020
Cash paid for amounts included in the measurement of lease liabilities			
Operating cash flows from operating leases	$ 57.4	$ 57.5	$ 57.1
Operating cash flows from finance leases	3.8	3.6	3.4
Financing cash flows from finance leases	29.5	27.6	24.3

Sale-leaseback transaction

During 2020, the Company recognized a net gain of $14.4 million associated with a sale-leaseback agreement related to a real estate property. The Company intends to lease the property for a period of 15 years and has classified the agreement as an operating lease.

20. Segments

The Company's operations are structured into four reportable segments that represent the geographic areas under which it operates and manages the business. Management, including the Chief Operating Decision Maker, monitors the operating results of its reportable segments separately for the purpose of making decisions about resource allocation and performance assessment. Management evaluates the performance of its reportable segments on the basis of Adjusted EBITDA. Adjusted EBITDA is defined as the sum of consolidated net income; depreciation; amortization; net interest expense; income tax expense; impairment charges; (gain) loss on sale of business; other operating expenses, net and other income (expense), net (for both, see "Note 8: Supplemental financial information"). For 2020, Adjusted EBITDA also included an adjustment to remove a Brazil VAT charge.

Transfer prices between reportable segments are set on an arms-length basis in a similar manner to transactions with third parties. Corporate operating expenses that directly benefit segments have been allocated to the reportable segments. Allocable operating expenses are identified through a review process by management. The allocable operating expenses are assigned to the reportable segments on a basis that reasonably approximates the use of services, which is generally measured based on a weighted distribution of margin, asset, headcount or time spent.

Financial information for the Company's reportable segments was as follows:

(in millions)	USA	EMEA	Canada	LATAM	Other/ Eliminations[1]	Consolidated
Net sales						
Year ended December 31, 2022						
External customers	$ 7,533.5	$ 2,064.7	$ 1,120.5	$ 756.6	$ —	$ 11,475.3
Inter-segment	129.5	10.6	10.5	0.5	(151.1)	—
Net sales	$ 7,663.0	$ 2,075.3	$ 1,131.0	$ 757.1	$ (151.1)	$ 11,475.3
Year ended December 31, 2021						
External customers	$ 6,024.0	$ 1,971.1	$ 930.0	$ 610.4	$ —	$ 9,535.5
Inter-segment	92.1	4.3	4.4	0.7	(101.5)	—
Net sales	$ 6,116.1	$ 1,975.4	$ 934.4	$ 611.1	$ (101.5)	$ 9,535.5
Year ended December 31, 2020						
External customers	$ 5,006.2	$ 1,697.1	$ 1,110.7	$ 451.0	$ —	$ 8,265.0
Inter-segment	81.2	3.1	2.5	—	(86.8)	—
Net sales	$ 5,087.4	$ 1,700.2	$ 1,113.2	$ 451.0	$ (86.8)	$ 8,265.0

(1) Other/Eliminations represents the elimination of intersegment transactions.

	Year ended December 31,		
(in millions)	2022	2021	2020
Adjusted EBITDA			
USA	$ 709.3	$ 498.2	$ 393.2
EMEA	176.6	170.6	142.7
Canada	119.7	104.2	89.7
LATAM	63.0	56.9	43.0
Other/Eliminations [1]	(22.7)	(32.2)	(32.8)
Consolidated	$ 1,045.9	$ 797.7	$ 635.8

(1) Other/Eliminations represents unallocated corporate costs consisting of items specifically related to parent company operations that do not directly benefit segments, either individually or collectively.

	December 31,	
(in millions)	2022	2021
Long-lived assets [1]		
USA	$ 788.6	$ 781.5
EMEA	191.9	191.2
Canada	173.4	153.3
LATAM	46.2	38.7
Other	55.6	30.6
Consolidated	$ 1,255.7	$ 1,195.3

(1) Long-lived assets consist of property, plant and equipment, net and operating lease assets.

The following table is a reconciliation of net income to Adjusted EBITDA for the years ended December 31, 2022, 2021 and 2020:

	Year ended December 31,		
(in millions)	2022	2021	2020
Net income	$ 545.3	$ 460.6	$ 52.9
Depreciation	131.7	150.9	162.9
Amortization	48.2	52.5	60.0
Interest expense, net	102.9	97.2	112.4
Income tax expense	210.9	124.6	6.1
EBITDA	1,039.0	885.8	394.3
Other operating expenses, net	32.8	107.5	89.4
Other (income) expense, net	(26.5)	(110.4)	61.0
Impairment charges	0.6	3.0	40.2
(Gain) loss on sale of business	—	(88.2)	50.6
Brazil VAT charge	—	—	0.3
Adjusted EBITDA	$ 1,045.9	$ 797.7	$ 635.8

Business line information

The Company’s net sales from external customers primarily relate to its chemical distribution business. Other sales to external customers primarily relate to services for collecting and arranging for the transportation of hazardous and non-hazardous waste.

Risks and Concentrations

No single customer accounted for 10% or more of net sales in any of the years presented.

The Company has portions of its labor force that are a part of collective bargaining agreements. A work stoppage or other limitation on operations could occur as a result of disputes under existing collective bargaining agreements with labor unions or government-based work counsels or in connection with negotiations of new collective bargaining agreements. As of December 31, 2022, approximately 24% of the Company’s labor force is covered by a collective bargaining agreement. As of December 31, 2022, approximately 1% of the Company’s labor force is covered by a collective bargaining agreement that will expire withinone year.

Other segment information

Information on total assets by segment is not disclosed, as the Company's Chief Operating Decision Maker does not evaluate reportable segments using total asset information.

21. Subsequent events

Kale Kimya Agreement

In January 2023, the Company entered into an agreement to acquire Kale Kimya, a leading regional specialty chemical distributor in Turkey, for cash consideration of €125 million, subject to closing adjustments, and contingent consideration of up to €7 million. The transaction is expected to close in the second quarter of 2023.

ChemSol Group Acquisition

In February 2023, the Company acquired ChemSol Group, a leading ingredients and specialty chemical distributor in Central America, for cash consideration of approximately $19 million, subject to closing adjustments, and contingent consideration of up to $5 million.

Schedule II - Valuation and qualifying accounts

(in millions)	Balance at beginning of period	Additions: Charged to costs and expenses	Additions: Charged to other accounts	Deductions	Balance at end of period
Year ended December 31, 2022					
Income tax valuation allowance	$ 8.6	$ 5.8	$ (1.4)	$ (0.3)	$ 12.7
Year ended December 31, 2021					
Income tax valuation allowance	$ 19.9	$ 0.7	$ (0.3)	$ (11.7)	$ 8.6
Year ended December 31, 2020					
Income tax valuation allowance	$ 87.5	$ 3.0	$ 1.0	$ (71.6)	$ 19.9

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

As of December 31, 2022, we carried out an evaluation, under the supervision and with the participation of our management, including our Chief Executive Officer ("CEO") and our Chief Financial Officer ("CFO"), of the effectiveness of the design and operation of our disclosure controls and procedures (as such term is defined in Rule 13a-15(e) under the Securities Exchange Act of 1934 as amended). Based upon that evaluation, our Chief Executive Officer and our Chief Financial Officer concluded that our disclosure controls and procedures are designed at a reasonable assurance level and are effective to provide reasonable assurance that information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized, and reported within the time periods specified in the rules and forms of the Securities and Exchange Commission ("SEC"), and that such information is accumulated and communicated to our management, including our CEO and CFO, as appropriate, to allow timely decisions regarding required disclosure.

Changes in Internal Control over Financial Reporting

There were no changes in our internal control over financial reporting identified in management's evaluation pursuant to Rules 13a-15(d) of the Exchange Act during the quarter ended December 31, 2022 that materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Management's Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting for the company. Internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of our financial reporting for external purposes in accordance with accounting principles generally accepted in the United States of America. Internal control over financial reporting includes maintaining records that in reasonable detail accurately and fairly reflect our transactions; providing reasonable assurance that transactions are recorded as necessary for preparation of our financial statements; providing reasonable assurance that receipts and expenditures of company assets are made in accordance with management authorization; and providing reasonable assurance that unauthorized acquisition, use, or disposition of company assets that could have a material effect on our financial statements would be prevented or detected on a timely basis. Because of its inherent limitations, internal control over financial reporting is not intended to provide absolute assurance that a misstatement of our financial statements would be prevented or detected.

Management conducted an assessment of the effectiveness of the Company's internal control over financial reporting based on the criteria set forth in Internal Control – Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework). Based on the Company's assessment, management has concluded that its internal control over financial reporting was effective as of December 31, 2022 to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements in accordance with GAAP. The Company's independent registered public accounting firm, Ernst & Young LLP, has issued an audit report on the Company's internal control over financial reporting, which appears in Part II, Item 8 of this Annual Report on Form 10-K.

ITEM 9B. OTHER INFORMATION

None.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

None.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

All information required by this Item will be included in our Proxy Statement relating to our 2023 Annual Meeting of Stockholders to be filed pursuant to Regulation 14A within 120 days after the end of the fiscal year ended December 31, 2022 ("2023 Proxy Statement") and is incorporated herein by reference.*

ITEM 11. EXECUTIVE COMPENSATION

All information required by this Item will be included in our 2023 Proxy Statement and is incorporated herein by reference.*

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

All information required by this Item will be included in our 2023 Proxy Statement and is incorporated herein by reference.*

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS, AND DIRECTOR INDEPENDENCE

All information required by this Item will be included in our 2023 Proxy Statement and is incorporated herein by reference.*

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

All information required by this Item will be included in our 2023 Proxy Statement and is incorporated herein by reference.*

**Except for information or data specifically incorporated herein by reference under Items 10 through 14, other information and data appearing in our 2023 Proxy Statement are not deemed to be a part of this Annual Report on Form 10-K or deemed to be filed with the SEC as part of this report.*

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

(a)(1)(2) Financial Statements and Financial Statement Schedules

Reference is made to the information set forth in Part II, Item 8 of this Annual Report on Form 10-K, which information is incorporated herein by reference.

(a)(3) Exhibits

Exhibit Number	Exhibit Description
2.1	Agreement and Plan of Merger, dated September 17, 2018, by and among Nexeo, Univar, Pilates Merger Sub I Corp and Pilates Merger Sub II LLC, incorporated by reference to Exhibit 2.1 to the Current Report on Form 8-K of the Company, filed on September 18, 2018.
2.2	Purchase and Sale Agreement, by and among Nexeo Solutions, Inc., Neon Holdings, Inc. and Univar Inc., dated as of February 8, 2019, incorporated by reference to Exhibit 2.1 to the Current Report on Form 8-K of the Company filed on March 1, 2019.
3.1	Restated Third Amended and Restated Certificate of Incorporation of the Company, incorporated by reference to Exhibit 3.1 to the Form 10-K of the Company filed on February 25, 2022.
3.2	Fourth Amended and Restated Bylaws of the Company effective May 6, 2021, incorporated by reference to Exhibit 3.5 to the Form 10-Q of the Company filed on August 3, 2021.
4.1	Form of Common Stock Certificate, incorporated by reference to Exhibit 4.1 to the Registration Statement on Form S-1 of the Company, filed on June 8, 2015.
4.2	Description of Univar Solutions Inc. Securities registered pursuant to Section 12 of the Securities Exchange Act of 1934, incorporated by reference to Exhibit 4.2 to the Form 10-K of the Company, filed on February 25, 2022.
4.3	Indenture, dated as of November 22, 2019, between Univar Solutions USA Inc., Univar Solutions Inc., the guarantors listed on the signature pages thereto and U.S. Bank National Association, incorporated by reference to Exhibit 4.1 to the Current Report on Form 8-K of the Company, filed on November 22, 2019.

4.4	First Supplemental Indenture, dated as of November 22, 2019, between Univar Solutions USA Inc., Univar Solutions Inc., the guarantors listed on the signature pages thereto and U.S. Bank National Association, incorporated by reference to Exhibit 4.2 to the Current Report on Form 8-K of the Company filed on November 22, 2019.
4.5	Form of 5.125% Senior Notes due 2027 (included in Exhibit 4.3 hereto), incorporated by reference to Exhibit 4.3 to the Current Report on Form 8-K of the Company, filed on November 22, 2019.
10.1	Credit Agreement, dated as of July 1, 2015 between Univar USA Inc., Univar Inc., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.1 to the Current Report on Form 8-K of the Company, filed on July 7, 2015.
10.2	First Amendment to Credit Agreement, dated as of January 19, 2017 between Univar USA Inc., Univar Inc., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.1 to the Current Report on Form 8-K of the Company, filed on January 20, 2017.
10.3	Second Amendment to Credit Agreement, dated as of November 28, 2017 between Univar USA Inc., Univar Inc., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.1 to the Current Report on Form 8-K of the Company, filed November 29, 2017.
10.4	Third Amendment to Credit Agreement, dated as of February 23, 2019, between Univar USA Inc., Univar Inc., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.5 to the Form 10-K of the Company, filed on February 25, 2020.
10.5	Fourth Amendment to Credit Agreement, dated as of February 28, 2019 between Univar USA Inc., Univar Inc., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.2 to the Current Report on Form 8-K of the Company, filed on March 1, 2019.
10.6	Fifth Amendment to Credit Agreement, dated November 22, 2019, between Univar Solutions USA Inc., Univar Solutions Inc., Univar Netherlands Holding B.V, the several banks and financial institutions from time to time party thereto, Goldman Sachs Bank USA and Bank of America, N.A., to the Credit Agreement dated July 1, 2015, between Univar Solutions USA Inc., Univar Solutions Inc., Univar Netherlands Holding B.V., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.1 to the Current Report on Form 8-K of the Company, filed on November 22, 2019.
10.7	Sixth Amendment to Credit Agreement, dated June 3, 2021, between Univar Solutions USA Inc., Univar Solutions Inc., Univar Netherlands Holding B.V., the several banks and financial institutions from time to time party thereto, JPMorgan Chase Bank, N.A. and Bank of America, N.A., to the Credit Agreement dated July 1, 2015, between Univar Solutions USA Inc., Univar Solutions Inc., Univar Netherlands Holding B.V., the several banks and financial institutions from time to time party thereto and Bank of America, N.A., incorporated by reference to Exhibit 10.1 to the Current Report on Form 8-K of the Company, filed on June 9, 2021.
10.8	Term Loan Guarantee and Collateral Agreement, dated as of July 1, 2015, made by Univar Inc., Univar USA Inc. and the guarantors listed on the signature pages thereto in favor of Bank of America, N.A, as collateral agent for the banks and other financial institutions that are parties to the Credit Agreement, incorporated by reference to Exhibit 10.2 to the Current Report on Form 8-K of the Company, filed on July 7, 2015.
10.9	Amendment No. 1 to Term Loan Guarantee and Collateral Agreement, dated as of November 22, 2019, made by Univar Solutions Inc., Univar Solutions USA Inc. and the guarantors listed on the signature pages thereto in favor of Bank of America, N.A, as collateral agent, incorporated by reference to Exhibit 10.9 to the Form 10-K of the Company, filed on February 25, 2020.
10.10	Second Amended and Restated ABL Credit Agreement, dated as of October 27, 2022, by and among Univar Solutions Inc. and certain of its subsidiaries, the several banks and financial institutions from time to time party thereto and Bank of America, N.A., as administrative agent, incorporated by reference to Exhibit 10.1 to the Current Report on Form 8-K of the Company, filed on October 31, 2022.
10.11*	Second Amended and Restated ABL Guarantee and Collateral Agreement, dated as of October 27, 2022, made by the Company and certain of its Domestic Subsidiaries in favor of Bank of America, N.A, as collateral agent.
10.12†	Form of Director Indemnification Agreement, incorporated by reference to Exhibit 10.56 to the Registration Statement on Form S-1 of the Company, filed on June 8, 2015.

10.13†	Form of Employee Stock Option Agreement, incorporated by reference to Exhibit 10.34 to the Registration Statement on Form S-1 of the Company, filed on August 14, 2014.
10.14†	2014 Form of Employee Stock Option Agreement, incorporated by reference to Exhibit 10.62 to the Registration Statement on Form S-1 of the Company, filed on May 26, 2015.
10.15†	Form of Employee Stock Option Agreement for awards granted between June 23, 2015 and February 1, 2017, 2015 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.5 to the Registration Statement on Form S-8 of the Company, filed on June 23, 2015.
10.16†	Form of Employee Stock Option Agreement for awards granted after February 1, 2017, 2015 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.67 to the Form 10-K of the Company filed on February 28, 2017.
10.17†	Form of Employee Stock Option Agreement for awards granted after April 13, 2017, 2015 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.1 to the Form 10-Q of the Company filed on May 5, 2017.
10.18†	Form of Employee Stock Option Agreement, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.3 to the Form 10-Q of the Company filed on May 5, 2017.
10.19†	Form of Employee Stock Option Agreement for awards granted on or after February 7, 2018, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.2 to the Form 10-Q of the Company, filed on May 10, 2018.
10.20†	Stock Option Agreement, dated as of February 7, 2018, by and between Univar Inc. and Stephen D. Newlin. 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.3 to the Form 10-Q of the Company, filed on May 10, 2018.
10.21†	Form of Employee Stock Option Agreement for awards granted on or after February 6, 2019, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.4 to the Form 10-Q of the Company, filed on May 9, 2019.
10.22†	Form of Employee Stock Option Agreement for awards granted on or after February 21, 2020, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.23 to the Form 10-K of the Company, filed on February 25, 2020.
10.23†	Form of Employee Stock Option Agreement for awards granted on or after June 26, 2020, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.1 to the Form 10-Q of the Company, filed on August 7, 2020.
10.24†	Form of Employee Restricted Stock Unit Agreement for awards granted on or after February 21, 2020, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.31 to the Form 10-K of the Company, filed on February 25, 2020.
10.25†	Form of Employee Restricted Stock Unit Agreement for awards granted on or after June 26, 2020, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.2 to the Form 10-Q of the Company, filed on August 7, 2020.
10.26†	Univar Solutions Supplemental Savings Plan (previously named Univar USA Inc. Supplemental Valued Investment Plan), effective June 1, 2017, incorporated by reference to Exhibit 10.32 to the Form 10-K of the Company, filed on February 25, 2020.
10.27†	First Amendment to the Univar Solutions Supplemental Savings Plan, dated October 9, 2018, incorporated by reference to Exhibit 10.33 to the Form 10-K of the Company, filed on February 25, 2020.
10.28†	Second Amendment to the Univar Solutions Supplemental Savings Plan, dated December 30, 2019, incorporated by reference to Exhibit 10.34 to the Form 10-K of the Company, filed on February 25, 2020.
10.29†	Univar USA Inc. Supplemental Benefits Retirement Plan, dated as of July 1, 2004, incorporated by reference to Exhibit 10.45 to the Registration Statement on Form S-1 of the Company, filed on August 14, 2014.
10.30†	First Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of May 17, 2005, incorporated by reference to Exhibit 10.30 to the Form 10-K of the Company, filed on March 3, 2016.

10.31†	Second Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of August 24, 2006, incorporated by reference to Exhibit 10.31 to the Form 10-K of the Company, filed on March 3, 2016.
10.32†	Third Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of June 11, 2007, incorporated by reference to Exhibit 10.32 to the Form 10-K of the Company, filed on March 3, 2016.
10.33†	Fourth Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of December 6, 2007, incorporated by reference to Exhibit 10.46 to the Registration Statement on Form S-1 of the Company, filed on August 14, 2014.
10.34†	Fifth Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of December 6, 2007, incorporated by reference to Exhibit 10.34 to the Form 10-K of the Company, filed on March 3, 2016.
10.35†	Sixth Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of December 19, 2007, incorporated by reference to Exhibit 10.35 to the Form 10-K of the Company, filed on March 3, 2016.
10.36†	Seventh Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of June 19, 2008, incorporated by reference to Exhibit 10.36 to the Form 10-K of the Company, filed on March 3, 2016.
10.37†	Eighth Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of December 23, 2008, incorporated by reference to Exhibit 10.37 to the Form 10-K of the Company, filed on March 3, 2016.
10.38†	Ninth Amendment to the Univar USA Inc. Supplemental Retirement Plan, dated as of December 21, 2009, incorporated by reference to Exhibit 10.38 to the Form 10-K of the Company, filed on March 3, 2016.
10.39†	Univar Inc. 2011 Stock Incentive Plan, effective as of March 28, 2011, incorporated by reference to Exhibit 10.32 to the Registration Statement on Form S-1 of the Company, filed on August 14, 2014.
10.40†	Amendment No. 1 to the Univar Inc. 2011 Stock Incentive Plan, dated as of November 30, 2012, incorporated by reference to Exhibit 10.33 to the Registration Statement on Form S-1 of the Company, filed on August 14, 2014.
10.41†	Univar Inc. 2015 Omnibus Equity Incentive Plan is incorporated by reference to Exhibit 10.3 to the Registration Statement on Form S-8 of the Company, filed June 23, 2015.
10.42†	Univar Inc. 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.6 to the Form 10-Q of the Company filed on May 5, 2017.
10.43†	First Amendment to Univar Inc. 2017 Omnibus Equity Incentive Plan dated as of December 6, 2019, incorporated by reference to Exhibit 10.50 to the Form 10-K of the Company, filed on February 25, 2020.
10.44†	Univar Solutions Inc. 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 4.5 to the Registration Statement on Form S-8 of the Company, filed on May 7, 2020.
10.45†	Letter Agreement, by and between Nick Powell and Univar Inc., dated as of February 27, 2019, incorporated by reference to Exhibit 5.1 to the Current Report on Form 8-K of the Company, filed on March 1, 2019.
10.46†	Letter Agreement between the Company and Nick Powell, incorporated by reference to Exhibit 10.1 to the Form 10-Q of the Company, filed on September 17, 2021.
10.47†	Letter Agreement, by and between Nick Powell and Univar Solutions Inc., dated as of September 12, 2022, incorporated by reference to Exhibit 10.1 to the Current Report on Form 10-Q, filed on November 2, 2022.
10.48*†	Form of Repayment Agreement, by and between Univar Solutions Inc. and Certain Executives.
10.49†	Letter Agreement between the Company and Jennifer McIntyre, incorporated by reference to Exhibit 10.2 to the Form 10-Q of the Company, filed on November 2, 2021.
10.50†	Form of Severance and Change in Control Agreement by and Between Univar Inc. and Certain Executives, incorporated by reference to Exhibit 10.3 to the Form 10-Q of the Company, filed on November 6, 2018.

10.51†	Severance and Change of Control Agreement, dated as of January 6, 2020, between the Company and Nicholas W. Alexos, incorporated by reference to Exhibit 10.2 to the Current Report on Form 8-K of the Company, filed on December 16, 2019.
10.52†	Form of Severance and Change in Control Agreement by and Between Univar Solutions Inc. and Certain Executives, incorporated by reference to Exhibit 10.1 to the Form 10-Q of the Company, filed on November 5, 2020.
10.53†	Amended and Restated Univar Solutions Inc. Employee Stock Purchase Plan, incorporated by reference to Exhibit 10.51 to the Form 10-K of the Company, filed on February 25, 2021.
10.54†	First Amendment to Univar Solutions Inc. Employee Stock Purchase Plan dated as of December 6, 2019, incorporated by reference to Exhibit 10.56 to the Form 10-K of the Company, filed on February 25, 2020.
10.55†	Second Amendment to Univar Solutions Inc. Employee Stock Purchase Plan executed as of October 28, 2020, incorporated by reference to Exhibit 10.53 to the Form 10-K of the Company, filed on February 25, 2021.
10.56†	Form of Employee Performance Based Restricted Stock Unit Agreement for awards granted on or after February 21, 2020, incorporated by reference to Exhibit 10.7 to the Form 10-Q of the Company, filed on August 7, 2020.
10.57†	Form of Employee Performance Based Restricted Stock Unit Agreement for awards granted on or after March 10, 2021, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.1 to the Form 10-Q of the Company, filed on May 10, 2021.
10.58†	Form of Director Deferred Share Unit Agreement for awards granted on or after February 7, 2018, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.8 to the Form 10-Q of the Company, filed on May 10, 2018.
10.59†	Form of Director Deferred Share Unit Agreement for awards granted on or after February 7, 2019, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.7 to the Form 10-Q of the Company, filed on May 9, 2019.
10.60†	Form of Director Deferred Share Unit Agreement for cash retainer granted on or after February 21, 2020, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.64 to the Form 10-K of the Company, filed on February 25, 2020.
10.61†	Form of Director Deferred Share Unit Agreement for equity awards granted on or after February 21, 2020, 2017 Omnibus Equity Incentive Plan, incorporated by reference to Exhibit 10.65 to the Form 10-K of the Company, filed on February 25, 2020.
10.62†	Form of Director Deferred Share Unit Agreement for cash retainer granted on or after June 26, 2020, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.3 to the Form 10-Q of the Company, filed on August 7, 2020.
10.63†	Form of Director Deferred Share Unit Agreement for equity awards granted on or after June 26, 2020, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.4 to the Form 10-Q of the Company, filed on August 7, 2020.
10.64†	Form of Director Restricted Stock Agreement for awards granted on or after June 26, 2020, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.5 to the Form 10-Q of the Company, filed on August 7, 2020.
10.65†	Form of Director Restricted Stock Unit Agreement for awards granted on or after June 26, 2020, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.6 to the Form 10-Q of the Company, filed on August 7, 2020.
10.66†	Univar Inc. Omnibus Waiver regarding Whistleblower Protections, dated as of May 3, 2017, incorporated by reference to Exhibit 10.8 to the Form 10-Q of the Company filed on May 5, 2017.
10.67†	Form of Indemnification Agreement by and Between Univar Solutions Inc. and Certain Executives, incorporated by reference to Exhibit 10.2 to the Form 10-Q of the Company, filed on November 5, 2020.
10.68†	Form of Employee Restricted Stock Unit Agreement for awards granted on or after February 7, 2022, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.75 to the Form 10-K of the Company, filed on February 25, 2020.

10.69†	Form of Employee Performance-Based Restricted Stock Unit Agreement for awards granted on or after February 7, 2022, 2020 Omnibus Incentive Plan, incorporated by reference to Exhibit 10.76 to the Form 10-K of the Company, filed on February 25, 2022.
21.1*	List of Subsidiaries
23.1*	Consent of Ernst & Young LLP, Independent Registered Public Accounting Firm
31.1*	Certification of Chief Executive Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
31.2*	Certification of Chief Financial Officer Pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
32.1**	Certification of Chief Executive Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
32.2**	Certification of Chief Financial Officer Pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
101.INS	Inline XBRL Instance Document - the instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document
101.SCH	Inline XBRL Taxonomy Extension Schema Document
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document
104	Cover Page Interactive Data File (formatted as Inline XBRL and contained in Exhibit 101)

† Identifies each management compensation plan or arrangement.
* Filed herewith.
** Furnished herewith.

ITEM 16. FORM 10-K SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

Univar Solutions Inc.

By: /s/ NICHOLAS W. ALEXOS

Nicholas W. Alexos, *Executive Vice President and Chief Financial Officer*

Dated February 22, 2023

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the date indicated.

By: /s/ DAVID C. JUKES

David C. Jukes, *President and Chief Executive Officer (Principal Executive Officer)*

By: /s/ CHRISTOPHER D. PAPPAS

Christopher D. Pappas, *Chairman of the Board*

By: /s/ JOAN BRACA

Joan Braca, *Director*

By: /s/ DANIEL P. DOHENY

Daniel P. Doheny, *Director*

By: /s/ RHONDA GERMANY

Rhonda Germany, *Director*

By: /s/ STEPHEN D. NEWLIN

Stephen D. Newlin, *Director*

By: /s/ ROBERT L. WOOD

Robert L. Wood, *Director*

By: /s/ NICHOLAS W. ALEXOS

Nicholas W. Alexos, *Executive Vice President and Chief Financial Officer (Principal Financial Officer and Principal Accounting Officer)*

By: /s/ MARK J. BYRNE

Mark J. Byrne, *Director*

By: /s/ RICHARD P. FOX

Richard P. Fox, *Director*

By: /s/ VARUN LAROYIA

Varun Laroyia, *Director*

By: /s/ KERRY PREETE

Kerry Preete, *Director*